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SYSTEM ARCHITECTURE & BUSINESS PROCESS RE-ENGINEERING REPORT



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LIST OF ACRONYMS AND ABBREVIATIONS

AFN — Afghan Afghani (National Currency)

ARD — Afghanistan Revenue Department

BPMN — Business Process Model and Notation

BPR — Business Process Re-engineering

BRT — Business Receipts Tax

CSV — Comma-Separated Values

ED — Executive Director

ERP — Enterprise Resource Planning

GRN — Goods Receipt Note

HQ — Headquarter

IAS — International Accounting Standards

IC — Internal Control (and Audit)

IFRS — International Financial Reporting Standards

ISO — International Organization for Standardization

RBAC — Role-Based Access Control

EXECUTIVE SUMMARY

The North Coal State Owned Corporation is undergoing a pivotal digital transformation to overcome systemic inefficiencies that have accumulated over decades of manual operations. As a massive, geographically dispersed enterprise, the corporation operates through a single central headquarter and multiple sub-offices, designated as "Second Units." These Second Units encompass active mining sites that are responsible for both coal extraction and direct sales, alongside regional hubs dedicated to distribution and revenue generation. Despite the massive scale of these operations, the organization currently relies on an outdated, paper-intensive legacy framework that heavily restricts growth, transparency, and agility.

To bridge this operational gap, Crest Point Consulting Group has been engaged to conduct a comprehensive system architecture and business process re-engineering overhaul. The core methodology of this engagement relies on defining the **"As-Is"** operational state, identifying its critical failures, and architecting a highly optimized **"To-Be"** digital infrastructure.

The "As-Is" Operational Environment: The current "As-Is" state of the North Coal State Owned Corporation is characterized by severe fragmentation. Departments operate in complete isolation, generating and storing data exclusively for their own localized use rather than feeding it into a unified corporate system. This lack of integration creates massive bottlenecks between the headquarter and the Second Units. Currently, the gathering of crucial operational data, specifically real-time revenue collection and coal stock inventory from the active mining and sales sites is virtually impossible. The transmission of financial and inventory reports from these dispersed units to the central office routinely takes between 6 to 12 months, and in many instances, extends far beyond that timeframe. This profound latency deprives executive leadership of the real-time visibility required to make informed financial, strategic, and operational decisions, while simultaneously exposing the corporation to significant risks of data loss, non-compliance, and revenue leakage.

The "To-Be" Digital Architecture: The objective of this engagement is to completely re-engineer these disjointed legacy processes into a streamlined, integrated "To-Be" state. By mapping out every existing workflow and restructuring it to eliminate redundancies, Crest Point Consulting Group is providing the exact blueprint required by the software developers to build a modern, centralized digital management system. The "To-Be" architecture will connect all Second Units directly to the headquarter, breaking down departmental barriers and enabling instantaneous, real-time reporting for revenue, inventory, procurement, and human resources.

Furthermore, to guarantee that the final product meets international quality management frameworks (ISO 9001), strict financial reporting standards (IFRS), and the precise operational needs of the corporation, Crest Point Consulting Group retains final oversight.

The digital system engineered by the development team is subject to strict, rigorous review and must receive formal approval from Crest Point before it can be officially launched and implemented across the organization.

1. INTRODUCTION

1.1. Project Background

The administrative architecture of the North Coal State Owned Corporation currently relies on physical ledgers, localized spreadsheets, and the manual transportation of hard-copy documents between the Second Units and the central headquarter. Because there is no shared digital infrastructure, each department has developed its own isolated methods for recording transactions, tracking inventory, and processing approvals.

This engagement was initiated to systematically audit these localized methods, strip away redundancies, and standardize every departmental process. By establishing a uniform set of rules and compliance gates across the organization, Crest Point Consulting Group is laying the required foundational logic that software developers will use to build a centralized, relational database and digital user interface.

1.2. Strategic Objectives

The primary objective of this engagement is to architect a system that transitions the North Coal State Owned Corporation from a reactive, paper-dependent entity into a proactive, data-driven enterprise. Specific objectives include:

- **Real-Time Data Integration:** Eradicate the 6-to-12-month reporting lag by establishing a unified digital framework where revenue and inventory data from all Second Units synchronize instantaneously with the headquarter.
- **Process Standardization:** Eliminate departmental isolation by enforcing uniform, standardized workflows across all operational units.
- **Regulatory & Quality Compliance:** Ensure the newly designed workflows strictly adhere to International Financial Reporting Standards (IFRS) and International Accounting Standards (IAS), national legal frameworks, and ISO 9001 quality management principles.
- **Developer Blueprinting:** Translate these optimized processes into precise, unambiguous graphical workflows using Business Process Model and Notation (BPMN), clearly displaying triggers, decision gates, data inputs, and Internal Control verification checkpoints for every departmental workflow.

1.3. Scope of Work

This business process re-engineering blueprint encompasses the workflow mapping, form standardization, and system logic design for the entire organizational network, including the headquarter and all Second Units. The departmental scope includes:

1. Finance and Accounting
2. Inventory and Warehouse Management
3. Payroll
4. Sales and Marketing
5. Properties
6. Procurement
7. Planning and Policy

Note: Internal Control and Audit is not treated as a standalone module but is rather embedded as a mandatory verification layer across all the above departments.

1.4. Scope Exclusions and Limitation of Liability

This Business Process Re-engineering (BPR) and System Architecture Blueprint has been exclusively prepared by Crest Point Consulting Group to define the operational logic, workflow standardization, and internal controls required for the North Coal State Owned Corporation.

The scope of Crest Point Consulting Group's responsibility is strictly limited to architectural design, process mapping, and strategic advisory. We are not responsible for the technical coding, software programming, server deployment, hardware installation, or the final execution of the software. Crest Point Consulting Group assumes no liability for any system bugs, data loss, security breaches, or operational failures that occur during or after the software development and implementation phases. The designated software development firm assumes total responsibility for building the system accurately according to this blueprint and ensuring its technical functionality and security.

2. METHODOLOGY AND APPROACH

To ensure the new digital system accurately reflected the operational realities of the North Coal State Owned Corporation while simultaneously eliminating legacy inefficiencies, Crest Point Consulting Group employed a rigorous, evidence-based three-phase methodology. This structured approach guaranteed a seamless transition from the manual environment to a fully integrated digital enterprise.

2.1. Phase 1: "As-Is" Discovery & Certification (Current State)

The foundation of the re-engineering process required a precise, ground-level understanding of how the company operated, both at the central headquarter and across all Second Units.

- **Comprehensive Staff Interviews:** In-depth consultations were conducted with individual staff members across all in-scope departments. During these interviews, the Crest Point team meticulously mapped out:
    - Their specific daily and weekly tasks and activities.
    - The exact reports, documents, and forms they were responsible for preparing.
    - The precise lifecycle of their workflows—identifying exactly where a task or document originated (inputs) and where it was routed to next (outputs).
- **Pain Point Identification & User Suggestions:** Staff members were asked to detail their current operational frustrations and the specific challenges they faced with the manual, paper-intensive system. Furthermore, the consulting team actively documented their suggestions and expectations regarding how the new digital system should assist them, ensuring the final architecture resolved real-world problems.
- **Direct Operational Observation & Site Verification:** Beyond verbal consultations, the consulting team executed physical inspections at a selected sample of Second Units and conducted active, on-site observations within each headquarter department. This direct shadowing approach allowed the team to witness the work firsthand, verifying reported daily activities against actual ground-level execution and ensuring a complete, unfiltered understanding of the physical workflows and operational environment.
- **Formal Certification:** To prevent scope creep and ensure absolute accountability, every interview concluded with a formal sign-off. The interviewed staff members and department heads signed a physical declaration certifying that all existing workflows, forms, challenges, and responsibilities had been fully and transparently disclosed to the Crest Point team.

2.2. Phase 2: Process Re-engineering ("To-Be" Future State)

Once the "As-Is" baseline was certified, Crest Point analyzed the data to strip away non-value-added steps and design the modernized "To-Be" digital infrastructure.

- **Workflow Optimization:** In the new architectural design, manual approvals and physical document transfers were replaced with automated, digital authorization gates. The architecture was designed to connect the Second Units directly to the headquarter, establishing real-time data transmission for revenue, sales, and inventory.

- **Compliance Integration:** The newly engineered workflows were structurally embedded with necessary controls to ensure strict adherence to ISO 9001 quality management principles, IFRS/IAS accounting standards, and national laws.
- **Role & Access Definition:** Clear user roles and permission levels were defined to ensure systemic accountability, dictating exactly who can view, enter, edit, or approve data within the system.

2.3. Phase 3: Developer Handoff, Visualization & Final Approval

This final phase dictates how the findings in this report will be utilized to build the software. To bridge the gap between management consulting and software development, Crest Point will coordinate directly with the development team to ensure the system is built exactly to specification.

- **Graphical Blueprinting:** The optimized "To-Be" processes documented in this report will serve as the standardized Business Process Model and Notation (BPMN) flowcharts for the developers, clearly displaying triggers, decision gates, and data inputs.
- **Final System Validation Protocol:** Prior to the official launch and deployment across the corporation, the digital system engineered by the development team will be subject to strict, rigorous testing by Crest Point Consulting Group. The system will only be authorized for live implementation once it receives formal approval from Crest Point, ensuring it perfectly matches this architectural design.

3. GENERAL SYSTEM ARCHITECTURE & INTERFACES

This section defines the universal components, access matrices, and core dashboards that govern the entire digital system across all departments and Second Units.

3.1. User Access Types & Control Log

The system architecture relies on strict Role-Based Access Control (RBAC) to ensure security and prevent unauthorized data manipulation. Every action within the system is recorded in an automated Control Log (Audit Trail) capturing the User ID, timestamp, and action performed.

Defined Access Levels:

Level 1 — System Administrator:

Full IT access for system maintenance, user creation, and password resets. No access to view or alter financial or operational data. Shares authority with the Finance Department Head to manage the chart of accounts structure.

Level 2 — Department Heads and Second Unit Managers:

Full access to view, review, and approve workflows strictly within their own assigned department or unit. Authority to approve or reject submissions from their Officers.

Authority to approve or reject correction requests raised by Officers, unlocking entries for correction if approved. No access to other departments' data.

Level 3 —Departmental Officers (Data Entry):

Restricted access to input data and submit forms within their specific module. Cannot approve their own submissions. Cannot modify or delete a submitted entry. Must raise a formal correction request through the system to their Department Head for any amendment. The system automatically sends a notification to the Department Head upon submission of a correction request.

Correction Protocol:

If a data entry error occurs after submission, Level 3 Officers cannot delete or modify the record themselves. They must submit a formal correction request within the system to their Department Head. The Department Head reviews the request and if approved unlocks the entry for the Officer to correct and resubmit. The original incorrect entry, the correction request, the Head's decision, and the corrected entry must all remain permanently recorded and visible in the system's Control Log.

Level 4 — Internal Control and Audit Department:

Read-only access across all departments, all Second Units, and the complete Control Log at all times. Cannot create, edit, approve, or delete any transaction or record in any module. Holds mandatory verification authority; every workflow across every department includes a defined Internal Control verification step that must be completed and recorded before the workflow is considered fully closed.

- **Verification & Rejection Authority:** The Internal Control team is responsible for verifying that all physical supporting documents are strictly in line with the ERP system entries, and for recalculating figures to ensure absolute accuracy. If any data, calculation, or document is not perfectly aligned, Internal Control will **Reject** the transaction.
- **The Rejection & Correction Loop:** Upon rejection, the transaction is not accepted into the final ledger. Instead, the system automatically sends a notification containing Internal Control's specific comments and reasons for rejection directly to the Head of the relevant department. Based on the correction protocol, the Department Head coordinates with the initiating Officer and directs them to adjust the entry. The Officer corrects, and resubmits the entry to the Department Head. Once the Department Head reviews and approves the adjusted entry, it is routed back to Internal Control for final re-verification.

ED and Directors — No System Access:

The Executive Director and all Department Directors do not interact with the ERP system directly. Their approvals for purchases, sales, payroll, and other matters requiring their authorization are obtained physically by the relevant department staff, documented in

the physical file, and recorded as confirmation entries in the system by the relevant department staff. No digital action is required from the ED or Directors at any point.

3.2. Secure Login Interface

The gateway to the system utilizes a high-contrast, Right-to-Left (RTL) split-screen design.

Core UI Elements:

- Corporate branding (North Coal State Owned Corporation).
- User ID and Password input fields.
- "Forgot Password" recovery link.



3.3. Main Operational Dashboard

Upon successful login, users are routed to a centralized dashboard providing a real-time overview of corporate operations, pulling live data from the headquarter and all Second Units.

Core Dashboard Elements by User Level

Department Head Dashboard:

- Pending approvals requiring the Head's action
- Correction requests raised by Officers awaiting the Head's decision

- Notifications of transactions Rejected by Internal Control, containing IC's specific comments and requiring coordination for adjustment.
- Department-specific performance metrics and status indicators
- Internal Control verification status for recent submissions

Officer Dashboard:

- Personal task queue — entries pending submission
- Status of previously submitted entries
- Notifications of correction requests approved or rejected by the Department Head
- Action alerts for rejected entries that have been unlocked and returned for correction and resubmission.

Internal Control Dashboard:

- Organization-wide transaction monitoring feed
- Department-Specific Notification Queues — Real-time notifications and dedicated queues for pending verifications, strictly segregated by each department (e.g., "Pending from Finance," "Pending from Payroll," "Pending from Sales," etc.) so auditors can manage their workload efficiently.
- Notifications of previously rejected entries that have been corrected and resubmitted for re-verification.
- Full Control Log access

3.4. Hybrid Connectivity & Offline Edge Functionality

Given the geographical remoteness of the active mining Second Units, a purely cloud-dependent architecture is unviable. The system must employ a Hybrid Edge-Computing model. While the HQ and regional hubs operate on a live central cloud, remote mining sites will utilize a localized application (Edge node). If internet connectivity drops, the Second Unit system must continue to function offline seamlessly, securely storing dispatch, inventory, and payroll data locally. Upon connection restoration, the system will perform an automated, encrypted batch-synchronization to the central HQ database.

4. DEPARTMENTAL PROCESS ARCHITECTURE

This section details the current operational baseline ("As-Is") and the newly engineered digital workflows ("To-Be") for each department. It serves as the definitive guide for both corporate management and the software development team.

4.1. Finance and Accounting

4.1.1. Current State ("As-Is") Analysis

Overview of Current Operations:

The Finance department currently operates on a highly reactive, manual basis. Financial recording is disconnected from real-time operations, heavily relying on delayed physical bank statements and delayed physical reporting from remote mining sites. There is a significant operational disconnect between the centralized headquarter (where cash is managed) and the Second Units (where revenue is physically realized), resulting in manual reconciliations that are perpetually out of date.

Key Daily/Weekly/Monthly Activities:

1. Recording financial transactions manually in the journal with a mandatory one-day delay, strictly dependent on receiving physical or delayed bank statements.
2. Executing internal fund transfers from the centralized revenue bank account to the corporate expense bank account to facilitate company payments and payroll.
3. Manually recording revenue entries based on physical dispatch reports received from Second Units only after the coal has been physically handed over to the customer.
4. Manual data entry of all transactions into physical or isolated journal ledgers.

Current Forms and Reports Used:

1. Physical Bank Statements
2. Physical Dispatch/Sales Reports (submitted by Second Units)
3. Manual Journal Ledgers

Identified Pain Points & Staff Suggestions:

1. **Extreme Delays in Sales Reporting:** Coal sales and delivery reports from the Second Units typically take 6 to 12 months and sometimes even longer than a year to reach the Headquarter. This delay means the finance team cannot see up-to-date revenue or remaining inventory levels.
2. **Unclear Customer Balances:** Because of the delayed paperwork and manual tracking, the finance team cannot easily check a specific customer's exact balance, remaining advance payment, or outstanding debt at any given moment.
3. **Unreconciled Fund Transfers:** New money transfers or advances are often executed before the previous transfers have been properly accounted for or reconciled in the system.

4. **Poor Payroll Segregation (Production vs. Operational Costs):** The salaries and statistics for technical, administrative, and service staff are mixed together rather than separated. This makes it a major challenge for the finance team to accurately split the Cost of Goods Sold (direct production costs) from general operational expenses.

4.1.2. Future State ("To-Be") Digital Architecture

Process Optimization

The Finance and Accounting module serves as the central financial intelligence hub of the ERP system. Rather than operating as a standalone department, it functions as the consolidation point for financial data flowing in from all other departments: Sales, Inventory, Procurement, Payroll, Properties, and all Second Units. Every departmental action that carries a financial consequence automatically generates the corresponding accounting entry within this module without requiring manual re-entry by finance staff.

Within this module, Internal Control and Audit verification is strictly positioned at the final disbursement stages to maximize efficiency and security. Mandatory Internal Control verification is required at two critical points: before any final payment is processed, and before the final payroll is disbursed. Transactions will not be considered fully closed until this verification is completed and recorded.

Chart of Accounts and Account Classification

The system maintains a fully structured chart of accounts organized into hierarchical classes and subclasses covering Revenue, Cost of Sales, Operating Expenses, Assets, Liabilities, and Equity. The chart of accounts is built and maintained directly by the Finance Department team after the system is developed and deployed it is not pre-configured by the development team. The system must therefore provide a fully flexible account creation interface that allows the Finance Department to build, edit, and manage the entire chart of accounts independently.

When creating any expense account within the chart of accounts, the system presents a mandatory classification checkbox requiring the user to designate the account as either a **Direct Cost** or an **Indirect Cost**. This classification drives the correct presentation in the Income Statement, direct costs appearing above the gross profit line and indirect costs below it in full compliance with IFRS financial statement presentation requirements.

All account classes and subclasses are fully customizable by the **Finance Department Head** and the **System Administrator** only. Finance Officers have no authority to create, edit, or deactivate any account in the chart of accounts.

Voucher Management and Double-Entry Principles

To ensure a fundamentally sound accounting engine, the ERP must be strictly governed by **Double-Entry Bookkeeping Principles**. The system must automatically validate that total Debits equal total Credits for every single transaction. If a manual entry does not balance perfectly, the system must hard-block the submission and prevent posting.

The system relies on three core, highly detailed voucher types for all financial recordings:

1. **Payment Vouchers (PV):** Used for all outgoing funds (cash or bank transfers) to suppliers, employees, or other entities.
2. **Receipt Vouchers (RV):** Used to record all incoming funds and customer advances.
3. **Journal Vouchers (JV):** Used for non-cash adjustments, depreciation postings, corrections, and internal transfers between accounts.

Voucher Printing and Hard-Copy Archiving: The system must generate highly detailed, printable PDF versions of every voucher. These printed vouchers will include signature blocks for the preparer, reviewer, and approver. Printing is a mandatory system feature because physical supporting documents (such as vendor invoices, bank deposit slips, and physical approvals) must be physically attached/stapled to the printed voucher and archived in hard files for external audit compliance.

Purchase Cycle Budget Control and Finance Notification

The purchase cycle is a shared operational flow between the Requesting Department, Finance, and Procurement. Before any purchase request can be acted upon by the Procurement department, it must pass through a strict financial gate:

1. **Budget Check Notification:** When a purchase request is submitted by any department, the system sends an automatic notification to the Finance team.
2. **Finance Verification:** The Finance team checks if there is an allocated budget for this specific item/category for the current fiscal year.
3. **Budget Exists:** If funds are available, Finance confirms it in the system, and the request is immediately transferred to the Procurement department to begin the purchasing process.
4. **No Budget Exists:** If there is no budget available, the system halts the purchase. The requesting department must then obtain a special, physical approval from the Executive Director (ED). Once this physical approval is obtained, the Finance Department records the ED's approval reference in the system. This acts as a budget override, allowing the request to proceed to Procurement.

Purchase Cycle and Payables Management

Once goods are received by Inventory, the Finance department receives a live "**Awaiting Payment**" notification showing that the purchase order, goods receipt, and supplier invoice are fully processed and ready for payment. Before any payment is processed, the Internal Control and Audit department conducts a mandatory verification of the payment request, verifying that all physical supporting documents are strictly in line with the ERP system entries and recalculating figures to ensure absolute accuracy.

- **Approval:** If verified, the Finance Officer proceeds to process the Payment Voucher (PV).
- **Rejection & Correction Loop:** If any data, calculation, or document is not aligned, Internal Control will **Reject** the transaction. The system automatically sends a notification containing IC's specific comments to the Finance Department Head. The Department Head unlocks the transaction in the system, which triggers an automatic notification to the Finance Officer. The Officer then adjusts the entry based on the IC comments and resubmits it to the Department Head, who approves it and routes it back to Internal Control for final re-verification.

Payroll Processing

Payroll entries are initiated by the Payroll department. Before the payroll Payment Voucher is processed and disbursed via bank transfer, the Internal Control and Audit department conducts a mandatory verification of the entire payroll record (confirming attendance, calculations, and deductions). This follows the exact same Rejection & Correction loop: if discrepancies are found, IC rejects the payroll, triggering a notification to the relevant Department Head with IC's comments, forcing a correction, resubmission, and re-verification before any funds are released.

Bank Account Integration and Statement Processing

The system supports two methods for capturing bank transaction data into the ERP:

Method 1 — Standardized Excel and CSV Import & Mapping:

The system requires a strict, standardized mapping template for bank statement imports. Before uploading, the Finance Officer ensures the bank's **Excel or CSV file** aligns perfectly with the system's exact column requirements (date, description, debit amount, credit amount, running balance). The system reads this mapped file and presents all transactions in a structured reconciliation screen for the Finance Officer to review. If the mapping format is incorrect, the system rejects the file.

Method 2 — Direct Bank Integration:

Where the bank's system supports it, the ERP connects directly to the bank and pulls transaction data automatically in real time or at defined intervals. The same reconciliation review process applies.

Reconciliation and Posting Process:

Importing the bank statement does not automatically post any accounting entry. All entries are held in a **Pending Reconciliation** queue until the Finance Officer manually reviews and matches each transaction:

- **Deposits (Credits in bank statement):** The Finance Officer reviews each deposit, identifies the corresponding customer and sales order, and confirms the match. Only after the Officer confirms the match does the system post the accounting entry: **Debit Bank Account / Credit Customer Advance (Current Liability)**. This reflects that cash received from customers represents an advance payment for coal not yet delivered.
- **Payments (Debits in bank statement):** The Finance Officer reviews each payment, identifies the corresponding approved purchase order or expense record, and confirms the match. Only after the Officer confirms the match does the system post the accounting entry: **Debit Relevant Purchase or Expense Account / Credit Bank Account**.
- **Unreconciled Transactions:** Any transaction that the Finance Officer cannot match to an existing record is flagged as **Unreconciled**. The system immediately sends a notification to the Finance Department Head. A live counter is displayed on the Finance dashboard at all times showing the total number and total value of unreconciled transactions outstanding. No period-end closing can be authorized while unreconciled transactions remain outstanding, the system blocks the closing workflow until all items are resolved or formally addressed by the Finance Department Head.

Revenue Recognition and Customer Advance Management

The corporation operates a designated bank account for all coal sale receipts. All deposits received into this account are initially recorded by the system as **Customer Advances (Current Liabilities)** at the point of reconciliation confirmation. This is because at the point of cash receipt, coal has not yet been physically delivered to the customer, the corporation holds a delivery obligation, not yet earned income.

Revenue is only recognized at the precise moment coal is physically transferred to the customer at the Second Unit. At that point, the Customer Advance liability is reduced and the Revenue account is increased by the corresponding value of coal delivered. This ensures that the revenue figures shown in the system at any given time represent only

coal that has physically left the corporation's premises and been received by the customer, not cash still held as a delivery obligation awaiting fulfillment.

Weighing System Integration and Real-Time Revenue Data

Each Second Unit (mining sites only) operates a computerized weighing system installed at the point of coal dispatch. When a customer presents the approved hard-copy coal allocation letter and coal is physically loaded and weighed for transfer to the customer's driver, the weighing system records:

- The exact quantity of coal dispatched in this transaction
- The cumulative quantity of coal received by this customer to date against their approved allocation letter
- The **remaining balance of coal** the customer is still entitled to receive based on the total quantity stated in their hard-copy approval letter

The ERP is integrated with this weighing system so that dispatch data is transmitted to the central ERP either in real time or at a defined end-of-day synchronization, even from geographically remote Second Units. Upon receipt and confirmation of the dispatch data, the system automatically posts the following accounting entries:

- **Debit:** Customer Advance (Current Liability) — reduced by the value of coal dispatched
- **Credit:** Revenue Account — increased by the same amount

Inventory and Expense Recognition

When the Inventory and Warehouse Management department issues items from stock, the ERP automatically reflects the correct accounting treatment:

- **Consumable items** issued from inventory are automatically expensed, debiting the relevant expense account and crediting the inventory account at the moment of confirmed issuance.
- **Non-consumable items** handed over to employees are recorded as asset movements capturing the employee's name, position, and date of handover and do not immediately hit the expense account.

Second Unit Finance Staff Access

Finance Officers based at Second Units have the same access level as HQ Finance Officers restricted to data entry within their assigned Second Unit. All Second Unit entries are immediately visible to the HQ Finance Department Head. The Finance Department Head at HQ retains review and approval authority over Second Unit submissions. Second Unit finance entries are subject to the same Internal Control verification workflow as all other entries.

Second Unit Finance Officers record all transactions relevant to their unit including local expenses and cash wage payments to mining site workers. Wages for site-based workers are paid in cash. All other payments are processed via bank transfer.

Bulk Transaction Import

The system supports bulk import of transactions via CSV or Excel files. A standardized import template is provided within the system. Finance Officers populate the template with transaction data and upload it directly into the ERP. The system validates each row against predefined rules before posting, flagging errors for correction before any data is committed to the ledger. This capability is particularly important for Second Units that may need to upload accumulated transaction data during periods of connectivity limitation.

Fiscal Year and Financial Period

The ERP system must include a dynamic "Fiscal Period Management" module to govern all financial reporting and data entry parameters.

- **Editable by Finance Head:** The Finance Department Head must have the exclusive authority to configure, define, and edit the exact start and end dates of the official Fiscal Year and its subset monthly or quarterly periods. The system must not hard-code these dates.
- **Period Control (Open/Close/Re-open Logic):** The module must allow the Finance Head to manually "Open" financial periods to permit data entry and formally "Close" or "Lock" historical periods. If adjustments are required after a period is closed (e.g., for year-end audit adjustments), the system must allow the Finance Head to temporarily "Re-open" the closed period, process the authorized adjusting entries, and then formally re-close and lock the period. All such status changes, closing, re-opening, adjusting, and re-closing, must be permanently captured in the system's Control Log for audit traceability.

Officer Correction Request Workflow

If a Finance Officer identifies an error in a submitted entry, the Officer cannot modify or delete it independently. The following workflow applies:

1. The Finance Officer raises a **formal correction request within the system** describing the error and the required correction
2. The system automatically sends a **notification to the Finance Department Head** with its justification**.**
3. The Finance Department Head reviews the request within the system
4. If the request is approved the system unlocks the entry and the Officer corrects and resubmits it
5. If the request is rejected the Head records the reason and the entry remains as originally submitted
6. All steps the original entry, the correction request, the Head's decision, and the corrected entry are permanently recorded in the Control Log and are visible to the Internal Control and Audit department

Single Currency (AFN)

The system operates exclusively in Afghan Afghani (AFN). No multi-currency functionality or exchange rate conversion is required. All transaction amounts, reports, and financial statements are denominated in AFN only.

Compliance and Standards

ISO 9001 Quality Management

Every financial transaction in the system follows a mandatory, documented approval sequence that cannot be bypassed. Mandatory field validation, budget checks, and multi-level authorization gates including the mandatory Internal Control verification step embedded in every workflow, ensure consistent process execution across HQ and all Second Units. The system's immutable Control Log provides the complete audit trail required under ISO 9001's document control and records management clauses.

IFRS and IAS Compliance

- **IFRS 15 — Revenue from Contracts with Customers:** Revenue is recognized only upon physical delivery of coal to the customer not at the point of cash receipt. All cash received in advance is held as a current liability (Customer Advance) until the delivery obligation is fulfilled at the Second Unit.
- **IAS 1 — Presentation of Financial Statements:** The system generates the four required financial statements: Income Statement, Balance Sheet, Cash Flow Statement, and Statement of Changes in Equity in the format required under IAS 1.
- **IAS 2 — Inventories:** Coal stock and consumable inventory are valued and reported in compliance with IAS 2 principles.
- **IAS 16 — Property, Plant and Equipment:** Fixed assets are recorded, classified, and depreciated in accordance with IAS 16. Depreciation is calculated and posted automatically on a monthly basis. Land is designated as a zero-depreciation asset class.
- **IAS 7 — Statement of Cash Flows:** The cash flow statement is generated automatically from the transaction data in the system, categorized into operating, investing, and financing activities.
- **IAS 12 — Income Taxes:** Payroll deductions and applicable tax obligations are computed and recorded in compliance with IAS 12 and applicable Afghan national tax law.

4.1.3. System Roles and Access Permissions

Role	User	Permissions
Data Entry / Initiator	Finance Officers (HQ and Second Units)	Can create Payment Vouchers (PV), Receipt Vouchers (RV), and Journal Vouchers (JV). Can upload Excel or CSV bank statements and bulk transaction import files. Can review and match bank transactions during reconciliation. Cannot approve or post any entry independently. Cannot unlock, modify, or delete a submitted entry; must raise a formal correction request to the Finance Department Head through the system. Receives system notifications when the Dept Head unlocks an entry for correction.
Department Head	Finance Department Head	Can review and approve all entries submitted by Finance Officers. Unlocks entries in the system for Officers to correct (either from a self-identified request or an IC rejection). Can authorize and formally close monthly bank reconciliation. Controls budget verification for purchase requests. Can initiate and authorize

Role	User	Permissions
		period-end closing. Has full authority to manage the chart of accounts. Prints relevant reports and printable PDF vouchers with attached physical documents for physical distribution to Directors and ED as required.
Internal Control Verification	Internal Control and Audit Department	Mandatory verification is strictly positioned at the final disbursement stages (Payment Vouchers and final Payroll processing). Read-only access to all transactions, vouchers, reconciliation records, and Control Log across all departments and Second Units. Can Verify or Reject transactions. Rejection triggers a notification to the Dept Head to unlock the entry and coordinate adjustments with the Officer.
ED and Directors	Executive Director and Department Directors	No system access. Physical approvals are obtained by the relevant department, documented in files, and recorded in the system by the relevant department staff.
System Administrator	IT Admin	Can create and deactivate user accounts and manage system settings. Can customize chart of accounts alongside Finance Department Head. Zero access to financial data or transactions.

4.1.4. Developer Specifications and Visual Assets

Voucher Creation Interface & Background Double-Entry Logic

The ERP must be designed to simplify data entry for Finance Officers while strictly enforcing Double-Entry Bookkeeping Principles (Debits must equal Credits) in the backend.

When a Finance Officer accesses the "Voucher Creation" module from their dashboard, they select one of three voucher types. The system interface dynamically adjusts based on their selection:

1. Payment Vouchers (PV):

- **User Action:** The Officer selects the Bank Account or Cash Box from which the payment is being made.
- **User Action:** The Officer selects the specific Expense, Asset, or Liability account being paid.
- **User Action (Paid To):** The Officer selects the recipient from a centralized dropdown menu linked to system databases (Suppliers, Customers, Employees, or "Other"). *Note: The interface must include a quick-add feature allowing the Officer to create a new Supplier or Customer profile directly from this screen if they do not already exist.*

- **User Action:** The Officer enters the Amount and Description.
- **Background System Action:** Upon submission, the system automatically constructs and posts the double-entry: *Debit [Selected Expense/Asset/Liability Account] / Credit [Selected Bank/Cash Account].*

2. Receipt Vouchers (RV):

- **User Action:** The Officer selects the Bank Account or Cash Box receiving the funds.
- **User Action:** The Officer selects the specific Revenue, Liability, or Asset account being credited.
- **User Action (Received From):** The Officer selects the payer from the centralized dropdown (Customers, Employees, etc.).
- **User Action:** The Officer enters the Amount and Description.
- **Background System Action:** Upon submission, the system automatically constructs and posts the double-entry: *Debit [Selected Bank/Cash Account] / Credit [Selected Revenue/Liability/Asset Account].*

3. Journal Vouchers (JV):

- **User Action:** Used for non-cash adjustments, corrections, or depreciation. The Officer manually selects the Debit Account(s) and Credit Account(s).
- **User Action:** The "Paid To / Received From," "Bank Account," and "Payment Method" fields are automatically hidden or marked "N/A".
- **System Check:** The system verifies that total Debits exactly equal total Credits. If they do not balance, the system hard-blocks the submission.

System-Generated & Bulk-Imported Vouchers For financial events that are not manually typed one-by-one into the voucher screen, the system must still **automatically generate viewable and printable vouchers (RV, PV, or JV)** in the background. This mandatory requirement applies to:

1. **Automated Entries:** Transactions triggered by other modules (e.g., Bank Reconciliation matching, Revenue Recognition via the coal weighing system, or automatic Payroll postings).
2. **Bulk Transaction Imports:** When a Finance Officer uploads a standardized Excel/CSV file, the system must parse the validated data and automatically generate a distinct, printable voucher in the system for the imported transactions.

This ensures that absolutely *every* financial transaction in the ledger, whether manually entered, system-automated, or bulk-imported has a standardized voucher available in the UI that can be viewed, printed, signed, and attached to physical supporting documents for audit compliance.

Required Data Fields — Vouchers

(Applicable to all manually created, system-generated, and bulk-imported vouchers)

Field	Type	Notes
Voucher Number	Auto-generated	Sequential — not editable
Voucher Date	Date picker	System strictly prevents backdating beyond the current open period. No override allowed.
Voucher Type	Dropdown	Payment Voucher (PV) / Receipt Voucher (RV) / Journal Voucher (JV)
Second Unit / HQ	Auto-populated	Based on logged-in user's unit assignment
Account Code(s)	Dropdown	Linked to chart of accounts (Opposing entries auto-posted for PVs and RVs)
Amount (AFN)	Numeric	Must balance. Real-time budget check on entry for PVs.
Paid To / Received From	Dropdown / Search	Linked to system databases (Suppliers, Customers, Employees, etc.). Includes "Create New" option. Mandatory for PVs and RVs. Marked "N/A" for JVs.
Payment/Receipt Method	Dropdown	Cash / Bank Transfer / Cheque. Marked "N/A" for JVs.
Bank/Cash Account	Dropdown	Pre-loaded authorized accounts only. Marked "N/A" for JVs.
Description / Narration	Text area	Minimum 10 characters
Internal Control Verification	Auto-populated	Pending / Verified / Rejected (Active for Final Payments and Payroll only)
Print Voucher	Action Button	Generates printable PDF with signature blocks for hard-copy archiving. (Available for manual, automated, and bulk-imported entries).
Status	Auto-populated	Draft / Submitted / Under Review / Approved / Pending IC Verification / Fully Complete

Required Data Fields — Bank Statement Reconciliation

Note: The system auto-posts double entries in the background upon confirmed match.

Field	Type	Notes
Bank Account	Dropdown	Select account being reconciled
Statement Period	Date range	From / To
Import Method	Toggle	Excel or CSV Upload / Direct Bank Connection
Excel or CSV File Attachment	File upload	System maps columns automatically. System rejects incorrect format.
Transaction Date	Mapped from file	
Description	Mapped from file	
Debit Amount (AFN)	Mapped from file	Payments — matched to approved PVs or expense records
Credit Amount (AFN)	Mapped from file	Deposits — matched to RVs, customers, and sales orders
Running Balance	Mapped from file	
Matched To	Dropdown / Search	Officer selects matching record from system
Accounting Entry	Auto-posted	System posts background double-entries (e.g., Dr Bank / Cr Customer Advance) based on the match
Reconciliation Status	Auto-generated	Matched / Unreconciled
Unreconciled Counter	Auto-calculated	Live count and total value of unreconciled items — visible on dashboard
Finance Head Approval	Required	Before reconciliation period is closed

Required Data Fields — Revenue Recognition Entry (Auto-generated from Weighing System)

Note: The system auto-posts double entries in the background upon dispatch confirmation.

Field	Type	Notes
Entry Reference	Auto-generated	Linked to weighing system dispatch record
Second Unit	Auto-populated	Source Second Unit
Customer Name	Auto-populated	From approved sales order
Approval Letter Reference	Auto-linked	From Sales module
Total Approved Qty (tonnes)	Auto-populated	From approved allocation letter
Prev Dispatched Qty (tonnes)	Auto-populated	Cumulative from prior dispatches
Qty Dispatched (tonnes)	Auto-populated	From weighing system
Remaining Entitlement (tonnes)	Auto-calculated	Total Approved minus Total Dispatched to date
Revenue Amount (AFN)	Auto-calculated	Quantity Dispatched × Approved Unit Price
Customer Adv Reduced (AFN)	Auto-calculated	Same as Revenue Amount (System auto-posts Dr Customer Advance / Cr Revenue)
Date and Time of Dispatch	Auto-populated	From weighing system timestamp
Status	Auto-posted	Posted automatically upon dispatch confirmation

Accounting Entries — Developer Reference

Trigger Event	Debit	Credit
Deposit matched and confirmed in bank reconciliation	Bank Account	Customer Advance (Current Liability)
Payment matched and confirmed in bank reconciliation	Relevant Expense or Purchase Account	Bank Account
Coal dispatched to customer (weighing system confirmation)	Customer Advance (Current Liability)	Revenue Account
Purchase invoice approved and received	Relevant Expense or Asset Account	Accounts Payable
Payment made to supplier	Accounts Payable	Bank Account
Consumable inventory issued to department	Expense Account	Inventory Account
Non-consumable item handed to employee	Asset — In Use Account	Inventory Account
Cash wages paid to site workers	Wages Expense	Cash Account
Payroll posted after ED physical approval recorded	Payroll Expense	Salaries Payable
Payroll paid via bank transfer	Salaries Payable	Bank Account
Depreciation posted monthly	Depreciation Expense	Accumulated Depreciation
Fixed asset purchased	Fixed Asset Account	Accounts Payable / Bank Account
Inventory loss or wastage recorded	Loss on Inventory Account	Inventory Account

Required Reports

Report	Frequency	Filter Options
Trial Balance	On demand	Custom date range, Second Unit, cost center
Income Statement	On demand	Custom date range, Consolidated or per Second Unit
Balance Sheet	On demand	Custom date range, Consolidated or per Second Unit
Cash Flow Statement	On demand	Custom date range, Consolidated
Statement of Changes in Equity	On demand	Custom date range, Consolidated
Budget vs Actual	On demand	Custom date range, Per cost center, per Second Unit

Report	Frequency	Filter Options
Customer Advance Ledger	On demand	Custom date range, Per customer
Revenue Recognition Report	On demand	Custom date range, Per Second Unit, per customer
Coal Entitlement Tracking	On demand	Custom date range, Per customer — total approved, dispatched, remaining
Unreconciled Transactions	On demand / Real-time	Custom date range, Per bank account — count and total value
Bank Reconciliation Report	On demand	Custom date range, Per bank account
Voucher Status Report	On demand	Custom date range, By status, unit, user
Correction Request Log	On demand	Custom date range, By department, user
Control Log / Audit Trail	On demand	Custom date range, By user, action type
Bulk Import Log	On demand	Custom date range, Accepted, flagged, and rejected rows
Internal Control Verification	On demand	Custom date range, By department, workflow type, verification outcome

All reports must display: Corporation name and logo, report title, date range, generation timestamp, and generating user name. All reports must be exportable to PDF and Excel formats.

4.1.5. BPMN Workflow Narratives — Finance and Accounting

Workflow 1: Bank Reconciliation and Revenue Receipt Workflow

Excel or CSV bank statement imported via direct connection or file upload → Transactions appear in Pending Reconciliation queue → Finance Officer reviews each deposit → Officer matches deposit to customer and sales order → System automatically posts background double-entry: Debit Bank Account / Credit Customer Advance → Finance Officer reviews each payment → Officer matches payment to approved purchase or expense → System automatically posts background double-entry: Debit Relevant Expense or Purchase Account / Credit Bank Account → Unmatched items flagged as Unreconciled → Notification sent to Finance Head → Finance Head resolves or formally addresses

unreconciled items → Finance Head approves reconciliation → Reconciliation marked fully complete.

Workflow 2: Revenue Recognition Workflow

Customer presents approved hard-copy allocation letter at Second Unit → Coal weighed and dispatched → Weighing system records quantity dispatched, cumulative total dispatched, and remaining entitlement balance → Dispatch data transmitted to ERP in real-time or via sync → System automatically posts background double-entry: Debit Customer Advance / Credit Revenue Account → Entry marked fully complete.

Workflow 3: Purchase Cycle Budget Control & Payment Workflow

Requesting department submits purchase request → System sends Budget Check Notification to Finance team → Finance checks allocated budget → If Budget Exists: Finance confirms in system, request immediately transferred to Procurement. If No Budget: ED physical override approval obtained, recorded in system by Finance, request transferred to Procurement → Procurement processes in hard copy → Goods received at Inventory → Finance receives "Awaiting Payment" notification → Finance Officer prepares Payment Voucher (PV) → Finance Head approves PV → Routed to Internal Control for mandatory final verification → IC verifies physical documents vs system and recalculates → If Rejected: System sends notification with IC comments to Finance Head → Finance Head unlocks the transaction → Automatic notification sent to Finance Officer → Officer corrects entry based on IC comments and resubmits to Head → Head re-approves and routes back to IC. If Verified: Payment processed → System automatically posts all background ledger entries → Workflow fully complete.

Workflow 4: Officer Correction Request Workflow

Finance Officer identifies error in an already submitted entry (prior to IC verification or final closing) → Officer raises formal correction request in system detailing the error → System sends notification to Finance Head → Finance Head reviews request → If Approved: Finance Head unlocks the entry in the system → Automatic notification sent to Finance Officer → Officer corrects and resubmits → All steps (original entry, request, Head's decision, and corrected entry) recorded permanently in Control Log.

Workflow 5: Period-End Closing Workflow

Finance Head initiates closing → System checks all pending PVs, RVs, and JVs are resolved → Unreconciled transactions counter confirms zero outstanding → Bank reconciliation confirmed approved → Finance Head authorizes close → Period locked → System strictly prevents any backdated entries into the closed period → If adjustment required post-closing: Finance Head initiates "Re-open" protocol → Period temporarily unlocked → Authorized adjusting entries processed → Finance Head formally re-closes and locks the period → All status changes (close, re-open, adjust, re-close) permanently captured in the Control Log for audit traceability.

4.2. Inventory and Warehouse Management

4.2.1. Current State ("As-Is") Analysis

Overview of Current Operations:

The Inventory and Warehouse Management department currently handles both the physical custody of goods and the manual accounting of all corporate assets. The department relies entirely on manual storage cards, physical ledgers, and paper-based vouchers. They manage everything from daily office stationary and vehicle fuel to the safekeeping of official Ministry of Mines dispatch booklets. The workflow is highly fragmented, requiring physical documents to be passed between the warehouse keepers and the inventory accountants, causing severe delays in updating stock balances.

Key Daily/Weekly/Monthly Activities:

- Physically receiving, storing, and issuing purchased goods, fuel (diesel/petrol), and stationary using manual inventory cards.
- Receiving, safeguarding, and distributing official Ministry of Mines coal dispatch booklets to the weighing stations at the Second Units.
- Processing manual "FS-9" purchase request forms and forwarding them to the Procurement department.
- Manually calculating and recording the depreciation of fixed assets.
- Tracking all movable and immovable properties (land, buildings, vehicles) manually in physical ledgers.
- Collecting, recording, and managing scrap or damaged items (tires, batteries, metals) and tracking items built internally in the corporate workshop.
- Recording incoming and outgoing inventory vouchers into manual debit/credit journal ledgers after obtaining multiple physical approvals.
- Organizing committees for manual, physical inventory counts at the end of the fiscal year.

Current Forms and Reports Used:

- Physical Inventory/Storage Cards
- FS-9 Purchase Request Forms
- Ministry of Mines Dispatch Booklets (Bar Nama)
- Manual Journal and Inventory Ledgers
- Physical Fuel Distribution/Circulation Slips
- Physical Incoming and Outgoing Inventory Vouchers

Identified Pain Points & Staff Suggestions:

- **No Visibility of Coal Stock:** There are no accurate, up-to-date records of the actual coal inventory sitting at the remote Second Units (mining sites).
- **Redundant Signature Bottlenecks:** A single document often requires multiple, repetitive physical signatures from different managers, leading to severe delays and wasted time when issuing or receiving goods.
- **Lost Documents:** Because there is no electronic tracking system for physical paperwork (requests, letters, and approvals), documents frequently go missing during the manual routing process.
- **Lack of Standardized Forms:** The department suffers from using non-standardized, inconsistent paperwork, which causes confusion and slows down data entry.
- **No Centralized Electronic Database:** The total reliance on paper cards and ledgers makes it impossible to quickly check stock levels, track asset locations, or generate accurate inventory reports without manual counting.

4.2.2. Future State ("To-Be") Digital Architecture

Process Optimization

The Inventory and Warehouse Management module serves as the central hub for all stock movement, asset tracking, and item issuance across the corporation. It acts as the bridge between the requesting departments, the Procurement department, and the Finance and Accounting module, ensuring that every movement of inventory or asset is systematically recorded, approved, and automatically reflected in the financial ledger without manual re-entry.

Item Request and Issuance Workflow

The inventory cycle is initiated when a staff member from any department submits a formal **Item Request Form** within the ERP system. The request must contain sufficient detail, item description, quantity required, purpose, and the requesting department. The system automatically routes the request to the relevant Department Head for approval.

Upon Department Head approval, the request is escalated for Director-level approval. The Director's physical approval is obtained by the relevant department staff, documented in the physical file, and the confirmation is recorded in the ERP by the relevant department staff. The Director does not interact with the system directly.

Once approvals are confirmed and recorded in the system, the request is routed to the Inventory Officer for processing. The Inventory Officer checks current stock levels within the system:

Scenario 1 — Item is Available in Stock:

The Inventory Officer confirms availability and proceeds to issue the item using a formal **Store Issuance Form** generated within the system. The issuance form captures: item description, quantity issued, requesting department, name and position of the receiving employee, and date of issuance.

The system then applies the correct accounting treatment automatically based on the item type:

- **Consumable Items:** Items that are used up in operation (stationery, fuel, cleaning materials, spare parts, etc.) are automatically expensed at the moment of confirmed issuance. The system posts: **Debit Relevant Expense Account / Credit Inventory Account** — without any manual journal entry required from Finance staff.
- **Non-Consumable Items:** Items that retain value beyond immediate use (computers, equipment, tools, furniture, etc.) are recorded as asset handovers. The system records the item as **Asset — In Use** and captures the employee's full name, position, department, and date of handover. This creates a fully traceable record of which items are with which employees at all times, allowing the Inventory Officer to track and verify asset locations across the organization.

Scenario 2 — Item is Not Available in Stock (Purchase Request Initiation)

If the item requested is not available in stock, the Inventory Officer records the unavailability in the system and the requesting department is notified. The requesting department then initiates a formal **Purchase Request Form** within the ERP system. The purchase request must contain: item description, quantity required, justification, estimated cost, and requesting department.

The purchase request is routed for approval as follows:

- **Department Head** approves the purchase request in the system
- **Director** physical approval is obtained by department staff, documented in the file, and recorded in the ERP by department staff
- Once all approvals are confirmed and recorded, the purchase request is formally transferred to the **Procurement Department** for further action

From this point forward, the Procurement department manages the supplier selection and purchasing process in hard copy the ERP system is not involved in the procurement procedures themselves. However, once the procurement process is complete and goods are delivered, the Inventory Department receives the goods and records the receipt in the system as described in the Goods Receipt section below.

Goods Receipt from Procurement

When goods purchased through the Procurement department are delivered, the Inventory Officer receives them physically and records the receipt in the ERP system using a formal **Goods Receipt Note (GRN)**. The GRN captures: supplier name, purchase order reference, item description, quantity ordered, quantity received, condition of goods, and date of receipt.

The system matches the GRN against the original approved purchase request and the purchase order raised by Procurement. Any discrepancy between ordered and received quantities is flagged automatically and must be resolved by the Inventory Officer before the GRN is finalized. Upon confirmed GRN, the system automatically updates the inventory balance and notifies the Finance department that the goods have been received triggering the Awaiting Payment notification in the Finance module.

The system posts the accounting entry automatically upon confirmed GRN: **Debit Relevant Expense or Asset Account / Credit Accounts Payable** reflecting the liability to pay the supplier.

Internal Control Verification:

Once the GRN is drafted, the Internal Control and Audit department conducts a mandatory verification. They confirm that what the corporation purchased actually entered the warehouse in the correct quantities and condition, matching the physical goods against the system entries and the purchase order.

- **Rejection & Correction Loop:** If discrepancies are found, IC **Rejects** the GRN. The system automatically sends a notification containing IC's specific comments to the Inventory Department Head. The Department Head unlocks the transaction in the system, which triggers an automatic notification to the Inventory Officer. The Officer adjusts the entry based on the IC comments and resubmits it to the Department Head, who approves it and routes it back to Internal Control for final re-verification.
- **Approval:** Once verified, the GRN is marked fully complete. The system automatically updates the inventory balance and notifies the Finance department that the goods have been received, triggering the "Awaiting Payment" notification in the Finance module. The system automatically posts the accounting entry: Debit Relevant Expense or Asset Account / Credit Accounts Payable.

Fixed Assets Received through Procurement

When the item received through procurement is a fixed asset other than land and buildings which are recorded and classified by the Properties department the Inventory Officer records it in the system as a new fixed asset entry. The system prompts the Officer to complete the following mandatory fields:

- **Asset Class:** selected from the predefined list: Motor Vehicles, Office Equipment, Machinery, Furniture, and other classes defined by the Finance Department Head
- **Acquisition Cost (AFN):** the total cost of the asset as per the purchase invoice
- **Date of Acquisition:** the date goods were received and the GRN was confirmed
- **Purchase Order Reference:** auto-linked from the confirmed GRN
- **Depreciation Rate (% per annum):** selected by the Inventory Officer based on the asset class. The system applies this rate to calculate and charge depreciation automatically on a monthly basis from the period following acquisition

The depreciation posting is made automatically by the system each month without any manual entry required: **Debit Depreciation Expense / Credit Accumulated Depreciation**

Depreciation Control — All Fixed Asset Classes

The Inventory and Warehouse Management department holds overall control and responsibility for depreciation charging across all fixed asset classes within the ERP system including buildings recorded by the Properties department. This means:

- When the Properties department records a new building in the system, the Inventory department is responsible for setting and managing the applicable depreciation rate for that building
- Land recorded by the Properties department is designated as a **zero-depreciation asset class** when Land is selected as the asset class, the depreciation rate field is automatically locked at zero and cannot be edited
- All monthly depreciation postings across all asset classes; Motor Vehicles, Office Equipment, Machinery, Buildings, and any other class are generated and managed through the Inventory and Warehouse Management module
- The Inventory Department Head has authority to review, adjust, and approve depreciation rates across all asset classes. Any change requires a formal adjustment record capturing the original rate, the new rate, the reason for change, and the effective date permanently recorded in the Control Log.

Residual Value and Inventory Write-Down

For inventory items that have partially deteriorated but retain some residual value, the system provides a write-down option allowing the Inventory Officer to reduce the recorded value of the item to its estimated net realizable value rather than writing it off

entirely. The write-down entry is subject to Inventory Department Head review and approval before it is posted.

Real-Time Inventory Visibility

The system maintains a live inventory balance for all items across HQ and all Second Units. Department Heads and the Inventory Officer can view current stock levels, recent movements, and pending requests at any time. A low-stock alert is generated automatically when any item falls below a defined minimum stock threshold, triggering a notification to the Inventory Officer to initiate a replenishment request.

Officer Correction Request Workflow

If an Inventory Officer identifies an error in an entry that they have already submitted (and the error has not been caught/rejected by IC at the GRN stage), the Officer cannot modify or delete the entry independently. The following workflow applies:

1. The Inventory Officer raises a formal correction request within the system detailing the error and the required correction.
2. The system automatically sends a notification containing the justification to the Inventory Department Head.
3. The Inventory Department Head reviews the request.
4. If approved, the Department Head unlocks the entry in the system.
5. This action triggers an automatic notification to the Inventory Officer, who corrects and resubmits the entry.
6. All steps (original entry, request, Head's decision, and corrected entry) are permanently recorded in the Control Log.

Compliance and Standards

ISO 9001 Quality Management

Every inventory movement in the system follows a mandatory, documented approval sequence request, Department Head approval, Director confirmation, Inventory Officer issuance, and Internal Control verification. No item can be issued, received, written off, or transferred without the complete approval chain being recorded in the system. The Control Log provides an immutable audit trail of every inventory action satisfying ISO 9001's requirements for document control, traceability, and corrective action management.

IAS 2 — Inventories

All inventory items are recorded and valued in accordance with IAS 2. Consumable items are expensed at the point of issuance. Non-consumable items are maintained as assets on the balance sheet until disposal or write-off. Inventory losses and write-downs are recognized in the period they are identified in compliance with IAS 2's net realizable value requirements.

IAS 16 — Property, Plant and Equipment

Fixed assets received through procurement and recorded by the Inventory department are classified, measured, and depreciated in accordance with IAS 16. The system enforces correct asset class selection and automatically applies the defined depreciation rate for each class. Land and buildings are excluded from the Inventory department's scope and are managed by the Properties department.

4.2.3. System Roles and Access Permissions

Role	User	Permissions
Data Entry / Initiator	Requesting Department Officers	Can create and submit Item Request Forms within their own department module. Cannot approve their own requests. Cannot modify a submitted request — must raise a formal correction request to their Department Head through the system.
Department Head — Requesting	Requesting Department Head	Can review and approve or reject item requests submitted by their department Officers. Can approve or reject correction requests from their Officers. Records Director-level approval confirmations in the system on behalf of the Director.
Inventory Officer	Inventory and Warehouse Officer	Can view all approved item requests routed to the Inventory module. Can check stock levels and confirm availability. Can generate and confirm Store Issuance Forms. Can record Goods Receipt Notes. Can record inventory losses, wastages, and write-downs subject to defined thresholds. Cannot approve their own entries — all issuance records must be reviewed by the Inventory Department Head.
Department Head — Inventory	Inventory Department Head	Can review and approve all Store Issuance Forms, Goods Receipt Notes, and inventory adjustment entries submitted by Inventory Officers. Can approve or reject correction requests raised by Officers. Can authorize inventory write-offs and write-downs. Can generate all inventory reports. Prints relevant inventory reports for physical distribution to Directors and ED as required.

Role	User	Permissions
Internal Control Verification	Internal Control and Audit Department	Mandatory verification is positioned at the Goods Receipt Note (GRN) stage. Can Verify or Reject the GRN. Rejection triggers a notification to the Inventory Dept Head to unlock the entry and coordinate adjustments with the Officer. Read-only access to all inventory records and the Control Log.
ED and Directors	Executive Director and Department Directors	No system access. Physical approvals are obtained by the relevant department staff, documented in physical files, and recorded in the system by the relevant department staff.
System Administrator	IT Admin	Can create and deactivate user accounts and manage system settings. Zero access to inventory data or financial records.

4.2.4. Developer Specifications — Inventory and Warehouse Management

Accounting Entries — Developer Reference

Trigger Event	Debit	Credit
Consumable item issued from inventory	Relevant Expense Account	Inventory Account
Non-consumable item handed to employee	Asset — In Use Account	Inventory Account
Goods received from procurement (expense items)	Relevant Expense Account	Accounts Payable
Goods received from procurement (asset items)	Inventory Account	Accounts Payable
Inventory loss or wastage recorded and verified	Loss on Inventory Account	Inventory Account
Inventory write-down to net realizable value	Loss on Write-Down Account	Inventory Account
Fixed asset received via procurement	Fixed Asset Account	Accounts Payable
Monthly depreciation — all asset classes (auto-posted)	Depreciation Expense	Accumulated Depreciation
Depreciation rate adjustment recorded Control Log entry only; no financial posting unless backdated correction required		

Required Reports — Inventory and Warehouse Management

Report	Frequency	Filter Options
Current Stock Level Report	On demand	Custom date range, Per item, per category, per Second Unit
Item Movement Report	On demand	Custom date range, Per item, per department
Store Issuance Report	On demand	Custom date range, Per department, per item
Goods Receipt Note Report	On demand	Custom date range, Per supplier, per purchase order
Asset Handover Report	On demand	Custom date range, Per employee, per department, per item category
Non-Consumable Asset Register	On demand	Custom date range, Per employee, per asset class — showing all items in use and with whom
Low Stock Alert Report	On demand / Real-time	Custom date range, Per item — items below minimum stock threshold
Inventory Loss and Wastage	On demand	Custom date range, Per item, per Second Unit
Purchase Request Status	On demand	Custom date range, Per department — showing pending, approved, transferred to Procurement
Control Log / Audit Trail	On demand	Custom date range, By user, action type
Internal Control Verification	On demand	Custom date range, By workflow type and verification outcome

All reports must display: Corporation name and logo, report title, date range, generation timestamp, and generating username. All reports must be exportable to PDF and Excel formats.

4.2.5. BPMN Workflow Narratives — Inventory and Warehouse Management

Workflow 1: Item Request and Issuance (Item Available in Stock)

Department Officer submits Item Request Form in system → System routes to Department Head → Department Head reviews and approves in system → Director physical approval obtained outside system → Relevant staff record physical approval confirmation in ERP → Request routed to Inventory Officer → Inventory Officer checks stock — item available → Inventory Officer generates Store Issuance Form → System automatically determines item type → If consumable: system automatically posts background double-entry: Debit Relevant Expense Account / Credit Inventory Account → If non-consumable: system records asset handover capturing employee name, position, and date → Issuance workflow fully complete.

Workflow 2: Item Request (Item Not Available — Purchase Request)

Department Officer submits Item Request Form → Department Head approves in system → Director physical approval recorded in system by staff → Request routed to Inventory Officer → Inventory Officer checks stock — item not available → Inventory Officer records unavailability in system → System notifies Requesting Department → Requesting department submits formal Purchase Request Form in system → Department Head approves → Director physical approval obtained and recorded in system by staff → Purchase Request transferred to Procurement Department → Purchase Request initiation workflow fully complete.

Workflow 3: Goods Receipt from Procurement

Goods delivered by supplier → Inventory Officer physically receives goods → Inventory Officer drafts Goods Receipt Note (GRN) in system → System automatically matches GRN against original approved purchase request and purchase order → Routed to Internal Control for mandatory verification → IC physically inspects goods and matches against system entries and purchase order → If Rejected: System sends notification with IC comments to Inventory Department Head → Department Head unlocks the transaction → Automatic notification sent to Inventory Officer → Officer corrects GRN based on IC comments and resubmits to Head → Head re-approves and routes back to IC. If Verified: GRN marked fully complete → System updates inventory balance → System sends "Awaiting Payment" notification to Finance → System automatically posts background double-entry: Debit Relevant Expense or Asset Account / Credit Accounts Payable → GRN workflow fully complete.

Workflow 4: Inventory Loss and Wastage

Inventory Officer identifies deteriorated, lost, or damaged items → Officer records details in system (item, quantity, reason, estimated or residual value) → Inventory Department Head reviews the entry → If write-off approved: System automatically posts background double-entry: Debit Loss on Inventory Account / Credit Inventory Account → If write-down to net realizable value approved: System automatically posts background double-

entry: Debit Loss on Write-Down Account / Credit Inventory Account → Workflow fully complete.

Workflow 5: Officer Correction Request Workflow

Inventory Officer identifies error in an entry they previously submitted (prior to any IC verification or finalization) → Officer raises formal correction request in system detailing the error → System sends notification containing justification to Inventory Department Head → Department Head reviews request → If Approved: Department Head unlocks the entry in the system → Automatic notification sent to Inventory Officer → Officer corrects and resubmits the entry → All steps (original entry, request, Head's decision, and corrected entry) recorded permanently in Control Log.

4.3. Payroll

4.3.1. Current State ("As-Is") Analysis

Overview of Current Operations:

The Payroll department operates through a highly complex, paper-based calculation process. At the end of each month, the department relies on physical attendance and statistics reports submitted by the HR department and the administrative offices of the remote mining units. Salary calculations involve intricate, multi-layered additions (basic pay, professional pay, overtime, food allowances) and deductions (rent, electricity, loans, absences, coal purchases) processed manually using standardized government paper forms (Form M-41 and M-40). Furthermore, the department manages entirely separate, highly detailed piece-rate payment structures for contract workers at the mining sites.

Key Daily/Weekly/Monthly Activities:

- Collecting physical employee attendance and statistics reports from HR and Second Unit administrative offices at the end of each month.
- Manually calculating salaries for official staff using Form M-41 and Form M-40. This involves determining gross pay (basic, professional, overtime, and food allowances) and manually subtracting a wide array of deductions (house rent, electricity, unexcused absences, advances, loans, and internal coal purchase deductions).
- Calculating complex piece-rate wages for contract mine workers based on specific daily activities (e.g., coal production paid per ton; repairs, excavations, and structural reinforcements paid per centimeter).
- Compiling manual page-by-page totals and a grand summary for the payroll sheets, which must then be physically verified by Internal Control.
- Preparing bank transfer lists (including net pay and bank commissions) and sending them to the bank both physically and via email.
- Maintaining separate manual tracking tables for employee salary advances and outstanding debts.
- Manually calculating employee travel allowances and transit fares.

Current Forms and Reports Used:

- Physical Employee Statistics/Attendance Reports
- Form M-41 (Standardized Comprehensive Payroll Form)
- Form M-40 (Secondary Payroll Form)
- Piece-Rate/Activity Tables for Mine Workers

- Bank Transfer Lists
- Manual Advance Payment and Debt Tracking Tables

Identified Pain Points & Staff Suggestions:

- **Need for System Flexibility:** In the sign-off section of the interview, the Payroll Manager explicitly requested that the new digital database be designed with high flexibility to accommodate future changes in salary norms and regulations, requiring close coordination with developers to prevent the system from becoming obsolete if government payroll rules change.
- **High Risk of Calculation Errors (Consultant Observation):** The heavy reliance on manual, piecewise calculations for mine workers (tracking excavations by the centimeter) and manual page-by-page summations on Form M-41 introduces a massive risk of mathematical errors, delayed salary disbursements, and a lack of real-time visibility into outstanding employee loans and advances.

4.3.2. Future State ("To-Be") Digital Architecture

Process Optimization

The Payroll module manages the complete employee compensation cycle from timesheet recording and leave management through to salary calculation, deduction processing, tax computation, and final payment authorization. The module covers three categories of staff: **Permanent employees**, **Temporary employees**, and **Wages-based workers** each with distinct treatment within the system.

The Internal Control and Audit department conducts a mandatory verification at the conclusion of the complete payroll sheet preparation covering all three staff categories simultaneously before any accounting entries are posted or any payment is processed. This single consolidated verification point ensures efficiency while maintaining full oversight integrity.

Employee Database

The system maintains a centralized employee database as the foundation of the entire payroll module. The database contains a record for every Permanent and Temporary employee across HQ and all Second Units. Each employee record contains the following mandatory fields:

- Full name and father's name
- Position and department
- Employment type: Permanent or Temporary
- Second Unit or HQ assignment
- Joining date

- Basic salary
- Bank account details, for Permanent and Temporary staff only
- Active or inactive status

Wages-based workers are not recorded in the employee database as named individuals. Their wages are recorded as a collective cash payment per Second Unit per period captured by the designated Second Unit person as a cash wages expense entry in the system.

Leave Management — Permanent Employees Only

The leave management module applies exclusively to Permanent employees. Temporary employees do not have a leave entitlement when they are present they are paid and when they are absent no payment is made. Wages-based workers are not subject to the leave system.

Each Permanent employee is allocated the following annual leave balance within the system upon joining and at the start of each new year:

Leave Type	**Annual Entitlement**
Annual Leave	20 days
Urgency Leave	10 days
Sick Leave	20 days
Other / Exceptional Leave	As approved
Total Standard	**50 days**

Leave Request Process:

When a Permanent employee wishes to take leave, the employee submits a physical hard copy leave request letter which is reviewed and approved by the relevant Director physically. Once the Director's physical approval is obtained and documented in the file, the Department Head records the approved leave in the system entering the employee name, leave type, start date, end date, and number of days based on the physically approved letter. The Director does not interact with the system directly.

The system automatically checks the employee's available balance for the requested leave type. If the balance is sufficient, the leave is recorded and the balance is updated in real-time.

Exceptional Leave Rule: If a Permanent employee requires leave beyond the standard 50-day annual entitlement, the system allows the Department Head to select the "Other / Exceptional Leave" category. When selected, the system enforces two mandatory rules: a written justification must be entered into the system, and a specific Director-level physical approval must be obtained, documented in the file, and recorded in the system before the system will bypass the standard balance limit and accept the extra days.

The finalized leave record is automatically reflected in the employee's timesheet for the relevant period.

At the end of each month, the system generates a leave balance summary for each Permanent employee showing opening balance, days taken by type, and closing balance which feeds directly into the monthly timesheet used for payroll processing.

Timesheet Recording

The timesheet is the primary record of each employee's working days, absences, leave taken, field work, official travel, and overtime for each month. The system handles timesheet recording differently depending on the employee's location and system access:

HQ Employees — With Individual System Access:

Each Permanent and Temporary employee with system access fills their own timesheet within the system. The timesheet can be filled on a daily basis or at any convenient time up to month end. It is not required to be submitted daily but must be complete before the month-end deadline.

HQ Employees — Without Individual System Access (Support Staff):

For employees who do not use computers as part of their daily role (e.g., drivers, cooks, cleaners, security, or certain administrative staff), a designated officer within their specific department is assigned to record their daily timesheets on their behalf.

Second Unit Employees — Single Designated Person:

At each Second Unit, one designated person is responsible for recording timesheets for all staff at that unit within the system including Permanent employees, Temporary employees, and Wages-based workers. This single-entry approach ensures ease of preparation and time efficiency at remote locations where individual system access for all staff is not practical.

Timesheet Fields:
The timesheet must capture the following for each employee for each working day:

Field	Options
Date	Calendar date
Status	Present / Absent / Annual Leave / Urgency Leave / Sick Leave / Field Work / Official Travel
Regular Hours	Number of regular working hours
Overtime Hours	Number of overtime hours worked
Notes	Optional — reason for absence or travel details

The Field Work and Official Travel status options ensure that employees engaged in off-site duties are correctly recorded as working not as absent and their overtime eligibility is preserved.

Timesheet Finalization by Department Head:

At month end, the Department Head reviews all timesheet data for their department including leave taken, absences, field work days, official travel days, and overtime hours. The Department Head formally submits the finalized timesheet record through the system. Once submitted by the Head, the timesheet is locked and routed to the Payroll department for salary processing.

Payroll Calculation & Mid-Month Pro-Rata Adjustments

Once the finalized timesheet records are received from all Department Heads, the Payroll Officer processes the payroll calculation for each employee category separately.

Mid-Month Salary and Position Changes (Pro-Rata Calculation):

The system must accommodate mid-month promotions or salary adjustments. When an employee's basic salary or position is formally changed in the Employee Database, the change must be "date-effective." During the monthly payroll calculation, the system must automatically proportion (pro-rata) the basic salary. For example, if the month has 30 days and the change occurs on the 21st, the system will automatically calculate 20 days at the old basic salary and 10 days at the new basic salary, merging them into a single correct Gross Basic Salary for that month.

Benefits entered by Payroll Officer each month:

The Payroll Officer enters the following inputs for each employee in the payroll module:

Benefits entered by Payroll Officer each month:

- **Overtime Pay** — calculated based on the overtime hours recorded in the timesheet and the applicable overtime rate. Overtime is part of gross taxable income.
- **Any other applicable allowances** — entered by Payroll Officer as applicable per employee.

Deductions entered by Payroll Officer each month:

- **Home Rent Deduction** — where the corporation provides housing to an employee, the agreed rental fee is entered by the Payroll Officer each month to be deducted from the employee's salary.

- **Electricity Deduction** — where the corporation covers electricity costs for an employee's accommodation, the electricity usage amount is entered by the Payroll Officer each month to be deducted from the salary.
- **Employee Advance Deduction** — the system notifies the Payroll Officer if an employee has an outstanding advance balance at the time of payroll preparation. The notification displays the total outstanding balance and the approved maximum deduction limit per month. The Payroll Officer manually enters the deduction amount for the current month within the approved limit. The system updates the outstanding advance balance accordingly after the deduction is applied.
- **Income Tax (Auto-Calculated)** — the system automatically calculates the applicable income tax for each employee based on the total gross taxable income in accordance with Afghan national tax law.

Net Pay Calculation:

Net Pay = Basic Salary + Overtime Pay + Other Allowances − Home Rent Deduction − Electricity Deduction − Employee Advance Deduction − Auto-Calculated Income Tax

The system calculates net pay automatically for each employee once all inputs are entered by the Payroll Officer.

Wages-Based Workers:

Wages-based worker payments are entered as a collective total per Second Unit by the designated Second Unit person. The Payroll Officer reviews and confirms the wages totals per unit before including them in the consolidated payroll record.

Payroll Sheet Finalization and Internal Control Verification

Once all payroll calculations are complete for Permanent employees, Temporary employees, and Wages-based workers (collective totals per Second Unit), the Payroll Officer submits the complete payroll record to the Payroll Department Head for review and system approval.

The Payroll Department Head reviews the complete payroll verifying all earnings, benefits, deductions, tax calculations, and net pay figures for every employee. Upon satisfaction, the Department Head formally approves the payroll record in the system.

The complete approved payroll record covering all three staff categories is then routed to the Internal Control and Audit department for a single consolidated verification. The Internal Control team:

- Checks timesheet data against other available records for consistency

- Verifies that leave days recorded are within approved balances
- Confirms that field work and official travel entries are properly recorded
- Reviews overtime hours and calculations
- Confirms that deduction entries; rent and electricity are accurately recorded and deducted from net pay
- Verifies that advance deductions are within approved limits and that outstanding balances are updated correctly
- Confirms income tax computation is accurate per applicable Afghan tax law
- Verifies wages-based worker totals per Second Unit

Upon successful Internal Control verification, the system automatically posts all accounting entries in the background without any manual input required from Finance staff. The payroll sheet is then available for printing. The printed hard copy is submitted to the relevant Director for physical review and signature this physical approval process is outside the ERP system and is documented in the physical file by Payroll staff. The confirmation of Director approval is then recorded in the system by Payroll staff.

If the Internal Control team identifies any discrepancy, IC Rejects the payroll. The system automatically sends a notification containing IC's specific comments to the Payroll Department Head. The Head unlocks the transaction, triggering a notification to the Payroll Officer. The Officer corrects the entries based on the IC comments and resubmits to the Head, who re-approves and routes it back to IC for final re-verification.

Accounting Entries Posted Automatically After IC Verification

Upon Internal Control verification of the complete payroll record, the system posts the following accounting entries automatically in the background:

Permanent and Temporary Staff — Payroll Posting:

- **Debit:** Salary Expense Account
- **Credit:** Salaries Payable Account

Tax Withheld:

- **Debit:** Salary Expense Account
- **Credit:** Tax Payable to Government Account

Advance Deduction Applied:

- **Debit:** Salaries Payable Account
- **Credit:** Employee Advance Account — reducing the outstanding advance balance

Salary Payment via Bank Transfer (after bank processing):

- **Debit:** Salaries Payable Account
- **Credit:** Bank Account

Wages-Based Workers — Cash Payment:

- **Debit:** Wages Expense Account
- **Credit:** Cash Account

Digital Employee Payslips

The system generates individual digital payslips for each Permanent and Temporary employee upon completion of the monthly payroll. Each payslip is available in PDF format and can be downloaded by the Payroll Department Head or Payroll Officer on demand. The payslip displays:

- Employee name, father's name, position, and department
- Month and year of payroll
- Basic salary
- All benefits — overtime, and other allowances
- Gross taxable income
- All deductions — rent, electricity, advance deduction, and income tax
- Net pay
- Corporation name and logo
- Payroll period reference number

Salary Advance Management

When an employee requires a salary advance, the Department Head submits the advance request in the system recording the employee name, amount requested, reason, and proposed monthly deduction limit. The request is routed for Department Head confirmation, Director physical approval (documented in file and recorded in system by staff), and Finance Department Head authorization before the advance is paid.

Once approved and paid, the advance is recorded against the employee's record in the system showing the total advance amount and the approved maximum monthly deduction limit. During each subsequent monthly payroll, the system notifies the Payroll Officer of the outstanding advance balance. The Payroll Officer manually enters the

deduction amount for that month within the approved limit and the system reduces the outstanding balance accordingly. The advance balance remains visible in the employee record at all times until fully recovered.

Officer Correction Request Workflow

If a Payroll Officer identifies an error in an entry that they have already submitted (and the error has not been caught/rejected by IC at the final payroll stage), the Officer cannot modify or delete the entry independently. The following workflow applies:

1. The Payroll Officer raises a formal correction request within the system detailing the error.
2. The system sends a notification to the Payroll Department Head.
3. The Payroll Department Head reviews the request.
4. If approved, the Department Head unlocks the entry in the system, triggering an automatic notification to the Officer, who corrects and resubmits the entry.
5. All steps are permanently recorded in the Control Log.

Compliance and Standards

ISO 9001 Quality Management

Every step of the payroll process follows a mandatory, documented sequence timesheet submission, Department Head finalization, Payroll Officer calculation, Department Head approval, Internal Control consolidated verification, accounting entry posting, and printing for Director physical approval. No payroll can be processed without the complete sequence being recorded in the system. The Control Log provides an immutable audit trail of every payroll action.

IAS 19 — Employee Benefits

All employee compensation salaries, benefits, and deductions is recorded and recognized in accordance with IAS 19. Salaries are recognized as an expense in the period they are earned not when paid in compliance with the accrual basis required under IAS 19. Internal corporation charges for housing and electricity are properly recognized as salary deductions rather than taxable compensation.

Afghan National Tax Law

Income tax deductions are calculated in accordance with applicable Afghan income tax law and the current tax slab rates issued by the Afghanistan Revenue Department (ARD). Tax withheld from employee salaries is recorded as Tax Payable to Government and must

be remitted to ARD within the legally required timeframe. The system maintains a running record of all tax withheld and remitted for each period.

4.3.3. System Roles and Access Permissions

Role	User	Permissions
Data Entry / Initiator	Staff / Payroll Officer / Designated Timesheet Officer	Permanent/Temp staff enter own timesheets. Designated department officers enter timesheets for support staff without system access. Payroll Officer enters payroll calculation variables (allowances, deductions). Cannot approve their own entries. Must raise a formal correction request to Department Head to fix submitted errors.
Department Head	Requesting Dept Heads & Payroll Dept Head	Dept Heads approve timesheets and leave requests. Payroll Dept Head approves the consolidated payroll sheet. Unlocks entries in the system for Officers to correct (either from a self-identified request or an IC rejection). Records Director-level physical approval confirmations in the system.
Targeted Verification	Internal Control and Audit Department	Mandatory verification is strictly positioned at the conclusion of the complete payroll sheet preparation. Can Verify or Reject the payroll. Rejection triggers a notification to the Payroll Dept Head to unlock the entry and coordinate adjustments. Read-only access to all payroll and timesheet records.
ED and Directors	Executive Director and Department Directors	No system access. Physical approvals obtained by relevant staff, documented in files, and recorded in the system.
System Administrator	IT Admin	Manages user accounts and settings. Zero access to payroll data or financial records.

4.3.4. Developer Specifications — Payroll

Accounting Entries — Developer Reference

Trigger Event	Debit	Credit
Permanent/Temp Staff Payroll Posting	Salary Expense Account	Salaries Payable Account
Income Tax Withheld	Salary Expense Account	Tax Payable to Government Account
Advance Deduction Applied	Salaries Payable Account	Employee Advance Account
Salary Payment via Bank Transfer	Salaries Payable Account	Bank Account
Wages-Based Workers (Cash Payment)	Wages Expense Account	Cash Account

Required Reports — Payroll

Report	Frequency	Filter Options
Consolidated Payroll Summary	On demand	Custom date range, By HQ/Second Unit, by department
Digital Payslips	On demand	Custom date range, Per employee
Leave Balance Report	On demand	Custom date range, Per employee, per department
Timesheet Summary Report	On demand	Custom date range, Per department, per Second Unit
Employee Advance Ledger	On demand	Custom date range, Per employee (showing outstanding balances)
Tax Withholding Report	On demand	Custom date range, Consolidated for tax authority submission
Control Log / Audit Trail	On demand	Custom date range, By user, action type (tracks all unlocks/corrections)
Internal Control Verification	On demand	Custom date range, By verification outcome (Verified/Rejected)

All reports must display: Corporation name and logo, report title, date range, generation timestamp, and generating username. All reports must be exportable to PDF and Excel formats.

4.3.5. BPMN Workflow Narratives — Payroll

Workflow 1: Leave Request (Permanent Employees Only)

Permanent employee submits physical hard-copy leave request letter → Relevant Director provides physical approval → Director's physical approval documented in file → Department Head records approved leave details in ERP system → System automatically checks employee's available leave balance → If standard leave: System accepts entry and deducts from balance → If Exceptional Leave requested: System enforces mandatory justification field and confirmation of Director-level physical approval → System accepts entry and bypasses standard balance limit → Approved leave is automatically reflected in the employee's timesheet → Leave workflow fully complete.

Workflow 2: Timesheet Recording and Finalization

For HQ Staff with access: Employees input daily timesheet data directly into system.

For HQ Support Staff: Designated department officer inputs daily timesheet data into system.

For Second Unit Staff: Designated Second Unit person inputs timesheet data for all unit staff into system.

Month-end reached → Department Head reviews all timesheet data (leave, absences, field work, travel, overtime) for their department → Department Head formally submits finalized timesheet record in system → System locks timesheet → Timesheet data automatically routed to Payroll module for salary processing → Timesheet workflow fully complete.

Workflow 3: Salary Advance Request and Processing

Employee requests salary advance → Department Head submits advance request in system (amount, reason, proposed monthly deduction) → Request routed for Department Head confirmation in system → Director physical approval obtained by staff and recorded in system → Request routed to Finance Department Head for final system authorization → Finance Department Head approves → Advance payment processed (via Finance PV) → System automatically records advance balance against employee record and sets approved monthly deduction limit → System triggers monthly notification to Payroll Officer during subsequent payroll calculations until balance is fully recovered.

Workflow 4: Payroll Calculation, Finalization, and Payment (IC Verification Gate)

Payroll Officer receives finalized timesheets from all departments → Officer enters applicable benefits (overtime, allowances) → Officer enters applicable deductions (rent, electricity, advance deductions) → System auto-calculates Income Tax based on Afghan tax law → System auto-calculates Net Pay (including any automated pro-rata adjustments for mid-month salary/position changes) → Designated Second Unit person enters collective Wages-Based worker totals → Payroll Officer compiles complete consolidated payroll record → Payroll Department Head reviews and formally approves in system → System routes complete payroll to Internal Control for mandatory consolidated verification → IC verifies timesheets, leave balances, deductions, tax computations, and wages totals → If Rejected: System sends notification with IC comments to Payroll Dept Head → Head unlocks transaction → Automatic notification sent to Payroll Officer → Officer corrects entry based on IC comments and resubmits to Head → Head re-approves and routes back to IC. If Verified: System automatically posts all background double-entry accounting entries (Salary Expense, Salaries Payable, Tax Payable, Advance Reductions, Wages Expense) → Payroll sheet becomes available for printing → Printed hard copy submitted to Director for physical signature → Physical approval recorded in system by Payroll staff → Payroll payments processed via bank transfer → System generates individual Digital Payslips → Payroll workflow fully complete.

Workflow 5: Officer Correction Request Workflow

Payroll Officer identifies error in an already submitted entry (prior to IC verification or finalization) → Officer raises formal correction request in system detailing the error → System sends notification to Payroll Department Head → Department Head reviews request → If Approved: Department Head unlocks the entry in the system → Automatic notification sent to Payroll Officer → Officer corrects and resubmits the entry → All steps (original entry, request, Head's decision, and corrected entry) recorded permanently in Control Log.

4.4. Sales and Marketing

4.4.1. Current State ("As-Is") Analysis

Overview of Current Operations:

The Sales department currently operates through manual paperwork and physical verifications. It acts as the coordination point between the customer, the Finance department, and the mining sites. The process relies heavily on checking physical bank slips and writing official hard-copy letters to authorize customers to collect their coal from the designated mines. All tracking and reporting are done manually.

Key Daily/Weekly/Monthly Activities:

- Receiving physical purchase requests from customers and forwarding them to the Finance department.
- Checking and verifying physical bank deposit slips to confirm payment before assigning a customer to a specific mine.
- Writing and issuing official hard-copy letters (Maktub) that authorize the customer to collect coal from the assigned mining site.
- Manually recording sales and payment data, separated by each specific mining site.
- Preparing manual sales reports on a monthly, quarterly, and annual basis.

Current Forms and Reports Used:

- Customer Purchase Requests
- Physical Bank Deposit Slips
- Official Hard-Copy Coal Collection Letters (Maktub)
- Manual Sales and Payment Ledgers
- Manual Monthly, Quarterly, and Annual Sales Reports

Identified Pain Points & Staff Suggestions:

- **No Customer Tracking System:** There is currently no system in place to easily identify or track the exact quantity of coal sold to, or the exact payments made by, each individual customer. This manual limitation makes it very difficult to manage customer accounts accurately.

4.4.2. Future State ("To-Be") Digital Architecture

Process Optimization

The Sales and Marketing module manages the complete coal sales cycle from tariff issuance after ED-approved orders through to payment confirmation, coal allocation authorization, and dispatch tracking. The module acts as the initiating point of the

revenue cycle all revenue recognition events described in the Finance and Accounting section originate from actions taken within this module in conjunction with the Inventory module at the Second Units.

Targeted Internal Control Verification

The Internal Control and Audit department is embedded as a mandatory verification layer strictly at the **Payment Verification** stage. This ensures that the financial receipt matches the authorized order before the corporation authorizes the release of coal.

Sales Order Process and ED Approval

The sales cycle begins outside the ERP system. When a customer wishes to purchase coal, the customer submits a physical sales order request to the corporation. This request is reviewed internally and submitted to the Executive Director for approval. The ED physically reviews and approves the sales order signing the hard copy document. The signed hard copy approval is filed physically by the relevant department staff.

The sales order is not entered into the ERP system until the ED's physical approval has already been obtained and is in hand. Only after the signed hard copy ED approval is confirmed does the Sales Officer record the order in the ERP system. This ensures that no order exists in the system without prior authorization.

The Sales Officer records the following details in the system when entering the ED approved order:

Field	Source
Sequential Order Number	Auto-generated by system
Date of Order Entry	System date
Customer Name	Entered by Sales Officer
Coal Type and Grade	Entered by Sales Officer
Quantity Ordered (tonnes)	Entered by Sales Officer
Second Unit for Collection	Entered by Sales Officer
ED Approval Reference	**Entered manually by Sales Officer** — referencing the hard copy signed approval
Date of ED Physical Approval	Entered by Sales Officer
Notes	Optional

Once the order is recorded in the system, the Sales Department Head reviews and confirms the entry verifying it accurately reflects the physically approved order. The order status is then updated to ED Approved Recorded and the workflow proceeds to tariff issuance.

Tariff Issuance

Once the sales order is confirmed by the Sales Department Head, the Sales Officer generates a formal Tariff Slip within the ERP system. The tariff slip is the official payment instruction issued to the customer directing them to pay the specified amount into the corporation's designated bank account before coal can be released.

The system generates the tariff slip with the following fields:

Field	Source
Sequential Tariff Number	Auto-generated — sequential, not editable
Date of Issue	System date
Customer Name	From recorded sales order
Coal Type and Grade	From recorded sales order
Quantity Approved (tonnes)	From recorded sales order
Unit Price (AFN per tonne)	From approved tariff schedule
Total Amount Payable (AFN)	Auto-calculated — Quantity × Unit Price
Payment Bank Account	Pre-loaded — corporation's designated bank account
Second Unit Location for Coal Receipt	From recorded sales order
ED Approval Reference	From recorded sales order — entered by Sales Officer
Tariff Validity Period	Defined by Sales Department Head

The tariff slip is printed and issued to the customer as a hard copy payment instruction. It is also stored digitally in the system linked to the sales order record.

Important: The issuance of the tariff slip does not constitute revenue. It is purely a payment instruction. Revenue is only recognized when coal is physically transferred to the customer at the mining site Second Unit confirmed by the Ministry of Mines weighing scale software.

Customer Payment and Finance Confirmation

Once the tariff slip is issued, the customer makes payment into the corporation's designated bank account. The customer then physically presents their bank payment slip to the Finance department at HQ as evidence of payment.

The Finance Officer verifies the payment in the bank through the bank reconciliation process described in the Finance and Accounting section matching the deposit to the corresponding customer and tariff slip reference. Upon confirmed match, the Finance Officer posts the accounting entry: **Debit Bank Account / Credit Customer Advance (Current Liability)** and the system updates the sales order status to **Payment Confirmed**.

The Finance department then formally notifies the Sales department within the system that payment has been confirmed for the relevant tariff slip reference.

Internal Control Payment Verification

Upon Finance confirmation of payment, the Internal Control and Audit department conducts its mandatory payment verification before the coal allocation authorization can be issued. The Internal Control team verifies:

- The bank payment slip presented by the customer matches the deposit confirmed by Finance in the bank statement
- The payment amount matches the total amount stated on the tariff slip exactly
- The quantity ordered on the ED-approved hard copy approval letter is consistent with the quantity on the tariff slip and the amount paid
- The Customer Advance has been correctly posted in the Finance module

Only after Internal Control verification is recorded in the system can the Sales department proceed to generate the coal allocation authorization. This three-way check (bank payment slip, tariff slip, and ED-approved hard copy order) ensures full integrity before any coal is authorized for release. If discrepancies are found, IC rejects the verification, forcing a formal correction loop between Finance and Sales to resolve the mismatch.

Coal Allocation Authorization

Upon confirmed payment and Internal Control verification, the Sales Officer generates the **Coal Allocation Authorization** within the ERP system. The authorization records and tracks the customer's entitlement to receive coal from the designated Second Unit. The following details are recorded:

Field	Source
Sequential Authorization Number	Auto-generated
Date of Authorization	System date
Customer Name	From sales order
Total Quantity Authorized (tonnes)	From ED-approved order
Unit Price (AFN)	From tariff slip
Total Amount Paid (AFN)	From confirmed payment
Second Unit for Collection	From sales order
Tariff Slip Reference	Linked from system
ED Approval Reference	From sales order entry
Authorization Status	Open

The coal allocation authorization is printed as a hard copy official letter on the corporation's official letterhead outside the ERP system and issued to the customer. This hard copy letter is the physical document the customer presents at the mining site Second Unit to collect their coal. The ERP records the authorization digitally and tracks all subsequent dispatch activity against it.

The ERP maintains the following live tracking fields for each authorization:

Tracking Field	Description
Total Quantity Authorized (tonnes)	From approved order
Total Quantity Dispatched (tonnes)	Updated in real time from weighing system
Remaining Quantity Entitlement (tonnes)	Auto-calculated — Authorized minus Dispatched
Number of Collection Visits	Count of separate dispatch events
Authorization Status	Open / Partially Fulfilled / Fully Fulfilled / Expired

Weighing System Integration and Dispatch Tracking — Mining Sites Only

The Ministry of Mines weighing scale software is installed exclusively at mining site Second Units, it is not present at other sub-offices. At mining sites, the weighing scale software is operated by the user responsible for that operation at each Second Unit. The Sales department's role in this stage is visibility and tracking only the physical dispatch process is managed through the weighing scale software by the responsible user at the site.

When the customer presents their hardcopy coal allocation letter at the mining site, the responsible user at that Second Unit operates the weighing scale software to record the coal dispatch. The weighing scale software records:

- Customer name and authorization reference
- Quantity of coal dispatched in this transaction (tonnes)
- Cumulative quantity dispatched to this customer to date against their authorization
- **Remaining coal entitlement balance**: the quantity still owed to the customer based on their total authorized amount minus all dispatches to date
- Date and time of dispatch

The North Coal ERP is integrated with this Ministry of Mines weighing scale software. Dispatch data is transmitted from the weighing software to the central ERP either in real time or through a defined end-of-day synchronization ensuring that even geographically remote mining site Second Units contribute dispatch data to the central system without delay.

Upon receipt of dispatch data from the weighing system, the ERP automatically performs the following actions simultaneously:

1. Sales Module Update:

The authorization record is updated cumulative dispatched quantity increased, remaining entitlement balance reduced, authorization status updated (Open / Partially Fulfilled / Fully Fulfilled).

2. Finance Module — Revenue Recognition:
The system automatically posts the revenue recognition entry:

- **Debit:** Customer Advance (Current Liability) — reduced by the value of coal dispatched
- **Credit:** Revenue Account — increased by the same amount

3. Inventory Module — Stock Reduction:

The coal stock at the relevant mining site Second Unit is automatically reduced by the dispatched quantity.

The weighing system dispatch record is the sole trigger for revenue recognition.

Sub-Office Second Units — Coal Dispatch Process

Second Units designated as Sub-Offices are primarily utilized for coordination, employee transit, and distribution, rather than large-scale coal dispatch. The Ministry of Mines weighing scale software is not installed at these locations.

If minor coal sales or distributions occur at a Sub-Office, the transaction is not recorded via a dispatch integration. Instead, the transaction is processed directly through the standard Voucher Creation interface (described in the Finance and Accounting module). The Finance Officer or designated staff generates the relevant Receipt Voucher (RV) or Journal Voucher (JV), selecting the specific customer, relevant accounts, and classes. Upon submission, the system auto-posts the correct financial impacts based on the voucher's account selections.

These manual dispatch entries at selling hubs are subject to the same Internal Control verification process as weighing system entries at mining sites.

Order and Dispatch Tracking and Reporting

The system provides comprehensive real-time tracking of all sales orders showing at any point in time how much coal has been ordered, how much has been paid for, how much has been authorized for collection, and how much has actually been dispatched from each Second Unit. Reports can be generated for any defined time period and filtered by:

- Individual customer
- Specific Second Unit — mining site or Sub office
- Date range
- Authorization status — Open, Partially Fulfilled, Fully Fulfilled, Expired

Officer Correction Request Workflow

If a Sales Officer identifies an error in an entry that they have already submitted (prior to IC verification or finalization), the Officer cannot modify or delete the entry independently.

1. The Sales Officer raises a formal correction request within the system detailing the error.
2. The system sends a notification to the Sales Department Head.
3. If approved, the Department Head unlocks the entry in the system.
4. This action triggers an automatic notification to the Sales Officer, who corrects and resubmits the entry.
5. All steps are permanently recorded in the Control Log.

Compliance and Standards

ISO 9001 Quality Management

Every stage of the sales cycle follows a mandatory documented sequence: ED physical approval obtained first, order recorded, tariff issuance, Finance payment confirmation, IC payment verification, allocation authorization, and dispatch tracking. No coal can be released without the complete chain being recorded. The Control Log provides an immutable audit trail of every sales action.

IFRS 15 — Revenue from Contracts with Customers

The sales module enforces IFRS 15 compliance. Revenue is recognized only at the point of physical coal delivery confirmed by the Ministry of Mines weighing system at mining sites or via standard voucher processing at sub-offices. All payments received in advance of delivery are held as Customer Advance liabilities until the delivery obligation is fulfilled.

4.4.3. System Roles and Access Permissions

Role	User	Permissions
Data Entry / Initiator	Sales Officers	Can record ED-approved sales orders in system after physical ED approval is obtained. Can generate tariff slips for recorded orders. Can generate coal allocation authorizations upon confirmed IC-verified payment. Can view real-time dispatch tracking for all active authorizations. Cannot approve sales orders. Cannot modify a submitted record — must raise formal correction request to Sales Department Head.
Department Head	Sales Department Head	Can review and confirm all sales orders recorded by Sales Officers — verifying accuracy against physical ED-approved document. Can approve or reject correction requests from Officers, unlocking entries for correction. Can generate all sales reports. Prints coal allocation authorization letters for physical issuance to customers.
Finance Verification	Finance Officer	Confirms payment in bank reconciliation module. System automatically posts Customer Advance entry. Notifies Sales department of payment confirmation. Processes manual RV/JV vouchers for minor sales/distributions at Sub-Offices.
Targeted Verification	Internal Control and Audit Department	Mandatory verification is strictly positioned at the three-way Payment Verification stage (bank slip vs tariff slip vs ED-approved order). Read-only access to sales orders, authorizations, and dispatch records. Can Verify or Reject the payment confirmation.
Responsible User	Second Unit (Mining Sites)	Operates Ministry of Mines weighing scale software at mining sites. Responsible for coal dispatch recording through the weighing software (which integrates with the ERP).
ED and Directors	Executive Director and Department Directors	No system access. All approvals are physical. Sales order approval is documented in physical file and reference is entered manually in system by Sales Officers.

Role	User	Permissions
System Administrator	IT Admin	Can create and deactivate user accounts and manage system settings. Zero access to sales data or financial records.

4.4.4. Developer Specifications — Sales and Marketing

Accounting Entries — Developer Reference

Trigger Event	Debit	Credit
Bank deposit matched to customer tariff — Finance reconciliation	Bank Account	Customer Advance (Current Liability)
Coal dispatched — weighing system confirmation (mining sites)	Customer Advance (Current Liability)	Revenue Account
Inventory reduced upon dispatch (Auto-posted simultaneously with revenue)	Cost of Coal Sold Account	Coal Inventory Account
Minor Sales or Distributions at Sub-Offices	Handled via standard Finance Voucher selections (RV/JV)	Handled via standard Finance Voucher selections (RV/JV)

Required Reports — Sales and Marketing

Report	Frequency	Filter Options
Sales Order Register	On demand	Custom date range, Per customer, per status
Tariff Slip Register	On demand	Custom date range, Sequential, per customer
Payment Confirmation Report	On demand	Custom date range, Per customer, per tariff reference
Coal Allocation Authorization Register	On demand / Real-time	Custom date range, Per customer, per Second Unit, per status
Dispatch Tracking Report — Per Customer	On demand / Real-time	Custom date range, Total authorized, total dispatched, remaining entitlement, visit count
Dispatch Tracking Report — Per Second Unit	On demand / Real-time	Custom date range, Per mining site or Sub-Office — total dispatched/sold

Report	Frequency	Filter Options
Revenue Recognition Summary	On demand	Custom date range, Per Second Unit, per customer — linked to dispatch events/vouchers
Outstanding Entitlement Report	On demand / Real-time	Custom date range, All active authorizations with remaining quantities
Expired Authorization Report	On demand	Custom date range, Authorizations past validity with unfulfilled quantities
Internal Control Verification	On demand	Custom date range, By verification outcome (Verified/Rejected)
Control Log / Audit Trail Report	On demand	Custom date range, By user, action type

All reports must display: Corporation name and logo, report title, date range, generation timestamp, and generating username. All reports must be exportable to PDF and Excel formats.

4.4.5. BPMN Workflow Narratives — Sales and Marketing

Workflow 1: Sales Order and Tariff Issuance

Customer submits physical sales order request → Request reviewed internally and physical ED signature obtained outside system → Signed hard copy filed by staff → Sales Officer inputs order details and ED physical approval reference into ERP system → Sales Department Head reviews system entry against hard copy and confirms → System marks order "ED Approved Recorded" → Sales Officer generates Tariff Slip in system → System auto-calculates total amount payable based on quantity and approved unit price → Tariff Slip printed and issued to customer as payment instruction → Workflow complete.

Workflow 2: Payment Confirmation and IC Verification Gate

Customer pays funds into designated corporate bank account → Customer presents physical bank payment slip to HQ Finance → Finance Officer reconciles deposit in system against Tariff Slip → System auto-posts background double-entry: Debit Bank Account / Credit Customer Advance → System updates Sales order to "Payment Confirmed" → System routes to Internal Control for mandatory 3-way payment verification → IC verifies Bank Slip vs Tariff Slip vs ED-Approved Order → If Rejected: System notifies Finance and Sales Heads to coordinate correction → If Verified: System marks payment fully verified and clears order for Coal Allocation → Workflow complete.

Workflow 3: Coal Allocation Authorization and Dispatch (Mining Sites)

IC Verification of payment completed → Sales Officer generates Coal Allocation Authorization in system → Sales Head prints and issues hard-copy authorization letter to customer → Customer travels to Mining Site Second Unit → Customer presents hard-copy letter to Responsible User at site → User operates Ministry of Mines Weighing Software to record dispatch → Weighing software transmits dispatch data to ERP (real-time or end-of-day sync) → ERP automatically updates Authorization tracking (increases dispatched, reduces remaining balance) → ERP automatically posts background Revenue Recognition (Dr Customer Advance / Cr Revenue) and Inventory Reduction (Dr Cost of Coal Sold / Cr Coal Inventory) → Workflow complete.

Workflow 4: Minor Sales / Distribution (Sub-Offices)

Customer or employee requests minor coal distribution at Sub-Office → Finance Officer (or designated staff) initiates Receipt Voucher (RV) or Journal Voucher (JV) in Finance Module → Officer selects Customer/Employee from database and assigns appropriate accounting classes → Officer inputs amount/quantity and description → Voucher submitted and approved per standard Finance workflows → System auto-posts financial impacts based on specific voucher selections → Workflow complete.

Workflow 5: Officer Correction Request Workflow

Sales Officer identifies error in an entry they previously submitted (prior to finalization or IC review) → Officer raises formal correction request in system detailing the error → System sends notification to Sales Department Head → Department Head reviews request → If Approved: Department Head unlocks the entry in the system → Automatic notification sent to Sales Officer → Officer corrects and resubmits the entry → All steps (original entry, request, Head's decision, and corrected entry) recorded permanently in Control Log.

4.5. Properties

4.5.1. Current State ("As-Is") Analysis

Overview of Current Operations:

The Properties department is responsible for managing all corporate real estate, including operational buildings, commercial spaces, agricultural lands, and orchards. The department handles property registration, leasing for revenue generation, strategic property development, and the legal recovery of encroached lands. Currently, the management of these assets is entirely manual, relying on physical deed storage and paper-based tracking, which makes it difficult to ascertain the exact location and financial value of the corporation's real estate portfolio.

Key Daily/Weekly/Monthly Activities:

- Registering and safekeeping physical property documents, including Sharia deeds, possession documents, and property maps for all corporate assets.
- Managing lease agreements and contracting commercial areas and agricultural lands to external parties to generate corporate revenue.
- Developing strategic plans for property utilization, such as growing timber on company-owned orchard lands for use in coal mine reinforcements.
- Planning the development of new commercial markets and revenue-generating facilities.
- Tracking and actively pursuing the recovery of usurped or encroached corporate properties through judicial, legal, and relevant government agencies.

Current Forms and Reports Used:

- Physical Property Deeds and Legal Documents (Sharia deeds, possession documents)
- Physical Property Maps and Sketches
- Manual Lease and Tenancy Contracts
- Hard-copy Letters, Proposals, and Legal Rulings

Identified Pain Points & Staff Suggestions:

- **No Centralized Property Value Database:** There is no complete digital system to easily identify, verify, and track the corporation's lands and properties along with their current monetary value.
- **Disorganized Document Management:** The department lacks an organized electronic filing system for securely storing and retrieving official letters, lease proposals, government decrees, and rulings.

- **Missing Geographic Location Data:** There is no systematic coordinate database (GPS tracking) to accurately pinpoint and record the exact physical locations and boundaries of the corporation's real estate assets.

4.5.2. Future State ("To-Be") Digital Architecture

Process Optimization

The Properties module serves as the central registry for all land and building assets owned or rented by the North Coal State Owned Corporation across HQ and all Second Units. The module is responsible for recording, classifying, and maintaining complete details of every property asset whether in use by the corporation or generating rental income and ensuring that all property-related financial impacts are automatically reflected in the Finance and Accounting module without manual re-entry.

The Properties module works in direct coordination with the Inventory and Warehouse Management module which holds overall responsibility for depreciation charging across all fixed asset classes including buildings. The Properties department records and classifies the asset, the depreciation rate and monthly depreciation posting are managed by the Inventory department as defined in Section 4.2.

The Properties department functions as the **credit controller** for all rental properties responsible for tracking, following up, and reporting on all tenant payment obligations across all rental property types including residential homes, supermarkets, and other commercial spaces.

Property Asset Classification

All properties recorded in the system are classified under one of the following categories:

Classification	Description
Land — In Use	Land owned by the corporation and used for operational purposes; mining sites, offices, storage yards
Land — Rental	Land owned by the corporation and leased or rented to external parties
Building — In Use	Buildings owned by the corporation and occupied for operational or administrative purposes
Building — Rental	Buildings owned by the corporation and rented out to external tenants generating rental income

The system allows the Properties Department Head to add new property classes if required; subject to Finance Department Head approval to ensure correct chart of accounts alignment.

Property Asset Registration

When a new property asset is acquired or identified for recording, the Properties Officer registers it in the system with the following mandatory details:

For Land Assets:

Field	Notes
Property Reference Number	Auto-generated — sequential
Property Name / Description	Entered by Properties Officer
Classification	Land — In Use / Land — Rental
Location	Full address and Second Unit assignment if applicable
Total Area (square metres)	Entered by Properties Officer
Date of Acquisition	Entered by Properties Officer
Acquisition Cost (AFN)	Entered by Properties Officer
Title Deed Reference Number	Entered by Properties Officer
Current Status	In Use / Rented / Vacant
Depreciation Rate	Automatically locked at zero — land is a non-depreciable asset class
Supporting Documents	File upload — title deed, acquisition documents

For Building Assets:

Field	Notes
Property Reference Number	Auto-generated — sequential
Property Name / Description	Entered by Properties Officer
Classification	Building — In Use / Building — Rental
Location	Full address and Second Unit assignment if applicable
Total Area (square metres)	Entered by Properties Officer
Date of Construction or Acquisition	Entered by Properties Officer
Acquisition or Construction Cost (AFN)	Entered by Properties Officer
Title Deed or Registration Reference	Entered by Properties Officer
Current Status	In Use / Rented / Vacant / Under Maintenance
Depreciation Rate	Set and managed by Inventory and Warehouse Management department — not editable by Properties department
Supporting Documents	File upload — title deed, valuation reports, construction certificates

Upon registration of a building asset, the system automatically notifies the Inventory and Warehouse Management department to assign the applicable depreciation rate. Monthly depreciation is then charged automatically by the system based on the rate set by the Inventory department as described in Section 4.2.

In-Use Property Management

For properties classified as In Use whether land or buildings; the system maintains a complete operational record showing:

- Which department or Second Unit is occupying the property
- The purpose of use: administrative office, mining site, warehouse, staff accommodation, storage yard, etc.
- Date the property was assigned to its current use
- Any maintenance or renovation history recorded against the property
- Current condition status; Good / Under Maintenance / Requires Attention

The Properties Officer is responsible for keeping all In Use property records current within the system. Any change in the use, occupying department, or condition of a property must be updated in the system by the Properties Officer and approved by the Properties Department Head.

Rental Property Management and Credit Control

For properties classified as Rental where the corporation earns rental income from external tenants including residential homes, supermarkets, and other commercial spaces the system maintains a complete tenancy record and serves as the corporation's credit control tool for all rental income. The Properties department is solely responsible for monitoring, following up, and reporting on all tenant payment obligations.

The system maintains the following tenancy record for each rental property:

Field	**Notes**
Tenant Name	Entered by Properties Officer
Tenant Type	Residential / Commercial — Supermarket / Commercial — Other
Tenancy Agreement Reference	Entered by Properties Officer
Rental Start Date	Entered by Properties Officer
Rental End Date	Entered by Properties Officer
Monthly Rental Amount (AFN)	Entered by Properties Officer
Payment Frequency	Monthly / Quarterly / Annual
Total Contract Value (AFN)	Auto-calculated — Monthly Amount × Contract Duration
BRT Rate	Fixed at 4% auto-applied by system on each rental income posting
Bank Account for Rental Receipts	Pre-loaded — corporation's designated account

Field	Notes
Tenancy Status	Active / Expired / Under Renewal / Terminated
Supporting Documents	File upload — signed tenancy agreement

Business Receipts Tax (BRT) on Rental Income

In strict compliance with accrual accounting principles, the ERP recognizes rental income and computes the associated Business Receipts Tax (BRT) based on occurrence (when the rent becomes due), regardless of whether the cash has been received from the tenant yet.

All rental income generated by the corporation is subject to Business Receipts Tax (BRT) at a fixed rate of 4% payable to the Government of Afghanistan.

For every rental income recognized and confirmed in the Finance module, the system applies the BRT calculation as follows:

Example:

- Rental Income: AFN 100,000
- BRT at 4%: AFN 4,000 payable to Government
- Net Rental Income Recognized: AFN 96,000

On the scheduled due date for each rental period, the system automatically posts the following accounting entries in the background:

Rental Income Recognition (Occurrence):

- *Debit:* Tenant Receivable Account (Asset)
- *Credit:* Rental Income Account

BRT Computation (Occurrence):

- *Debit:* BRT Expense Account (computed at 4% of the rental income due)
- *Credit:* BRT Payable to Government Account (Liability)

The BRT Payable to Government account accumulates all outstanding BRT obligations across all rental properties and all periods; providing a running balance of the total tax liability due to the government at any point in time. The Finance department is responsible for the actual remittance of BRT to the government the Properties module records the liability and the Finance module processes the payment.

Tenant Payment Receipt (Internal Control Verification Gate):

When a tenant makes a rental payment into the corporation's bank account, they present their physical bank deposit slip to the Finance Department.

1. The Finance Officer logs the payment receipt in the system against the specific tenant's ledger.
2. **Internal Control Verification:** Before this payment is finalized and posted to the ledger, it is routed to the Internal Control and Audit department. IC verifies that the physical bank deposit slip perfectly matches the recorded receipt in the system and corresponds to the correct tenant's outstanding receivable balance.
3. **Rejection & Correction:** If discrepancies exist, IC Rejects the receipt, triggering a notification to Finance and Properties to investigate and correct the entry.
4. **Approval & Posting:** Upon confirmed IC verification, the system automatically posts the following background entry to clear the debt:
    - *Debit:* Bank Account
    - *Credit:* Tenant Receivable Account

Tenant Ledger and Payment Tracking

For each active tenant, the system maintains a running **Tenant Ledger** showing:

Ledger Field	Description
Total Rental Receivable to Date (AFN)	Cumulative rental due from contract start to current date
Total Received from Tenant to Date (AFN)	Cumulative confirmed rental income received and posted in Finance module
Outstanding Balance (AFN)	Total Receivable minus Total Received — live at all times
Total BRT Payable to Date (AFN)	Cumulative BRT computed on all rental income
Next Payment Due Date	Auto-calculated based on payment frequency and last payment received
Total Remaining Rental Until Contract Expiry (AFN)	Auto-calculated — remaining periods × rental amount
Number of Payments Received	Count of confirmed payments
Number of Payments Overdue	Count of missed payments
Last Payment Date	Date of most recent confirmed payment

This ledger gives the Properties department full visibility of each tenant's payment position at any moment functioning as a live credit control dashboard across all rental properties.

Automated Rent Due Notifications and Follow-Up

The system generates automated notifications and follow-up alerts to the Properties department for every active tenancy:

Alert Type	Trigger	Recipient
Rent Due — Advance Notice	5 days before each scheduled rental payment date	Properties Department Head and Properties Officer
Rent Due — On Due Date	On the scheduled payment date if not yet received	Properties Department Head
Rent Overdue	Day after due date if payment not confirmed in system	Properties Department Head
Escalating Overdue Alert	Every 7 days while payment remains outstanding	Properties Department Head
Tenancy Expiry — Advance Notice	60 days before contract end date	Properties Department Head
Tenancy Expiry — Final Notice	14 days before contract end date	Properties Department Head

When rental income is confirmed by the Finance Officer in the bank reconciliation module, the system automatically clears the relevant outstanding alert updating the Tenant Ledger and resetting the next payment due date counter. If payment is not confirmed by the due date, the overdue alert remains active and escalates until resolved.

The Properties Officer can log follow-up actions against each overdue alert within the system recording the date of follow-up, the method of contact (phone call, letter, in-person visit), and the tenant's response. This follow-up log is permanently recorded in the system and visible to the Properties Department Head and Internal Control team at all times.

Link with Finance and Accounting Module

The Properties module is directly linked to the Finance and Accounting module through the following automatic financial impacts; all posted without manual intervention from Finance staff:

Property Event	Debit	Credit
New building asset registered and approved	Building Asset Account	Accounts Payable / Bank Account
New land asset registered and approved	Land Asset Account	Accounts Payable / Bank Account
Monthly building depreciation — auto-posted by Inventory dept rate	Depreciation Expense	Accumulated Depreciation — Buildings
Land depreciation	No entry — zero rate locked	—

Property Event	Debit	Credit
Rental income received and confirmed in Finance reconciliation	Bank Account	Rental Income Receivable Account
BRT computed and posted on each rental income	BRT Expense Account	BRT Payable to Government Account
Property disposed or written off	Accumulated Depreciation + Loss on Disposal Account	Building / Land Asset Account

The Finance Department Head and Internal Control team have read-only access to all property records within the Properties module at all times ensuring full financial visibility without requiring manual reporting from the Properties department.

Officer Correction Request Workflow

If a Properties Officer identifies an error in a submitted record (prior to any finalization), the Officer cannot modify or delete the entry independently:

1. The Properties Officer raises a formal correction request within the system detailing the error.
2. The system sends a notification to the Properties Department Head.
3. If approved, the Department Head unlocks the entry in the system.
4. This action triggers an automatic notification to the Properties Officer, who corrects and resubmits the entry.
5. All steps are permanently recorded in the Control Log.

Compliance and Standards

ISO 9001 Quality Management

Every property record in the system follows a mandatory registration, review, and verification sequence. No property asset is added, modified, or disposed of without the complete approval chain being recorded. The Control Log provides an immutable audit trail of every Properties action across HQ and all Second Units.

IAS 16 — Property, Plant and Equipment

All land and building assets are recorded and maintained in accordance with IAS 16. Land is classified as a non-depreciable asset the depreciation rate is locked at zero in the system. Buildings are depreciated at the rates defined by the Inventory and Warehouse Management department and applied monthly by the system automatically. Any revaluation or impairment of property assets is recorded in the system with supporting documentation and subject to Finance Department Head and Internal Control approval before posting.

IAS 40 — Investment Property

Properties classified as Rental; where the corporation holds land or buildings to earn rental income rather than for operational use are recognized and presented in accordance with IAS 40. Rental income is recognized on a systematic basis over the tenancy period in compliance with IAS 40 requirements.

Afghan Tax Law — Business Receipts Tax (BRT)

All rental income earned by the corporation is subject to Business Receipts Tax at a rate of 4% per occurrence in accordance with applicable Afghan tax law and Afghanistan Revenue Department (ARD) regulations. The system automatically computes and records BRT on every rental income event. The accumulated BRT Payable to Government balance is maintained in real time and is available for review by the Finance department for timely remittance to ARD.

4.5.3. System Roles and Access Permissions

Role	User	Permissions
Data Entry / Initiator	Properties Officers	Registers land and building assets. Updates In-Use occupancy records. Enters tenancy records. Logs tenant follow-up actions against overdue alerts. Cannot approve records. Must raise formal correction requests to Properties Head for errors.
Department Head	Properties Department Head	Approves all property registrations and tenancy records. Authorizes property status changes. Approves correction requests, unlocking entries for Officers. Receives automated rent/overdue alerts.
Depreciation Assignment	Inventory Department Head	Receives notification upon registration of new building assets. Assigns applicable depreciation rates in the system.
Finance Department	Finance Officers / Head	Logs tenant bank deposit slips into the system. Remits accumulated BRT to ARD. Read-only access to property ledgers and tenancy records.
Targeted Verification	Internal Control and Audit Department	Mandatory verification is **strictly positioned at the Tenant Payment Receipt stage**. IC verifies bank deposit slips against the tenant ledger before the payment clears the receivable. Read-only access to all property records, ledgers, and BRT computations.

Role	User	Permissions
ED and Directors	Executive Director and Department Directors	No system access. Physical approvals for property acquisitions/disposals are documented in physical files and recorded by Properties staff.
System Administrator	IT Admin	Manages user accounts and settings. Zero access to property data or financial records.

4.5.4. Developer Specifications — Properties

Accounting Entries — Developer Reference

Trigger Event	Debit	Credit
New building asset registered and approved	Building Asset Account	Accounts Payable / Bank Account
New land asset registered and approved	Land Asset Account	Accounts Payable / Bank Account
Monthly depreciation on buildings (Auto-posted)	Depreciation Expense	Accumulated Depreciation — Buildings
Rental period occurs (Rent becomes due)	Tenant Receivable Account	Rental Income Account
BRT computation on rental occurrence	BRT Expense Account	BRT Payable to Government Account
Tenant payment receipt (After IC Verification)	Bank Account	Tenant Receivable Account
BRT remitted to Government by Finance	BRT Payable to Government Account	Bank Account
Property disposed	Accumulated Depr. + Loss on Disposal	Building / Land Asset Account

Required Reports — Properties

Report	Frequency	Filter Options
Full Property Asset Register	On demand	Custom date range, Per classification (Land In Use, Land Rental, Building In Use, Building Rental)
In Use Property Report	On demand	Custom date range, Per department, per Second Unit
Tenant Ledger Report	On demand	Custom date range, Per tenant, per property — total receivable, total received, outstanding balance
Rental Occurrence & Collection Summary	On demand	Custom date range, Total rent billed vs. total received per period
Outstanding Rental Balance Report	On demand / Real-time	Custom date range, Sorted by overdue duration
Rent Due / Overdue Alerts Report	On demand / Real-time	Custom date range, All active alerts
Tenant Follow-Up Log Report	On demand	Custom date range, Per tenant (actions & outcomes)
Tenancy Expiry Report	On demand	Custom date range, Expiring within defined period
BRT Summary Report	On demand	Custom date range, Total BRT accrued per period
BRT Payable to Government Ledger	On demand	Custom date range, Running balance of outstanding BRT obligations
Internal Control Verification Report	On demand	Custom date range, By verification outcome (Verified/Rejected)
Control Log / Audit Trail Report	On demand	Custom date range, By user, action type

All reports must display: Corporation name and logo, report title, date range, generation timestamp, and generating username. All reports must be exportable to PDF and Excel formats.

4.5.5. BPMN Workflow Narratives — Properties

Workflow 1: Property Asset Registration

Properties Officer inputs new property details (Land or Building) into system and uploads title deeds → Properties Department Head reviews and approves entry → System marks property as Active → If Building: System automatically notifies Inventory Department Head → Inventory Head enters applicable depreciation rate into system → System automatically posts monthly depreciation in background → Registration workflow complete.

Workflow 2: Rental Income and BRT Auto-Posting

System reaches scheduled rent due date for active tenancy (based on payment frequency) → System automatically calculates rental amount due → System automatically computes 4% BRT on the rental amount → System automatically posts background double-entries: (1) Debit Tenant Receivable / Credit Rental Income; and (2) Debit BRT Expense / Credit BRT Payable → System updates Tenant Ledger to reflect new Outstanding Balance → System triggers "Rent Due" notification to Properties Department → Occurrence workflow complete.

Workflow 3: Tenant Payment Receipt and IC Verification Gate

Tenant pays rent into corporate bank account and presents physical bank slip to HQ Finance → Finance Officer logs payment receipt details into system against the specific Tenant Ledger → System routes receipt to Internal Control for mandatory verification → IC matches physical bank slip to system entry and tenant ledger → If Rejected: System notifies Finance and Properties to coordinate correction → If Verified: System automatically posts background double-entry: Debit Bank Account / Credit Tenant Receivable → System clears outstanding balance on Tenant Ledger → System clears active "Rent Overdue" alerts for that tenant → Payment workflow complete.

Workflow 4: Officer Correction Request

Properties Officer identifies error in an entry they previously submitted → Officer raises formal correction request in system detailing the error → System sends notification containing justification to Properties Department Head → Department Head reviews request → If Approved: Department Head unlocks the entry in the system → Automatic notification sent to Properties Officer → Officer corrects and resubmits the entry → All steps (original entry, request, Head's decision, and corrected entry) recorded permanently in Control Log.

4.6. Procurement

4.6.1. Current State ("As-Is") Analysis

Overview of Current Operations:

The Procurement department manages all purchasing activities for the corporation, ranging from small daily purchases to large national contracts. The current process is highly manual, paper-heavy, and relies heavily on physical committees for gathering market rates, inspecting goods, and securing approvals. Procurement rules are dictated by strict monetary thresholds that determine whether cash or bank transfers are used, and whether simple quotes, small contracts, or national tenders are required. Because there is no central database, tracking these different contract types and vendor deliveries is entirely manual.

Key Daily/Weekly/Monthly Activities:

- Receiving manual warehouse request forms (FS-9) and converting them into official procurement forms (M-3), which require multiple physical signatures from different department heads before any purchase can begin.
- Deploying physical Purchasing Committees to the local market to gather supplier quotations based on strict financial rules (e.g., three quotes required for purchases over 20,000 AFN).
- Coordinating with a three-person Inspection Committee to physically verify the quality and quantity of purchased goods before handing them over to the Warehouse.
- Routing physical purchasing documents through Internal Control and Finance (Treasury) to clear cash advances and process payments (cash for under 50,000 AFN, bank transfers for amounts above).
- Processing and escalating high-value contracts based on authority limits (up to 5 million AFN for the Executive Director, up to 35 million AFN for the General Directorate, and above 35 million AFN to the National Procurement Authority).
- Compiling the annual procurement plan by collecting physical requirement lists from all departments and Second Units.

Current Forms and Reports Used:

- Warehouse Purchase Request Forms (FS-9)
- Official Procurement Forms (M-3)
- Physical Supplier Quotations and Invoices
- Manual Small Contracts and Memorandums of Understanding (MOUs)
- Manual Archiving Ledgers for Contracts and Purchases

- Manual Annual Procurement Plan Spreadsheets

Identified Pain Points & Staff Suggestions:

- **Cumbersome Pre-Procurement Approvals:** The initial approval process for request forms (FS-9) is too slow and bureaucratic before it even reaches the Procurement department, causing unnecessary delays.
- **Lack of Digital Contract and Purchase Tracking:** The department desperately needs specific digital databases to track small retail purchases, large national contracts, and the ongoing delivery of goods by contractors.
- **No Vendor Management System:** There is currently no digital database to record and flag blacklisted or barred companies, which creates compliance risks during the bidding process.
- **Manual Tracking of Official Documents:** Managing incoming and outgoing letters, official government decrees, and cross-departmental inquiries (such as checking tenant balances with Finance) is entirely manual and prone to lost paperwork.

4.6.2. Future State ("To-Be") Digital Architecture

Process Optimization

The Procurement department receives formally approved purchase requests directly from the requesting departments and is responsible for executing the purchasing process in accordance with the corporation's procurement procedures. The actual procurement procedures; supplier selection, tendering, quotation collection, and negotiation are conducted outside the ERP system in hard copy. The ERP system's role is to receive approved requests, record the procurement method and classification, manage contract deadlines, track security deposits, and pass the completed procurement record to Inventory (for goods receipt) or the relevant department (for project handover).

The Internal Control and Audit department is embedded as a mandatory verification layer strictly at the Goods Receipt Note (GRN) or Project Handover stage before any procurement cycle is considered fully complete.

Purchase Request Receipt

When a requesting department transfers an approved purchase request to Procurement following the full approval chain described in Section 4.2, the Procurement Officer receives a formal notification within the ERP system. The purchase request appears in the Procurement module's **Pending Purchase Requests** queue showing:

- Purchase request reference number
- Requesting department
- Item description and quantity required

- Date of request
- Approval status: Department Head approved and Director physical approval recorded
- ED physical approval status: obtained outside the system and recorded in the ERP by the relevant department staff before the request reaches Procurement

No purchase request can appear in the Procurement module's active queue without the ED physical approval confirmation already being recorded in the system. The system enforces this gate automatically purchase requests without confirmed ED approval recording are held in a Pending ED Approval status and are not accessible to the Procurement Officer for processing.

Procurement Method Selection, Classification, and Bid Tracking

Once the purchase request is verified and accessible, the Procurement Officer selects and records the procurement method used for that purchase within the ERP system. The three procurement methods available are:

Procurement Method	Description
Tender	Open or restricted competitive tendering process used: for high-value purchases above the defined tender threshold
Quotation-Based	Minimum three quotations obtained from suppliers and compared: used for medium-value purchases
Direct Purchase	Single supplier purchase without competitive process: used for low-value purchases within the defined direct purchase threshold

In addition to the method, the Procurement Officer must select the **Procurement Classification**. The system includes default classifications such as Goods, Consultancy, Non-Consultancy, and Construction. The system allows the Procurement Department Head to add, edit, or disable additional classification categories in the system settings to accommodate future needs.

Bid Tracking and Supplier Selection Justification:

For procurement methods requiring competition (Tenders and Quotation-Based purchases), the system requires the Procurement Officer to record the competing bids before finalizing the purchase order. The Officer must input the names of the competing suppliers and their submitted bid prices (AFN). Furthermore, the Officer must provide a mandatory written justification in the system explaining why the winning supplier was selected (e.g., lowest evaluated price, technical compliance). The system stores and displays this bid comparison and justification data for transparency and audit visibility.

All procurement procedures supplier contact, tender issuance, quotation collection, evaluation, and supplier selection are conducted by the Procurement department in hard

copy outside the ERP system. These hard copy documents are filed physically by the Procurement department and referenced in the system by document reference number.

Purchase Order & Contract Recording

Once the procurement process is complete and a supplier or contractor is selected, the Procurement Officer records the Purchase Order or Contract in the ERP system. The record captures:

Field	Notes
Purchase Order / Contract Ref Number	Auto-generated — sequential
Date of Purchase Order	System date
Purchase Request Reference	Linked from approved purchase request
Procurement Method	Tender / Quotation-Based / Direct Purchase
Supplier Name	Entered by Procurement Officer
Item / Project Description	From approved purchase request
Quantity Ordered	From approved request — any variance must be noted
Unit Price (AFN)	Entered by Procurement Officer
Total Value (AFN)	Auto-calculated — Quantity × Unit Price
Contract Deadline / Completion Date	Entered by Procurement Officer — exact date goods must be delivered or project completed
Custom Notification Threshold	Entered by Procurement Officer — defines when the system sends approaching deadline alerts (e.g., 1 week, 2 months)
Security Deposit Required	Checkbox (Yes/No) and required deposit amount (AFN)
Delivery / Project Location	Entered by Procurement Officer
Hard Copy Document Reference	Reference to physical documentation filed by Procurement
ED Approval Reference	Auto-linked from purchase request

Field	Notes
Status	Raised / Sent to Supplier / Goods Received / Completed

The purchase order is reviewed and approved by the Procurement Department Head before it is finalized in the system. Once approved, the purchase order is sent to the supplier in hard copy outside the ERP system. The ERP retains the digital record of the purchase order and its status.

Supplier Security Deposit Management

For contracts requiring a security deposit, the system enforces automated financial tracking:

1. **Receipt of Deposit:** The supplier deposits the required amount into the corporation's bank account. The Finance Officer confirms this receipt in the bank reconciliation module, linking it to the specific Procurement Contract. The system automatically posts a background double-entry: *Debit Bank Account / Credit Supplier Security Deposit (Current Liability)*.
2. **Refund of Deposit:** Upon successful completion of the contract, the Procurement Department Head authorizes the release of the deposit. Finance processes the payment, and the system posts: *Debit Supplier Security Deposit / Credit Bank Account*.
3. **Forfeiture of Deposit:** If the supplier breaches the contract, the Procurement Department Head can authorize the forfeiture of the security deposit. Upon approval, the system automatically posts the following background double-entry to recognize the penalty revenue: *Debit Supplier Security Deposit (Current Liability) / Credit Other Income Account*.

Project Implementation Tracking and Contract Extensions

To ensure large contracts (like Construction or Consultancy) are monitored effectively, the ERP includes a Project Implementation Tracker:

- **Implementation Percentage (%):** The Procurement Officer periodically updates the implementation status (e.g., 25%, 50%, 75%) based on physical site reports or milestone certificates.
- **Automated Deadline Alerts:** Based on the customized alert threshold set during contract creation, the system automatically sends a notification to the Procurement Officer and Procurement Head if the contract is approaching its deadline and the implementation % is not at 100%. If the deadline passes without completion, the system flags the contract as "Overdue/Breached."

- **Contract Extension Workflow:** If a supplier requires more time, the system allows for formal extensions. The Procurement Officer must initiate an "Extension Request" in the system, entering the proposed new deadline and a mandatory written justification. This request requires explicit system approval from the Procurement Department Head. Once approved, the system updates the Contract Deadline and resets the automated alerts.

Goods Delivery, Project Handover, and IC Verification Gate

When the supplier delivers the goods, the physical delivery is received by the Inventory and Warehouse Management department not the Procurement department. The Inventory Officer records the Goods Receipt Note in the system as described in Section 4.2. The Procurement module automatically updates the purchase order status to **Goods Received** upon the Inventory department's confirmation of the GRN.

For non-goods contracts (such as Consultancy or Construction), a "Project Completion or Handover" is recorded in the system by the relevant requesting department.

Before any accounting entries are posted or payments are cleared, the GRN or Handover document is routed to the Internal Control and Audit department. The Internal Control team conducts a mandatory verification to ensure that the delivered goods or completed project milestones exactly match the physical reality, the supplier invoice, and the approved contract terms.

Rejection and Correction Loop:

If the Internal Control team identifies any discrepancy, they reject the GRN or Handover. The system automatically sends a notification containing specific comments from Internal Control to the relevant Department Head (the Inventory Head for goods, or the Requesting Department Head for services/projects). The Head unlocks the transaction, triggering a notification to the respective Officer. The Officer corrects the entries based on the comments and resubmits it to the Head, who re-approves and routes it back to Internal Control for final re-verification.

Upon successful Internal Control verification, the system automatically posts the relevant accounting entries. The Procurement module automatically updates the purchase order or contract status to "Completed". At this point, the Procurement department's role in the purchase cycle is complete. The Finance department then receives the Awaiting Payment notification to process the final supplier payment as described in Section 4.1.

Procurement Activity Tracking and Reporting

The system provides the Procurement department with comprehensive tracking and reporting of all procurement activity across any defined time period. The Procurement Department Head can generate reports showing:

- **Bid Comparison and Selection:** Details of competing bids, prices, and the mandatory justifications for winning supplier selections.
- **Procurement by Classification:** Volume and value broken down by Goods, Consultancy, Non-Consultancy, Construction, or any custom classes.
- **Project Implementation and Deadlines:** Live tracking of implementation percentages, overdue contracts, and formally extended contracts.
- **Security Deposit Ledger:** Active deposits held, refunded deposits, and forfeited deposits transferred to Other Income.
- Total number of purchases by procurement method; Tender, Quotation-Based, Direct Purchase in any selected time period
- Total value of purchases by procurement method in any selected time period
- Status of all active purchase orders and contracts (Raised, Sent to Supplier, Goods Received, Completed).
- Purchases by requesting department
- Purchases by supplier
- Purchases by Second Unit or HQ delivery location
- Comparison of procurement method usage across different periods

This reporting capability gives management full visibility of how the corporation's procurement activity is distributed across methods and categories, supporting compliance with procurement thresholds and enabling identification of any bottlenecks or lapsed deadlines.

Officer Correction Request Workflow

If a Procurement Officer identifies an error in a submitted record (such as an incorrect contract deadline, notification threshold, or item description) prior to finalization, the Officer cannot modify or delete the entry independently. To ensure data integrity, the system enforces the following correction loop:

- The Procurement Officer raises a formal correction request within the system, providing a written justification detailing the error.
- The system sends a notification to the Procurement Department Head.

- If approved, the Department Head unlocks the specific entry in the system.
- This action triggers an automatic notification to the Procurement Officer, who then corrects and resubmits the entry.
- All steps (including the original entry, the correction request, the Department Head's approval, and the corrected entry) are permanently recorded in the Control Log for audit purposes.

Compliance and Standards

ISO 9001 Quality Management

Every procurement action in the system follows a mandatory documented sequence: purchase request receipt with confirmed approval chain, procurement method and classification selection, recording of competing bids and selection justifications, purchase order or contract generation with Department Head approval, and Internal Control verification strictly at the Goods Receipt or Project Handover stage. The system enforces quality control through automated deadline alerts and mandatory justifications for contract extensions. No purchase order can be finalized without the complete chain being recorded. The Control Log provides an immutable audit trail of every procurement action.

Afghan Procurement Law

The procurement method thresholds and procedures applied by the Procurement department comply with applicable Afghan public procurement regulations for state-owned corporations. The ERP system records the method used for every purchase, the competing bids, and the justification for supplier selection, providing the documentary evidence required for compliance audits by the Supreme Audit Office of Afghanistan and other oversight bodies. All physical procurement documentation (including tender documents, quotations, evaluation reports, and supplier contracts) are retained in hard copy files by the Procurement department and referenced in the system by document reference number.

Financial Accounting Standards

The system ensures strict compliance with financial reporting standards regarding Supplier Security Deposits. Deposits received are automatically classified and held as Current Liabilities. Any authorized forfeiture of a deposit is automatically accrued to Other Income, ensuring that penalty revenues are captured accurately without requiring manual journal entries by the Finance department.

4.6.3. System Roles and Access Permissions

Role	User	Permissions
Data Entry / Initiator	Procurement Officers	Can view approved purchase requests. Records procurement method, classification, competing bids, and selection justifications. Drafts purchase orders/contracts including deadlines, custom notification thresholds, and security deposit requirements. Updates Project Implementation %. Initiates Contract Extension requests with written justifications. Must raise formal correction requests to Procurement Head for errors. Cannot approve own entries.
Department Head	Procurement Department Head	Approves purchase orders and contracts. Approves Contract Extension requests. Can add, edit, or disable custom Procurement Classifications in system settings. Authorizes Supplier Security Deposit refunds and forfeitures. Receives automated deadline alerts. Approves correction requests, unlocking entries for Officers. Generates all procurement and bid tracking reports.
Receiving / Handover	Inventory Officers / Requesting Department Staff	Inventory Officers record Goods Receipt Notes (GRN). Requesting Department staff record Project Completion/Handover for services. Corrects GRN/Handover entries if rejected by Internal Control.
Receiving Department Head	Inventory Head / Requesting Department Head	Receives Internal Control rejection notifications with comments regarding GRNs or Project Handovers. Unlocks the transaction in the system for the receiving officer to correct.
Finance Processing	Finance Officers / Head	Confirms receipt of Supplier Security Deposits in the bank reconciliation module. Processes authorized deposit refunds. Processes final supplier payments upon contract completion.
Targeted Verification	Internal Control and Audit Department	Mandatory verification is strictly positioned at the GRN or Project Handover stage. Verifies physical delivery/project completion against the supplier invoice and contract terms. Can Verify or Reject (with mandatory comments). Read-only

Role	User	Permissions
		access to all procurement tracking, bids, classifications, implementation %, and security deposit ledgers.
ED and Directors	Executive Director and Department Directors	No system access. ED physical approval for purchases is obtained outside the system in hard copy and recorded in the ERP by the relevant department staff before reaching Procurement.
System Administrator	IT Admin	Manages user accounts and settings. Zero access to procurement data, bids, or financial records.

4.6.4. Developer Specifications — Procurement

Accounting Entries — Developer Reference

While the Procurement department manages the contract logic, the system automatically posts double entries in the background upon confirmed actions (either by Finance for deposits or through IC-verified Goods Receipts/Project Handovers). The relevant accounting entries are:

Trigger Event	Debit	Credit
Supplier Security Deposit Received (Confirmed by Finance)	Bank Account	Supplier Security Deposit Account (Current Liability)
Supplier Security Deposit Refunded (Processed by Finance)	Supplier Security Deposit Account (Current Liability)	Bank Account
Supplier Security Deposit Forfeited (Approved by Proc Head)	Supplier Security Deposit Account (Current Liability)	Other Income Account
Goods received / Project completed (IC-Verified GRN or Handover)	Relevant Expense or Fixed Asset Account	Accounts Payable
Final supplier payment processed by Finance	Accounts Payable	Bank Account

Required Reports — Procurement

Note: Every report listed below must be generated dynamically in real-time. The system must provide a flexible "Custom Date Range" filter.

Report	Frequency	Filter Options
Bid Comparison and Selection Report	On demand	By contract, showing competing bids, prices, and winning justifications
Procurement Classification Report	On demand	By Classification (Goods, Consultancy, Non-consultancy, Construction, etc.)
Project Implementation & Deadline Tracker	Real-time	By implementation %, Overdue contracts, Extended contracts
Security Deposit Ledger	On demand	Active deposits, Refunded, Forfeited (transferred to Other Income)
Procurement Activity Report by Method	On demand	Tender / Quotation-Based / Direct Purchase, per period, per department, per Second Unit
Purchase Order / Contract Register	On demand	By status, by supplier, by Second Unit
Pending Purchase Requests Report	Real-time	All requests awaiting Procurement action
Purchase Orders Awaiting Goods Receipt	Real-time	All finalized purchase orders pending IC-verified GRN/Handover
Completed Procurements Report	On demand	All fully completed purchase cycles (per period, per supplier)
Procurement Value by Method Report	Monthly / Annual	Total AFN value per procurement method in defined period
Supplier Purchase History Report	On demand	Per supplier (all purchase orders, values, and delivery records)
Internal Control Verification Report	Monthly	By verification outcome (Verified/Rejected) at GRN/Handover stage
Control Log / Audit Trail Report	On demand	By user, date, and action type (including extension approvals and unlocks)

All reports must display: Corporation name and logo, report title, date range, generation timestamp, and generating username. All reports must be exportable to PDF and Excel formats.

4.6.5. BPMN Workflow Narratives — Procurement

Workflow 1: Purchase Request Receipt and Contract Setup

Approved purchase request transferred from the Requesting Department to Procurement → System checks ED approval confirmation is recorded → Request moves to Procurement active queue → Procurement Officer selects method (Tender / Quotation-Based / Direct Purchase) and Classification (Goods / Consultancy / Construction / etc.) → Procurement procedures conducted in hard copy outside ERP → Procurement Officer inputs competing supplier names, bid prices, and mandatory written justification for the winning selection into system → Procurement Officer records contract details in system → Officer inputs Contract Deadline, sets Custom Notification Threshold (e.g., alert 1 month before), and flags if Security Deposit is required → Procurement Department Head reviews and approves → Contract finalized and sent to supplier outside ERP.

Workflow 2: Security Deposit Management

Path A (Receipt): Supplier pays security deposit into corporate bank → Finance Officer reconciles deposit in system → System auto-posts background double entry: Debit Bank Account / Credit Supplier Security Deposit (Current Liability).

Path B (Refund): Contract successfully completed → Procurement Department Head authorizes refund in system → Finance processes payment → System auto-posts background double entry: Debit Supplier Security Deposit / Credit Bank Account.

Path C (Forfeiture): Supplier breaches contract → Procurement Department Head authorizes forfeiture in system → System auto-posts background double entry: Debit Supplier Security Deposit / Credit Other Income Account → Deposit workflow complete.

Workflow 3: Project Implementation Tracking and Contract Extension

Contract active → Procurement Officer periodically logs Implementation % into system based on field reports → If deadline approaches: System triggers automated alert to Officer and Head based on custom notification threshold.

If extension required: Procurement Officer initiates Extension Request → Officer inputs new deadline and mandatory written justification → Request routed to Procurement Department Head → Procurement Head approves in system → System updates Contract Deadline and resets automated alerts → Tracking workflow complete.

Workflow 4: Goods Receipt, Project Handover, and IC Verification Gate

Supplier delivers goods or completes project → Inventory Officer (for goods) or Requesting Department Staff (for services) records GRN or Handover in system → System routes document to Internal Control for mandatory verification → Internal Control verifies physical delivery and project completion against contract terms.

If Rejected: System sends notification with IC comments to Receiving Department Head → Head unlocks transaction in system → Receiving Officer corrects entries based on comments and resubmits → Head re-approves and routes back to Internal Control.

If Verified: System updates Purchase Order status to Completed → System auto-posts GRN accounting entries → Finance receives notification to process final supplier payment → Procurement cycle fully complete.

Workflow 5: Officer Correction Request Workflow

Procurement Officer identifies error in a submitted record prior to finalization → Officer raises formal correction request detailing error and justification → System notifies Procurement Department Head → Head reviews request → *If Approved:* Head unlocks entry → Notification sent to Officer → Officer corrects and resubmits entry → All steps (original entry, request, approval, corrected entry) recorded permanently in Control Log → Correction workflow fully complete.

4.7. Planning and Policy

4.7.1. Current State ("As-Is") Analysis

Overview of Current Operations:

The Planning and Policy department is the strategic core responsible for setting operational targets, monitoring mine production, and generating mandatory performance reports for external government ministries. Currently, the department relies entirely on manual data gathering from the various mining units. Staff must manually process complex daily production, excavation, and repair data to verify contractor payrolls and calculate government royalties. The absence of a centralized digital database causes severe delays in management reporting, forces redundant data entry, and makes real-time monitoring of corporate goals impossible.

Key Daily/Weekly/Monthly Activities:

- Receiving and processing manual daily reports on coal production, structural repairs, and excavations from the remote mining sites.
- Verifying monthly production and excavation data to authorize piece-rate payroll entitlements for contract mine workers before forwarding the data to the Finance department.
- Calculating and submitting monthly coal royalty and payment reports to the Finance department, the Ministry of Mines and Petroleum, and the General Directorate of State-Owned Corporations.
- Preparing and submitting mandatory quarterly operational, evaluation, and financial realization reports to various external government bodies (e.g., Baghlan Statistics Directorate, Ministry of Mines, and the Directorate of Administrative Affairs).
- Evaluating and analyzing the monthly planned targets of the mining departments against actual manual results.
- Drafting the annual production, financial, and commercial plans based on physical topographical (survey) maps during the third quarter of each year.
- Managing the contracting process to hand over informally mined local areas to private companies for standardized mining operations.
- Manually logging all incoming and outgoing official correspondence into physical ledgers and archiving them in physical cartons.

Current Forms and Reports Used:

- Daily and Monthly Production, Repair, and Excavation Reports
- Quarterly Profit & Loss Realization Forms

- Physical Topographical/Survey Maps
- Manual Incoming/Outgoing Correspondence Ledgers (Carton Archives)
- Standardized Government Reporting Forms
- Annual Financial, Commercial, and Operational Plan Spreadsheets

Identified Pain Points & Staff Suggestions:

- **Heavy Reliance on Paper & Redundant Data Entry:** Critical operational information is manually written multiple times across different ledgers, forms, and physical archives. This massive volume of paperwork heavily increases the risk of data loss and human calculation errors.
- **Severe Reporting Delays:** Manually gathering data from multiple dispersed mining units and hand-calculating the results takes an excessive amount of time, delaying critical management decisions and government reporting.
- **No Real-Time Plan Monitoring:** Because operational data is scattered across physical documents, it is impossible for management to track plan realization or mine progress accurately in real time.
- **Lost Documents & Slow Retrieval:** The physical filing system makes tracking active official correspondence highly difficult and makes retrieving historical plans or reports incredibly time-consuming.
- **Urgent Need for Centralization:** The department explicitly highlighted that the manual reporting, redundant data entry, and physical archiving workflows urgently need to be digitized and centralized into a single integrated database to reduce administrative costs and improve accuracy.

4.7.2. Future State ("To-Be") Digital Architecture

Process Optimization

The Planning and Policy module serves as the strategic planning and performance monitoring hub of the ERP system. It enables the corporation to set comprehensive annual operational, financial, and capital expenditure plans for each Second Unit and HQ, and then track actual performance against those plans in real time using live data flowing from all other modules. The module provides a complete top-down view of the corporation's planned vs. actual profitability, operational efficiency, and asset acquisition.

Dynamic Plan Item Creation, Class Assignment, and CoA Linking

To ensure the Planning module is highly flexible and perfectly synchronized with actual financial performance, the system allows the Planning and Policy Department to dynamically create new plan items as business needs evolve.

When creating a new item for the annual plan (for example, adding a new requirement like "wood for excavation holes"), the Planning Officer must define its parameters to ensure accurate, automated variance tracking:

1. **Item Name / Description:** The specific name of the planned item or activity.
2. **Class Assignment (Location):** The Officer must specify which entity this plan item applies to by selecting the relevant Class (e.g., HQ, Second Unit A, Second Unit B). This ensures that planned budgets and operational targets are strictly isolated and tracked per location.
3. **Financial Classification:** The Officer selects the core category for the item:
    - Main Core Revenue (e.g., Coal Sales)
    - Non-Core Revenue (e.g., Rental Income, Scrap Sales)
    - Direct Costs (e.g., Excavation materials, direct mining labor)
    - Indirect Costs / General Operational Costs (e.g., HQ admin expenses, office supplies)
    - Fixed Assets / CAPEX (e.g., Purchasing new mining trucks or IT servers)
4. **Chart of Accounts (CoA) Linkage:** To enable automated variance analysis, the Officer must link the new plan item directly to the relevant account in the Finance module's Chart of Accounts. **The Chart of Accounts is structurally designed to include specific sub-accounts dedicated to each Second Unit and HQ.** For example, if "wood for excavation holes" is classed as a Direct Cost for Second Unit A, it is linked to Second Unit A's specific "Mining Consumables Expense" sub-account. The system will then automatically pull any actual money spent in that specific sub-account to compare against the plan.
5. **Non-Financial Metric Exemption (N/A):** If the planned item is strictly operational and does not follow the money measurement principle (such as tonnes of coal extracted or cubic meters of earth excavated), the Officer selects **"N/A"** for the CoA linkage. Because these are non-financial metrics, the system does not automatically pull this data. Instead, the Planning Officer must manually enter the actual achieved volumes into the system on a periodic basis. Every manual entry of actual operational data must be supported by verified physical documents (e.g., signed site extraction reports), and the specific document reference number must be recorded in the system alongside the entry.

Operational Plan (Excavation and Extraction) — Second Units

At the beginning of each fiscal year, the Planning Officer enters the operational physical targets for each mining site. Because these are physical volumes, the CoA linkage is marked as "N/A". The system captures:

- **Excavation Plan:** The planned volume of overburden/soil removal (in cubic meters) required to expose the coal seams, broken down by month and Second Unit.
- **Extraction Plan:** The planned volume of raw coal to be extracted (in tonnes), broken down by month and Second Unit.
- *Actuals Tracking:* As the year progresses, actual achievements against these operational targets are entered manually by the Planning Officer based strictly on verified field documents.

Comprehensive Financial, P&L, and CAPEX Plan

Alongside the operational volumes, the Planning Officer enters the financial targets for the year across all Classes (Second Units and HQ). By utilizing the dynamic item creation and CoA linkage described above, the system captures:

- **Planned Revenue:** Month-by-month targets for Main Core Revenue (coal sales) and Non-Core Revenue (rental income, scrap sales, etc.).
- **Planned Expenditure:** Month-by-month budgets for Direct Costs (directly tied to excavation/extraction) and Indirect Costs (HQ administration, general operations).
- **Planned Capital Expenditure (CAPEX):** Approved budgets for purchasing Fixed Assets during the year.

Automated Planned vs. Actual Profit or Loss (P&L)

Because the Planning module captures all revenues, direct costs, and indirect costs categorized by Class and linked to the Chart of Accounts, the ERP system automatically generates a **Planned Profit or Loss Statement**.

As the fiscal year progresses, the system continuously pulls live financial data from the Finance, Sales, and Inventory modules. This allows the Planning Department Head to instantly view a real-time **Actual vs. Planned P&L**, showing exactly where the corporation or a specific Second Unit is generating unexpected profits, overspending on direct costs, or falling behind on core revenues.

Variance Analysis — Plan vs Actual

The Planning and Policy module's most critical function is the real-time Plan vs Actual Comparative Variance Analysis. Once plans are approved and active, the system handles the comparison automatically.

- **For Financial Metrics (Revenues, Costs, CAPEX):** The system automatically pulls live actual data from the Finance, Inventory, Sales, and Procurement modules based on the mapped Chart of Accounts sub-accounts, comparing actual monetary values against planned monetary values.
- **For Non-Financial Metrics (Excavation/Extraction Volumes):** The system compares the planned physical targets against the actual physical volumes manually entered by the Planning Officer (which are backed by verified site reports).

The system auto-calculates the variance in absolute terms (AFN, tonnes, cubic meters) and as a percentage. The variance analysis is available in real time at any point during the year. The Planning Department Head can view the comparative position and the Actual vs. Planned P&L for any Second Unit, any department, any time period, and any plan category without waiting for manual reports to be compiled.

Automated Variance Alerts:

The system generates automatic alerts to the Planning Department Head when actual figures deviate from the plan by more than a defined threshold percentage. For example, if actual coal extraction falls below the planned volume by more than 10%, or if actual direct costs exceed the planned budget by more than a defined limit, the system triggers a notification. These alerts enable proactive management intervention before variances severely impact profitability.

Plan Amendment Protocol

Where circumstances require a change to an approved plan during the year for example, a significant change in coal market conditions, a natural event affecting a Second Unit, or a corporate restructuring, formal approval for the plan amendment must first be obtained from the Executive Director (ED) outside the system in hard copy.

Once physical ED approval is secured, the Planning Officer can submit a formal Plan Amendment Request within the ERP system. The amendment request records:

- Original plan reference
- Nature and reason for amendment
- Revised figures (Financial or Physical)
- ED Physical Approval Reference / Document Number (Mandatory)

- Supporting justification

The system amendment request is then reviewed and approved by the Planning Department Head to verify it accurately reflects the physical ED approval. Once approved in the system by the Head, the revised plan figures replace the original in the comparative analysis, while the original figures are retained in the system for historical reference and audit purposes. All plan amendments are permanently recorded in the Control Log.

Officer Correction Request Workflow

If a Planning Officer identifies an error in a submitted plan entry (prior to finalization) or an error in a manually entered actual volume (e.g., mistyping 5,000 tonnes as 50,000 tonnes), they cannot modify or delete the entry independently. To ensure data integrity, the system enforces the following correction loop:

- The Planning Officer raises a formal correction request within the system, providing a written justification detailing the error.
- The system sends a notification to the Planning Department Head.
- If approved, the Department Head unlocks the specific entry in the system.
- This action triggers an automatic notification to the Planning Officer, who then corrects and resubmits the entry.
- All steps (original entry, request, approval, corrected entry) are permanently recorded in the Control Log.

Compliance and Standards

ISO 9001 Quality Management

The Planning and Policy module enforces ISO 9001's requirements for planning and performance monitoring. All plans are documented, approved, and systematically compared against actual performance providing the evidence base required for management review and continual improvement processes. Plan amendments are formally controlled and permanently recorded satisfying ISO 9001's requirements for document control and change management. The Control Log provides an immutable audit trail of all planning actions.

IAS 1 — Presentation of Financial Statements

The planning data maintained in this module supports the preparation of comparative financial statements in accordance with IAS 1. The automated Plan vs Actual P&L enables management to present current period financial performance alongside planned budgets and prior period actuals in a structured, transparent, and highly accurate format.

4.7.3. System Roles and Access Permissions

Role	User	Permissions
Data Entry / Initiator	Planning and Policy Officers	Creates dynamic plan items (assigns Class, financial category, and CoA linkage or "N/A"). Enters annual operational (physical) and financial (P&L/CAPEX) plans for all Second Units and HQ. Manually enters actual non-financial operational data (e.g., extracted volumes) based on verified hard-copy documents. Initiates plan amendments recording the physical ED approval reference. Raises formal correction requests for errors. Cannot approve their own plans or amendments.
Department Head	Planning and Policy Department Head	Approves all plans and plan amendment system entries (verifying they match physical ED approvals). Approves correction requests and unlocks entries for Officers to modify. Has full access to the real-time Actual vs. Planned P&L and variance analysis. Receives automatic variance threshold alerts. Generates all planning and variance reports, and prints them for physical distribution to Directors and ED.
ED and Directors	Executive Director and Department Directors	No system access. Review and provide physical approval for all annual plans and mid-year plan amendments outside the system. Receive physical variance and P&L reports generated by the Planning Department Head.
System Administrator	IT Admin	Manages user accounts and system settings. Zero access to planning data, plan structures, or financial records.

4.7.4. Developer Specifications — Planning and Policy

Accounting Entries & System Logic — Developer Reference

The Planning and Policy module does not generate accounting entries. It is a strategic planning and comparative analysis module.

- **Financial Data:** The module reads live actual data from the Finance, Inventory, Sales, and Procurement modules based on the Chart of Accounts (CoA) linkages defined during dynamic plan item creation. No financial general ledger postings originate from this module.

- **Operational Data:** For non-financial plan items (where CoA linkage is marked "N/A"), the module accepts manual data entry of actual volumes (e.g., tonnes, cubic meters) by the Planning Officer, which must require a mandatory "Reference Document" field, but these entries do not impact the financial general ledger.

Required Reports — Planning and Policy

Note: Every report listed below must be generated dynamically in real-time. The system must provide a flexible "Custom Date Range" filter.

Report	Frequency	Filter Options
Annual Operational Plan Report	On demand	By Class (Second Unit), per physical metric (Excavation m^3 / Extraction tonnes), per month/year
Annual Financial Plan Report	On demand	By Class (Second Unit/HQ), by Category (Direct Cost, Indirect Cost, Revenue, CAPEX), per month/year
Actual vs Planned Profit & Loss (P&L)	Real-time	By Class (Specific Second Unit or Consolidated), showing all Revenues minus Costs with variance %
Operational Variance Report (Volumes)	Monthly / On demand	Per Second Unit, per period — physical variance (tonnes/m^3) and percentage
Financial Variance Report (AFN)	Monthly / On demand	Per Class, per CoA mapping, per period — AFN variance and percentage
Consolidated Variance Summary	Quarterly / Annual	All Second Units and HQ — consolidated operational and financial variance across all categories
Variance Alert Log	Real-time	All active variance alerts exceeding defined threshold parameters
Plan Amendment Log	On demand	All amendments — original figures, revised figures, reason, ED Approval Reference, and Head approval date
Control Log / Audit Trail Report	On demand	By user, date, and action type (including all officer correction request loops and unlocks)

All reports must display: Corporation name and logo, report title, fiscal year, date range, generation timestamp, and generating username. All reports must be exportable to PDF and Excel formats.

4.7.5. BPMN Workflow Narratives — Planning and Policy

Workflow 1: Dynamic Plan Item Setup and Annual Plan Entry

Planning Officer creates a dynamic plan item in the system -> Officer assigns the item to a specific Class (HQ or a specific Second Unit) -> Officer selects the Financial Category (Core Revenue, Non-Core Revenue, Direct Cost, Indirect Cost, or CAPEX) -> Officer links the item to a specific Chart of Accounts sub-account, OR selects "N/A" for non-financial physical metrics (e.g., extraction tonnes) -> Officer enters the monthly and annual target figures -> Plan is submitted to the Planning Department Head -> Department Head reviews the plan for completeness -> Department Head approves the plan in the system -> Plan is marked Active and tracking begins.

Workflow 2: Actuals Tracking and Real-Time Variance Monitoring

Path A (Financial Metrics): System continuously and automatically pulls live actual financial data from the Finance, Inventory, Sales, and Procurement modules based on the established CoA linkages.

Path B (Non-Financial Metrics): Planning Officer receives verified physical site reports -> Officer manually enters actual achieved volumes into the system -> Officer inputs the mandatory physical document reference number alongside the entry.

Combined Variance Engine: System auto-calculates variance for each plan line (actual vs. planned in absolute values and percentage) -> System generates the live Actual vs. Planned Profit & Loss (P&L) statement -> If a variance exceeds the defined threshold: System generates an automatic alert to the Planning Department Head -> Department Head reviews variance and takes management action.

Workflow 3: Plan Amendment Protocol

Circumstances require a revision of an active approved plan -> Formal approval for the amendment is obtained from the Executive Director (ED) in hard copy outside the system -> Planning Officer submits a formal Plan Amendment Request in the system -> Officer inputs original figures, revised figures, reason, and the mandatory ED Physical Approval Reference number -> Planning Department Head reviews the system request against the physical ED approval -> Department Head approves the amendment in the system -> Revised plan figures are activated -> Original figures are retained for historical reference and audit purposes -> Control Log is permanently updated.

Workflow 4: Officer Correction Request

Planning Officer identifies an error in a submitted plan entry or a manually entered actual volume -> Officer raises a formal correction request in the system -> Officer provides a written justification detailing the error -> System notifies the Planning Department Head -> Head reviews the request -> If Approved: Head unlocks the entry -> System notifies the Officer -> Officer corrects and resubmits the entry -> All steps (original entry, request, Head approval, and corrected entry) are permanently recorded in the Control Log.

5. FORMAL ACCEPTANCE AND SIGN-OFF

By signing below, the authorized representatives of the North Coal State Owned Corporation acknowledge formal receipt of this System Architecture & Business Process Re-engineering Report. The client confirms that the operational workflows, internal control mechanisms, and system logic detailed within this document have been reviewed and accurately reflect the accepted "To-Be" future state of the corporation. This document is hereby approved to serve as the foundational blueprint for the software development phase.

The formal acceptance and sign-off of this document marks the successful conclusion of Crest Point Consulting Group's business process re-engineering and architectural design mandate for this project phase. Upon execution of this sign-off, Crest Point Consulting Group's deliverables are considered fully rendered and accepted. Crest Point Consulting Group assumes no liability or responsibility for the third-party software development company's coding execution, technical deployment, timeline management, or subsequent software performance.

Prepared and Submitted By:

Fayaz Yasini

Engagement Lead Crest Point Consulting Group

Date: _____

Received and Accepted By:

Approved By:

APPENDIX A: BUSINESS PROCESS MODEL AND NOTATION (BPMN) FLOWCHARTS

A.1. Finance and Accounting Workflows

Figure A.1.1: Bank Reconciliation and Revenue Receipt Workflow

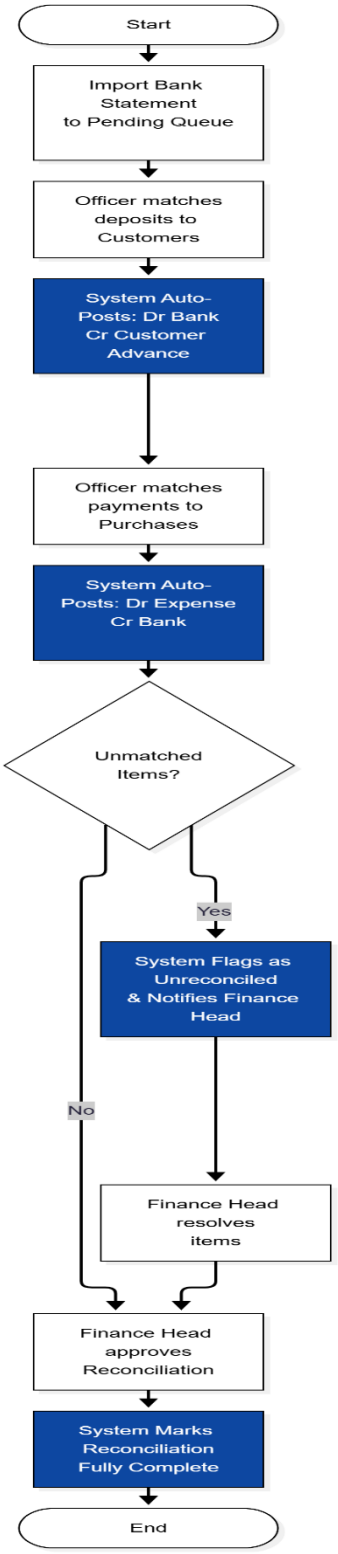


Figure A.1.2: Revenue Recognition Workflow

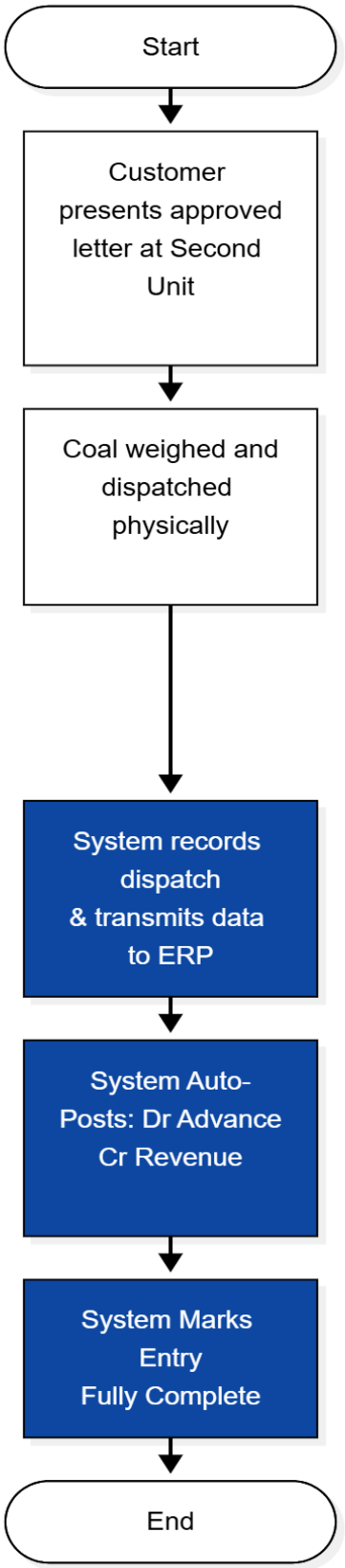


Figure A.1.3: Purchase Cycle Budget Control & Payment Workflow

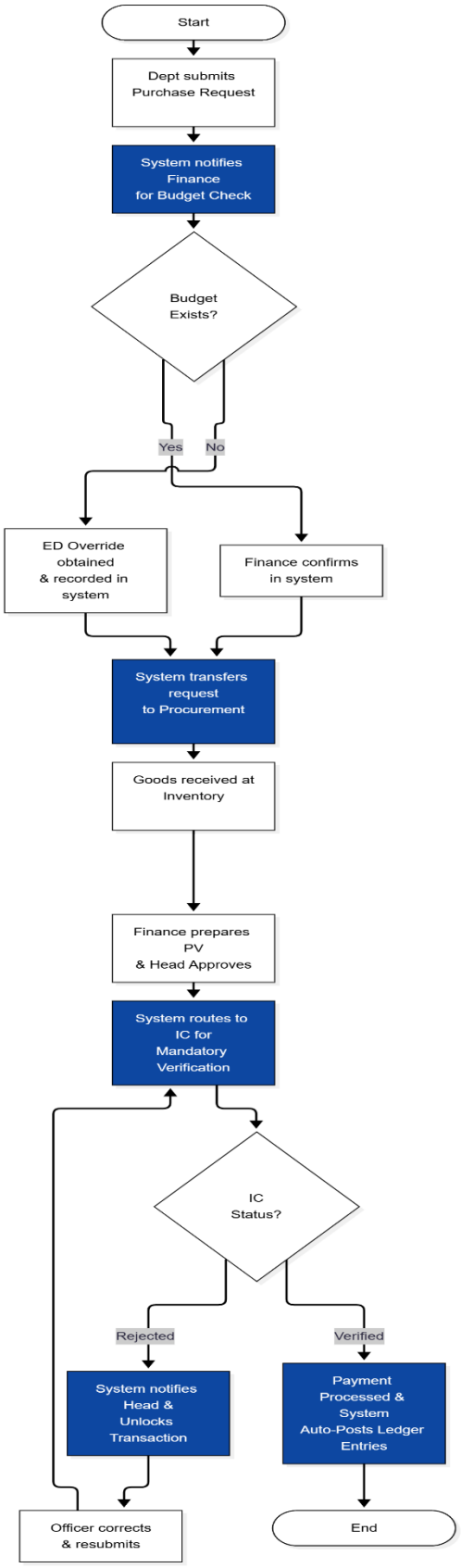


Figure A.1.4: Officer Correction Request Workflow

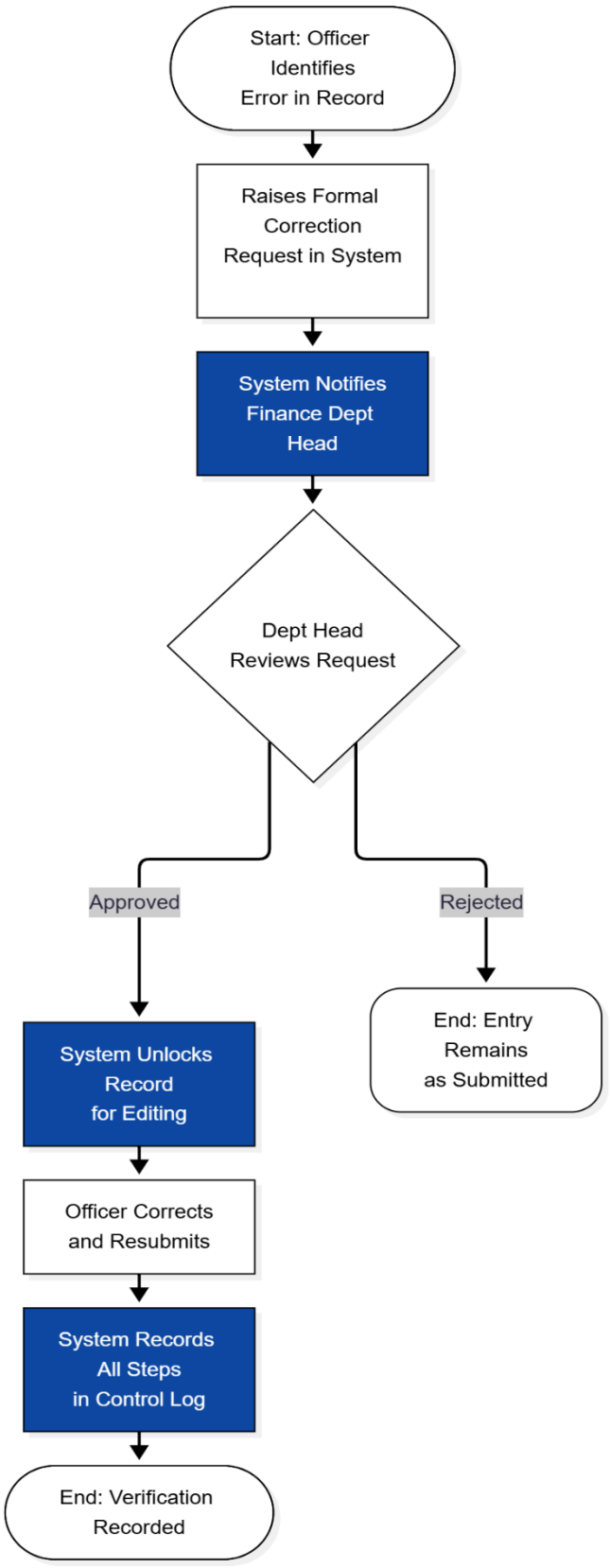
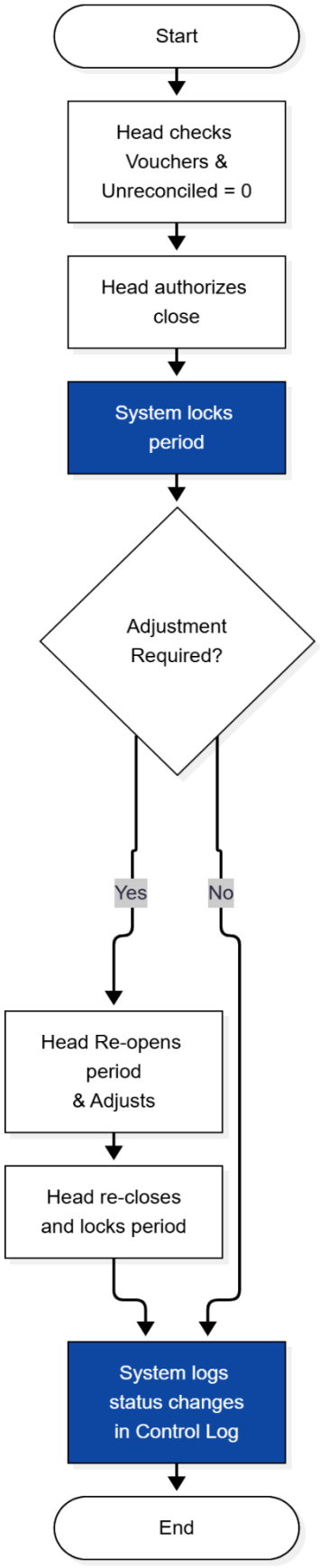


Figure A.1.5: Period-End Closing Workflow



A.2. Inventory and Warehouse Management Workflows

Figure A.2.1: Item Request and Issuance (Item Available in Stock)

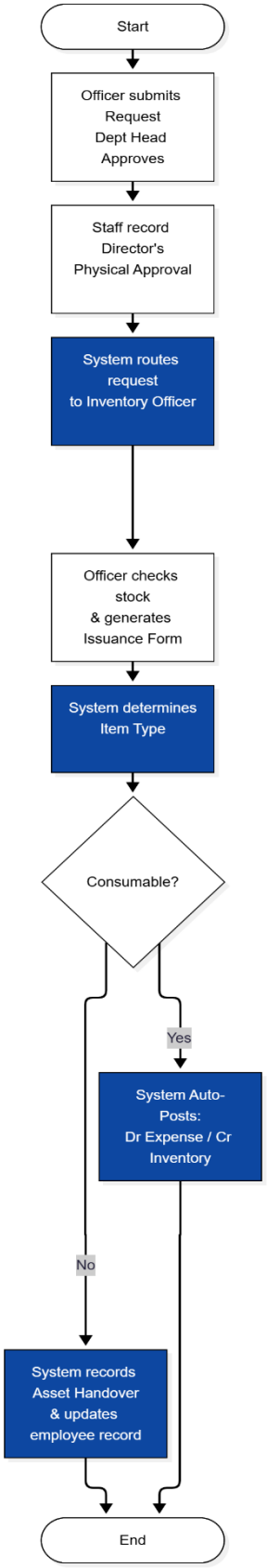


Figure A.2.2: Item Request (Item Not Available — Purchase Request)

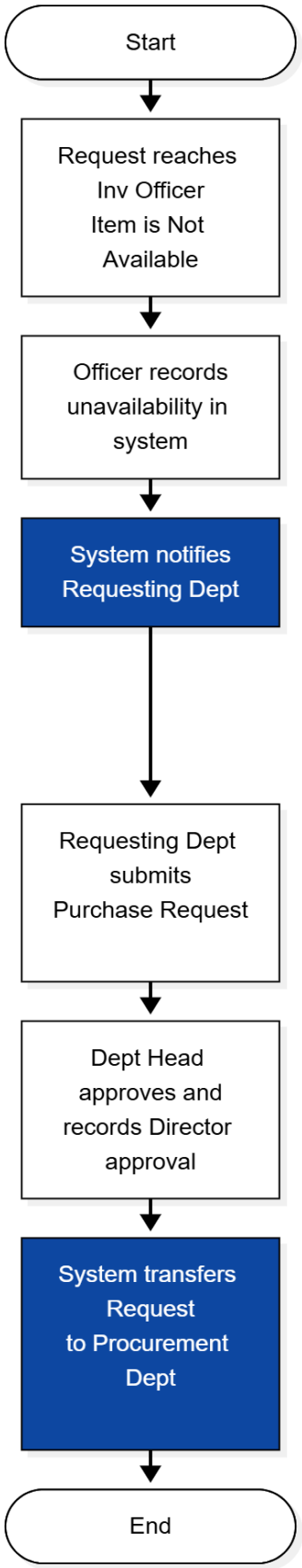


Figure A.2.3: Goods Receipt from Procurement

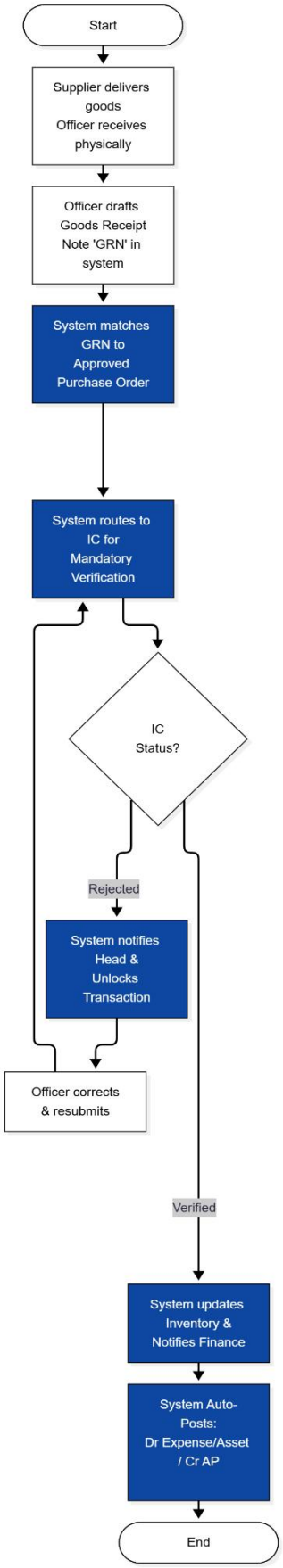


Figure A.2.4: Inventory Loss and Wastage

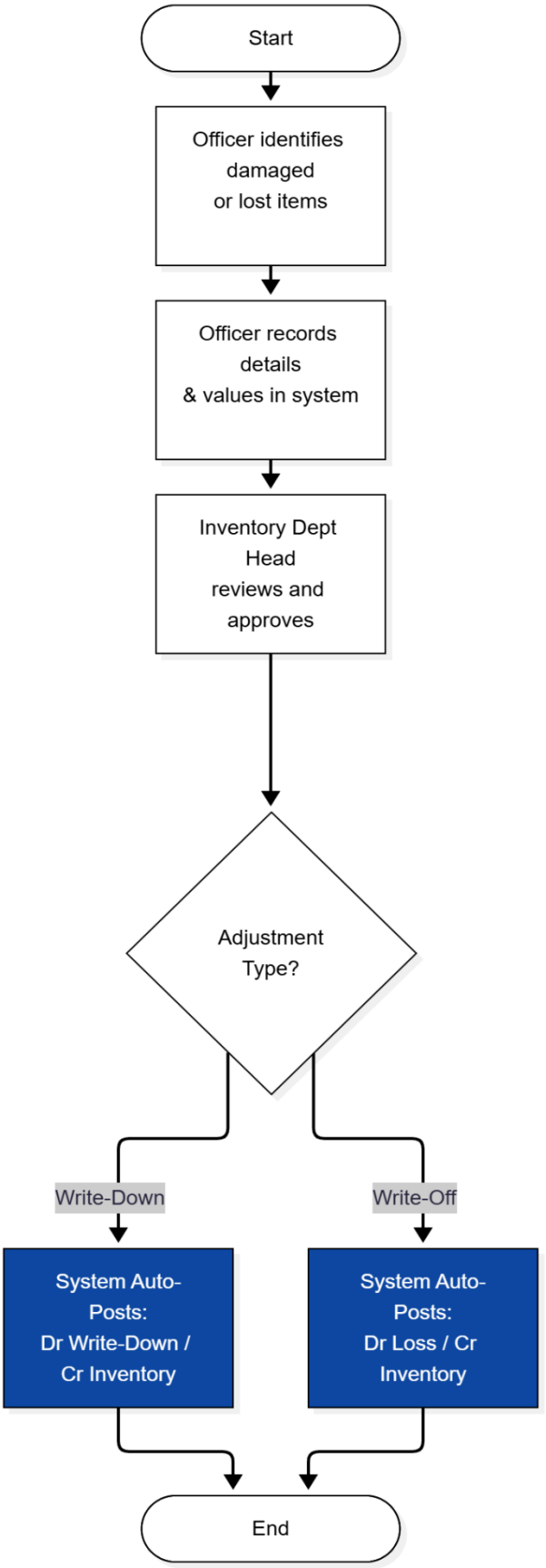
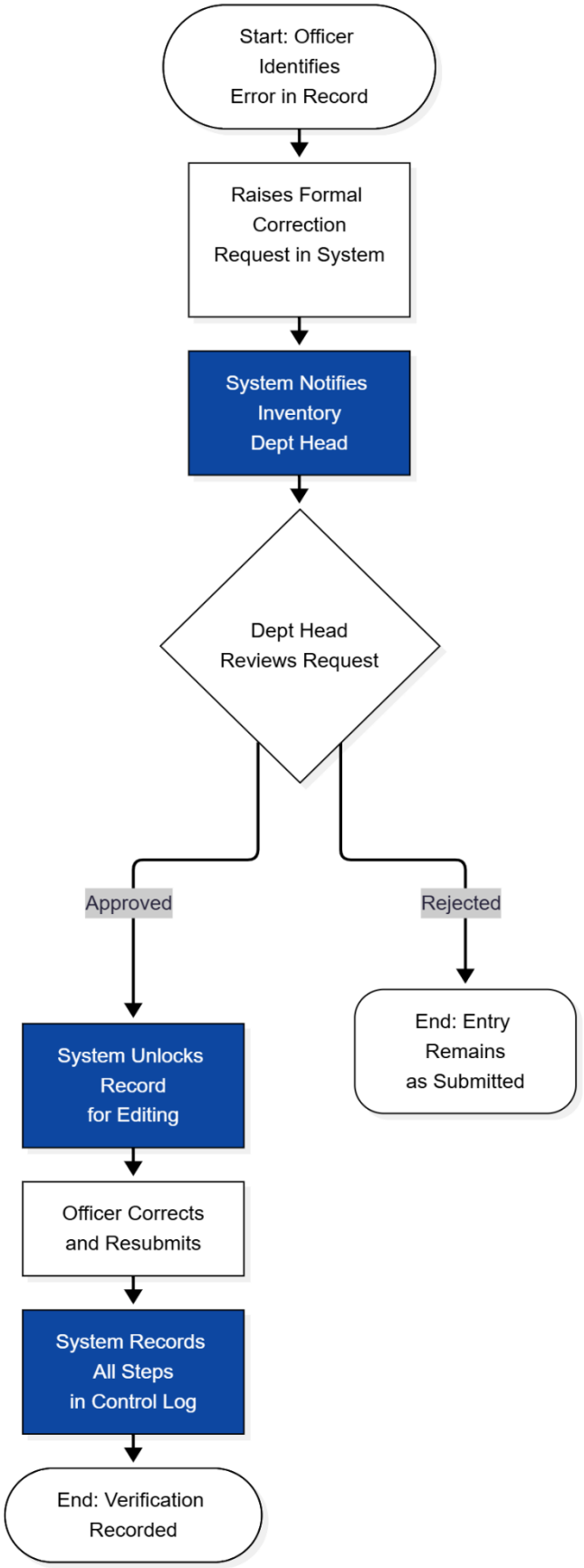


Figure A.2.5: Officer Correction Request



A.3. Payroll Workflows

Figure A.3.1: Leave Request (Permanent Employees Only)

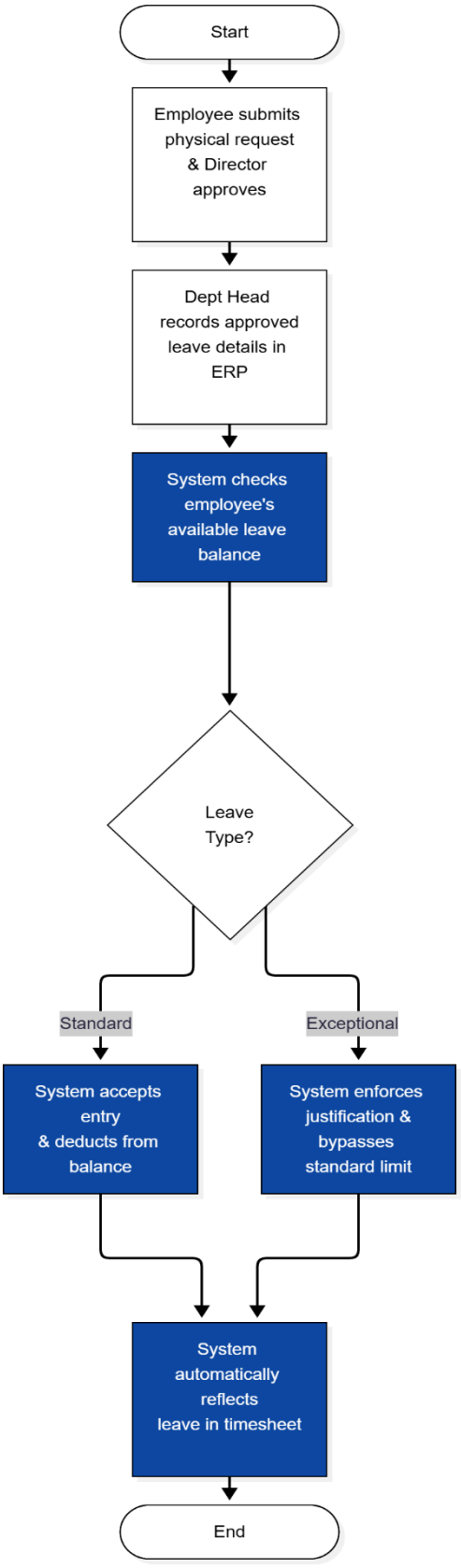


Figure A.3.2: Timesheet Recording and Finalization

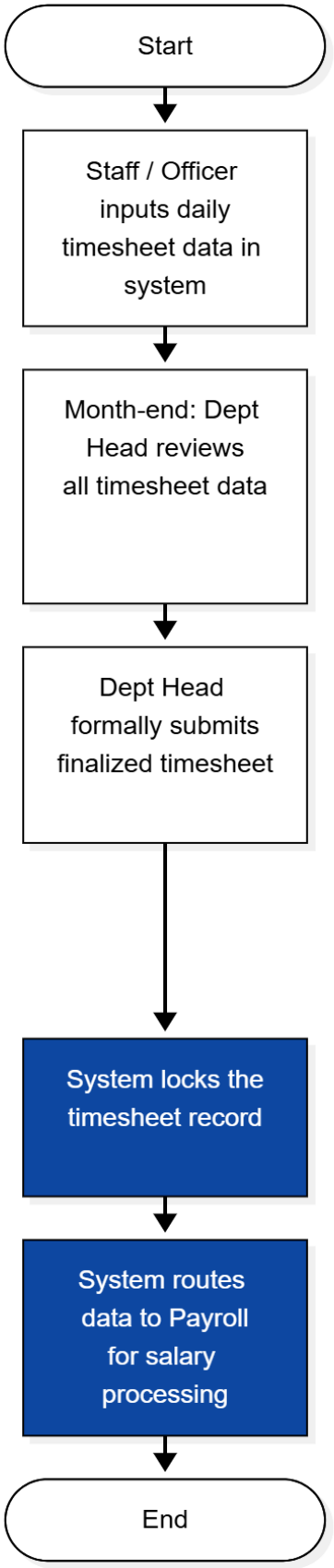


Figure A.3.3: Salary Advance Request and Processing

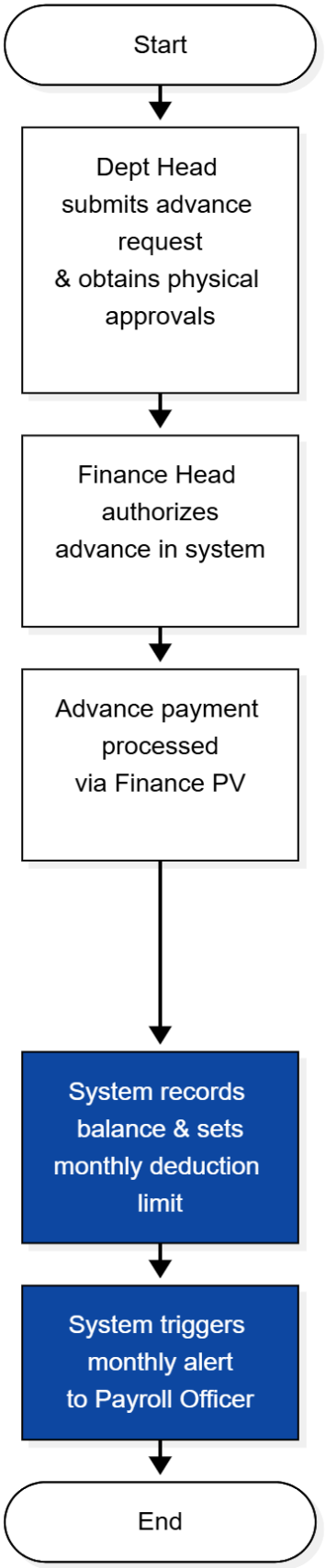


Figure A.3.4: Payroll Calculation, Finalization, and Payment (IC Verification Gate)

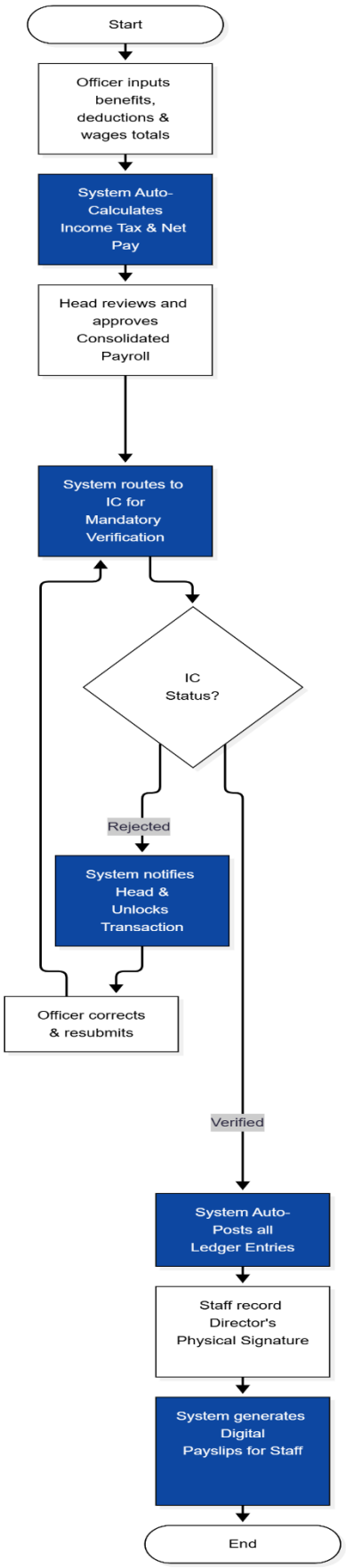
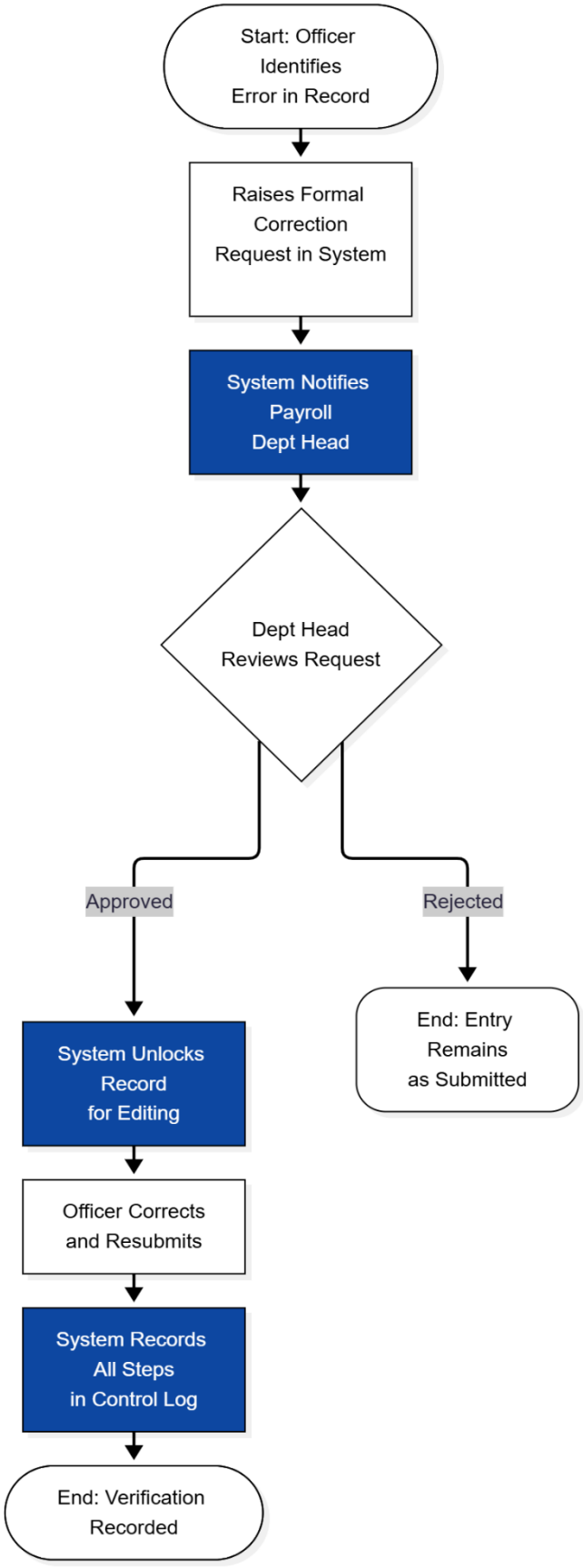


Figure A.3.5: Officer Correction Request Workflow



A.4. Sales and Marketing Workflows

Figure A.4.1: Sales Order and Tariff Issuance

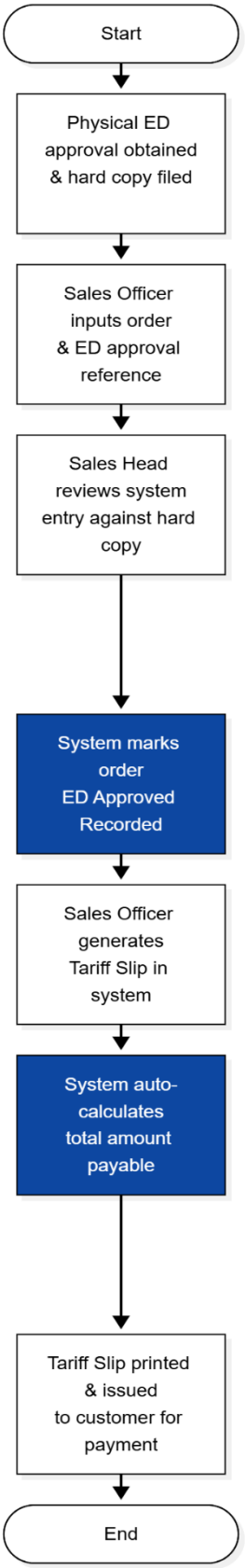


Figure A.4.2: Payment Confirmation and IC Verification Gate

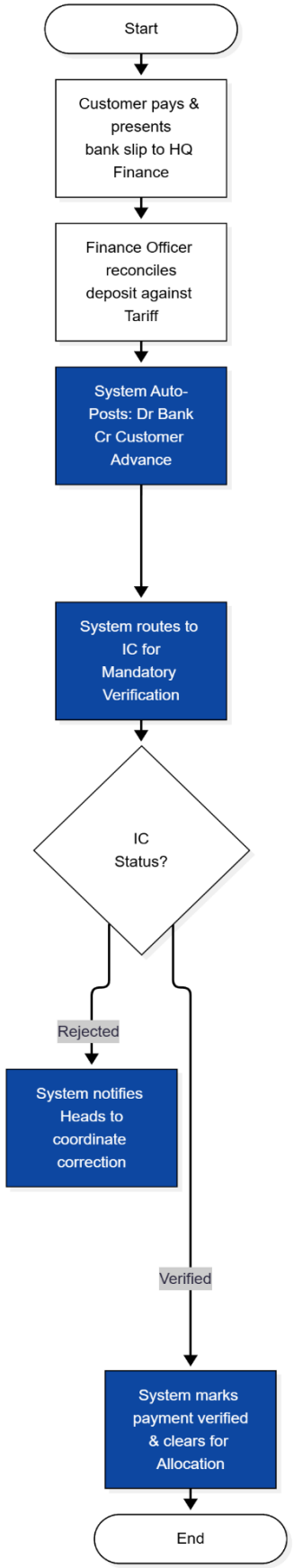


Figure A.4.3: Coal Allocation Authorization and Dispatch (Mining Sites)

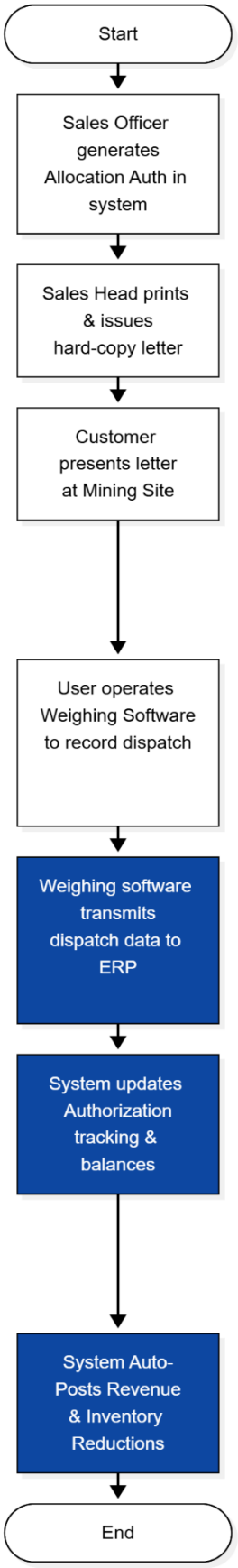


Figure A.4.4: Minor Sales / Distribution (Sub-Offices)

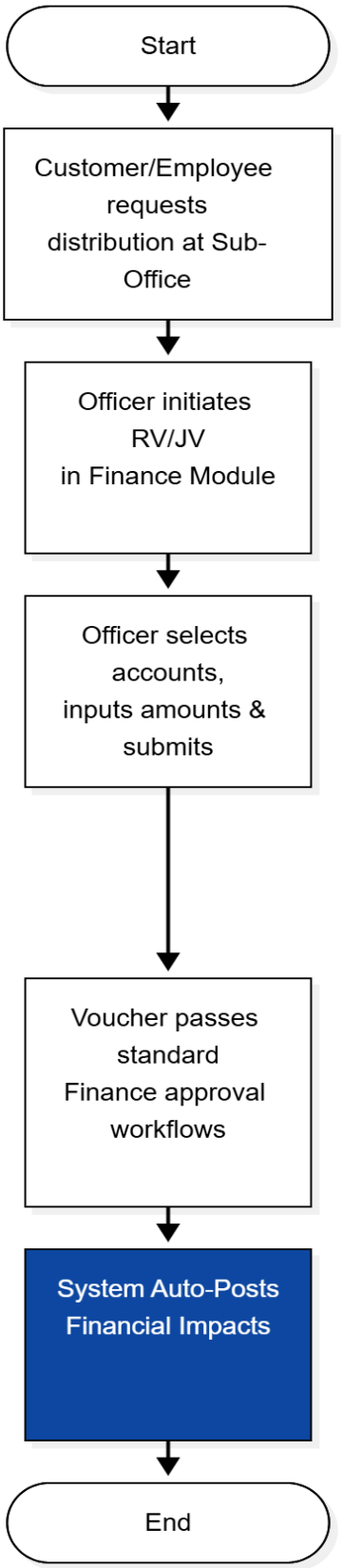
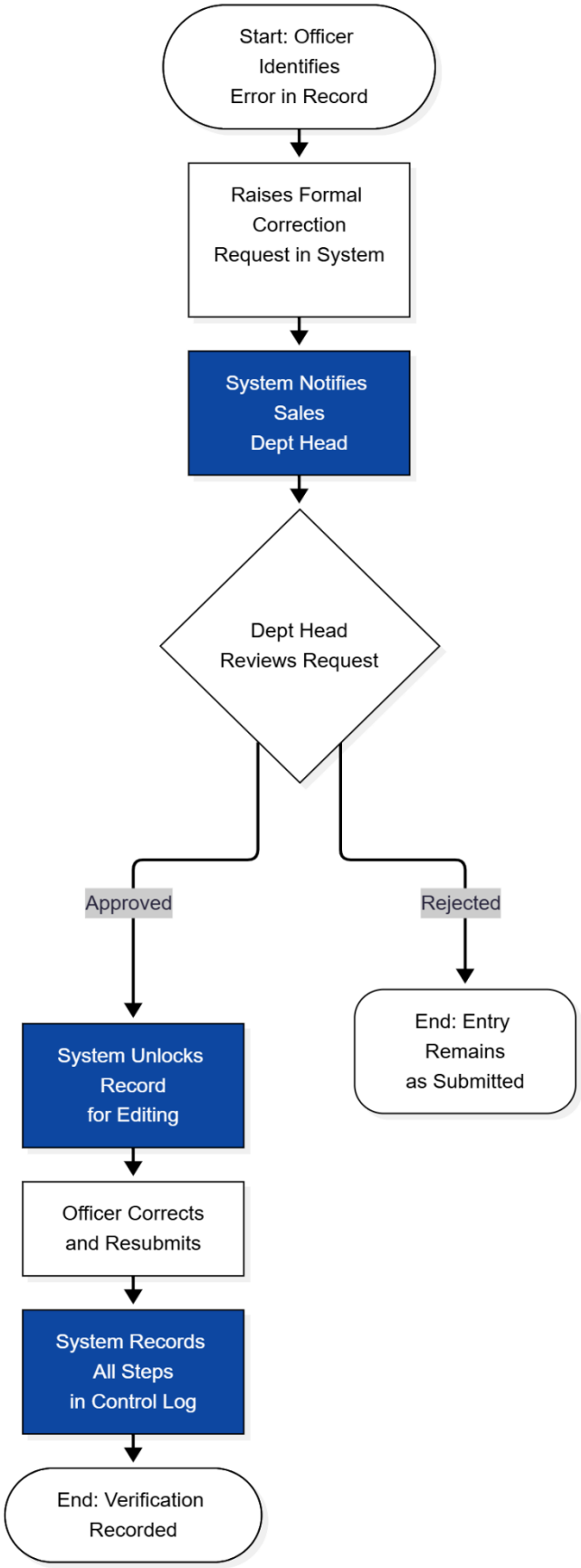


Figure A.4.5: Officer Correction Request Workflow



A.5. Properties Workflows

Figure A.5.1: Property Asset Registration

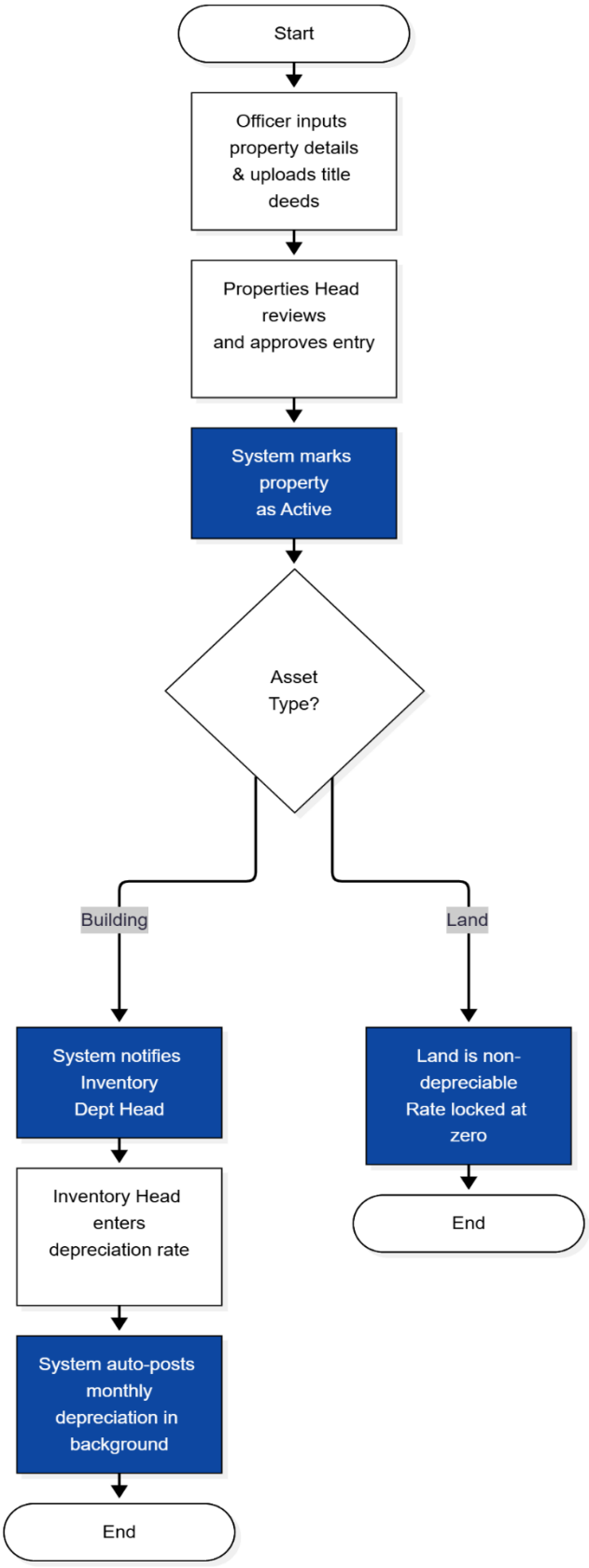


Figure A.5.2: Rental Income and BRT Auto-Posting

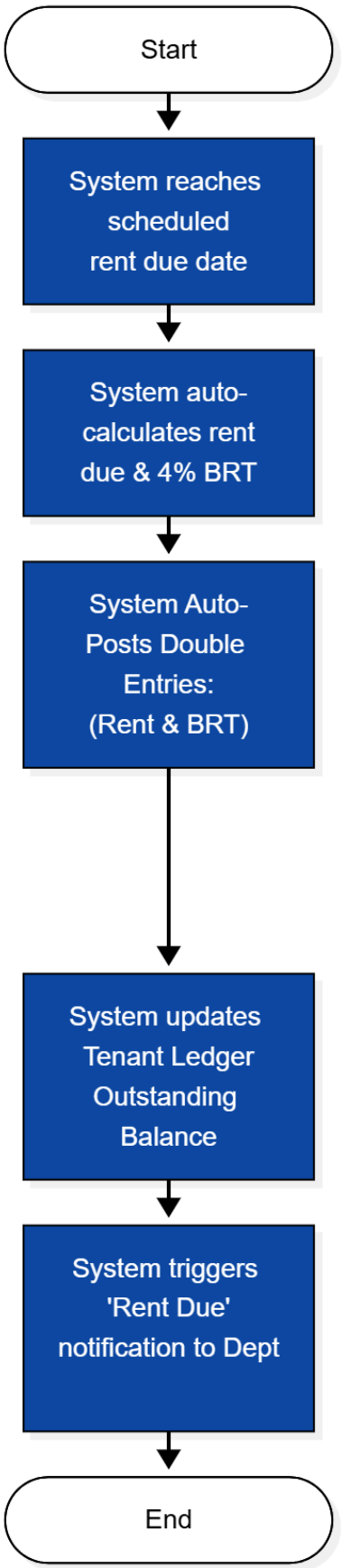


Figure A.5.3: Tenant Payment Receipt and IC Verification Gate

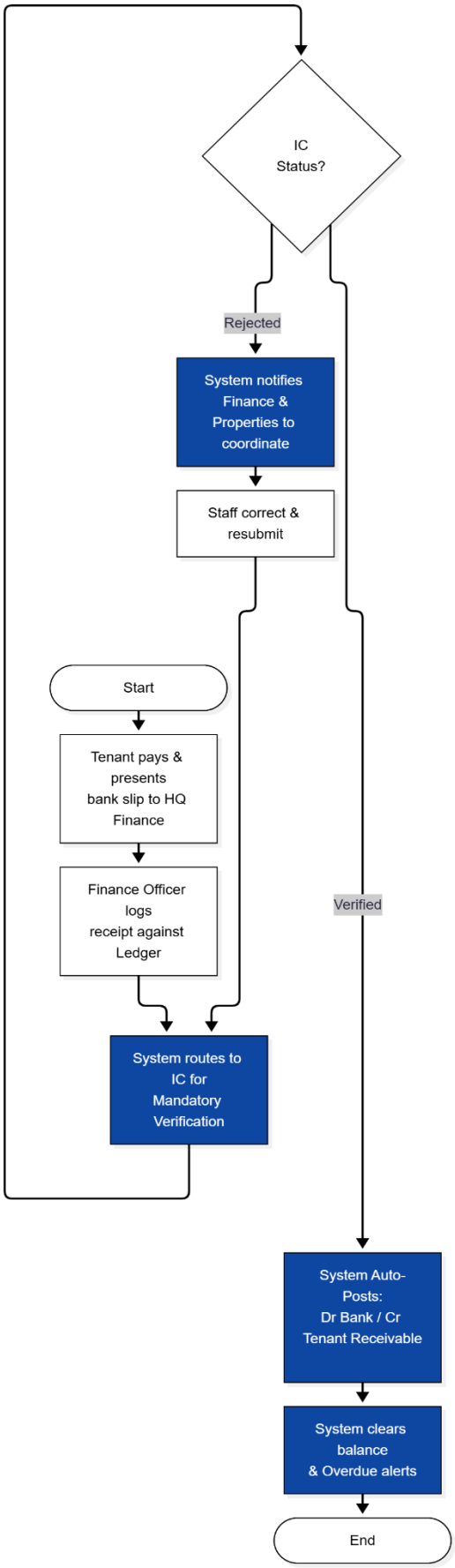
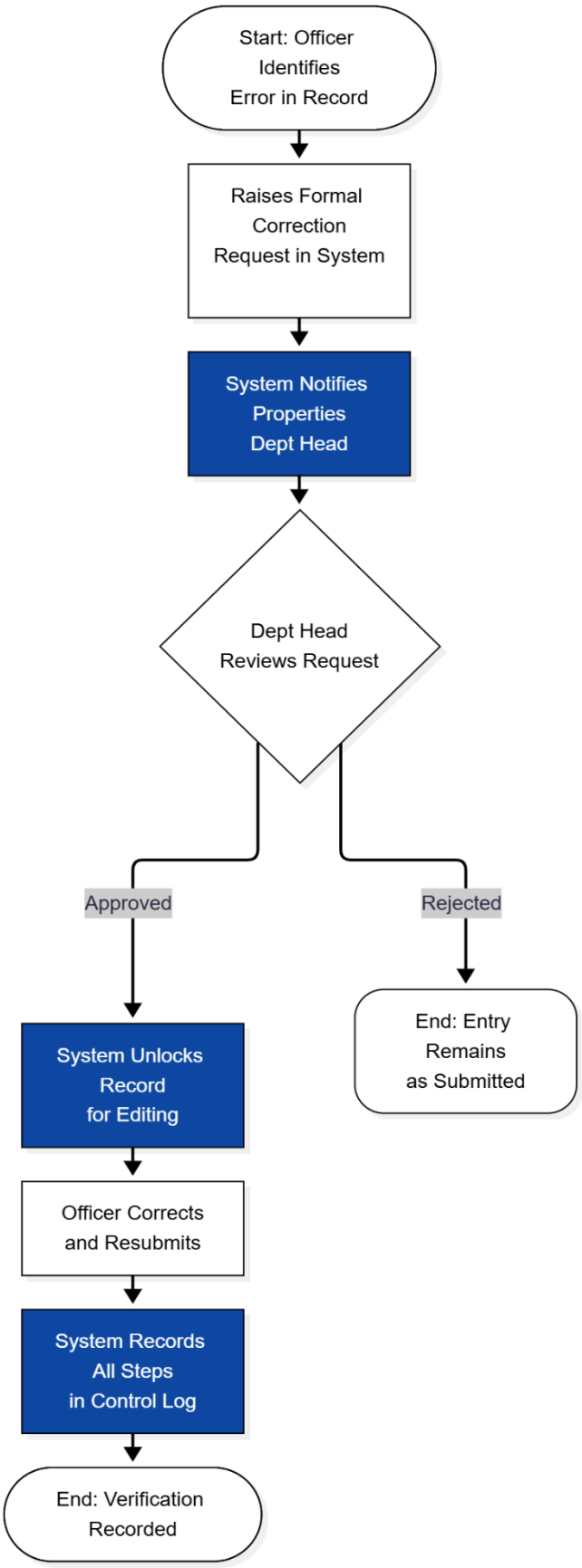


Figure A.5.4: Officer Correction Request



A.6. Procurement Workflows

Figure A.6.1: Purchase Request Receipt and Contract Setup

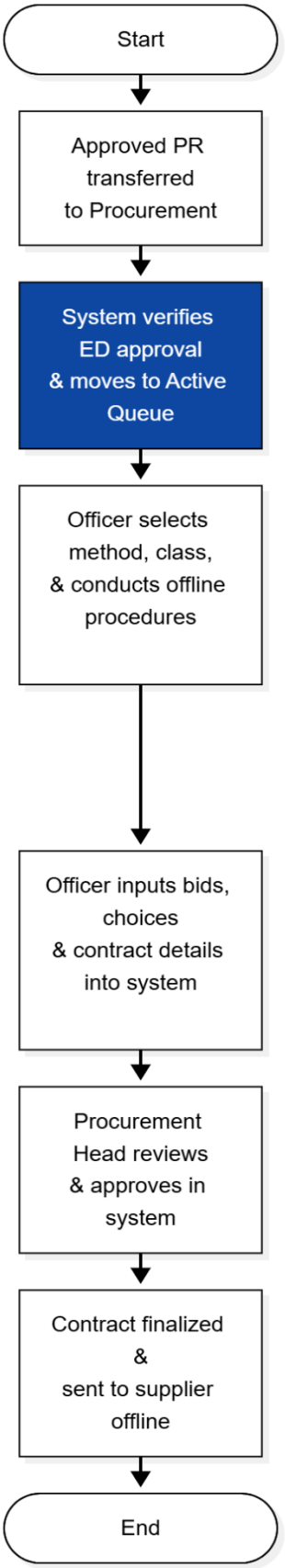


Figure A.6.2: Security Deposit Management

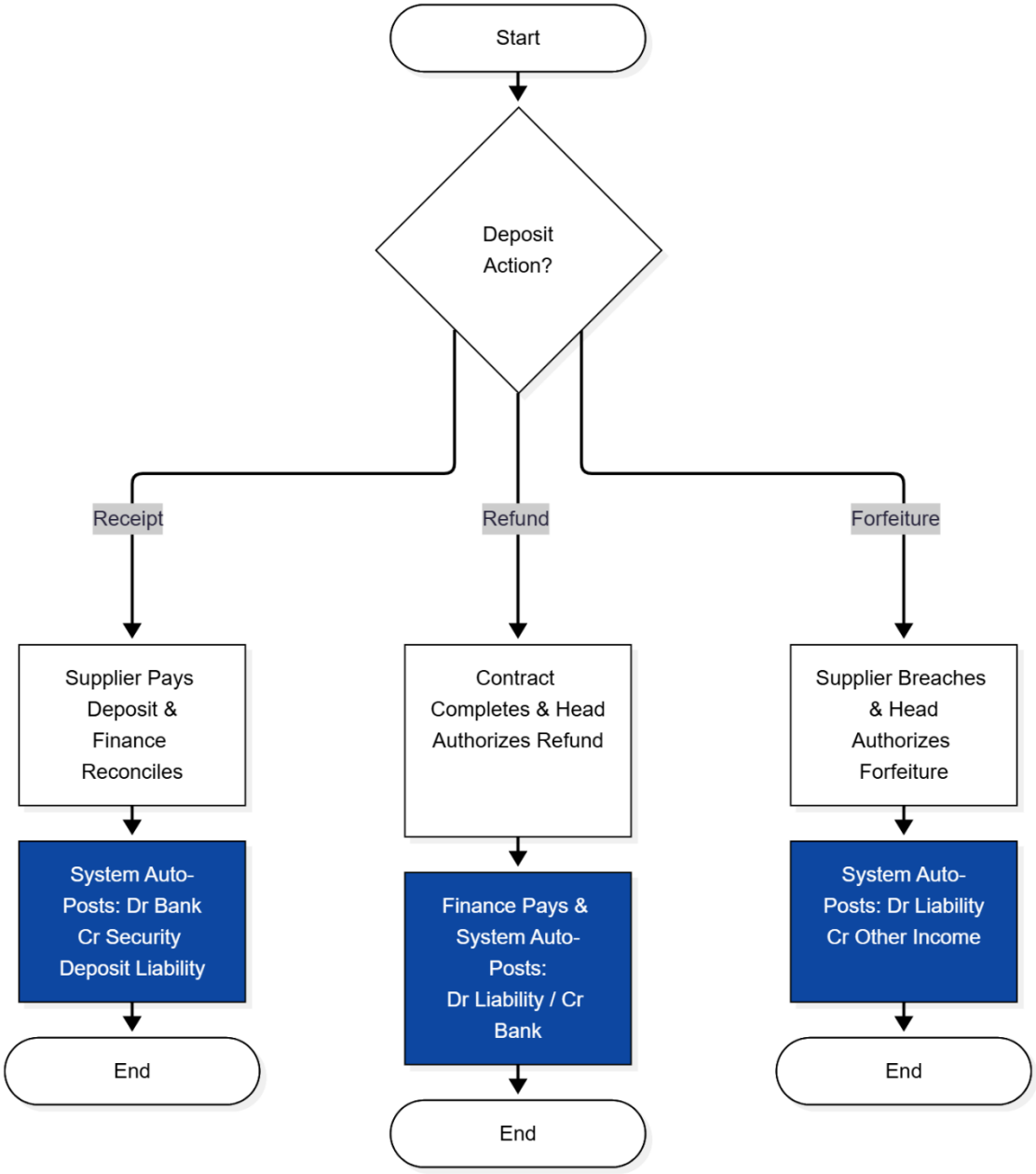


Figure A.6.3: Project Implementation Tracking and Contract Extension

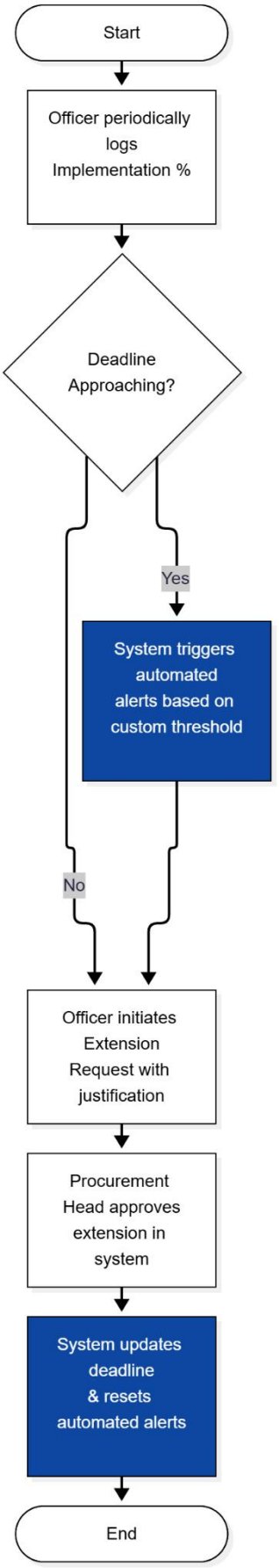


Figure A.6.4: Goods Receipt, Project Handover, and IC Verification Gate

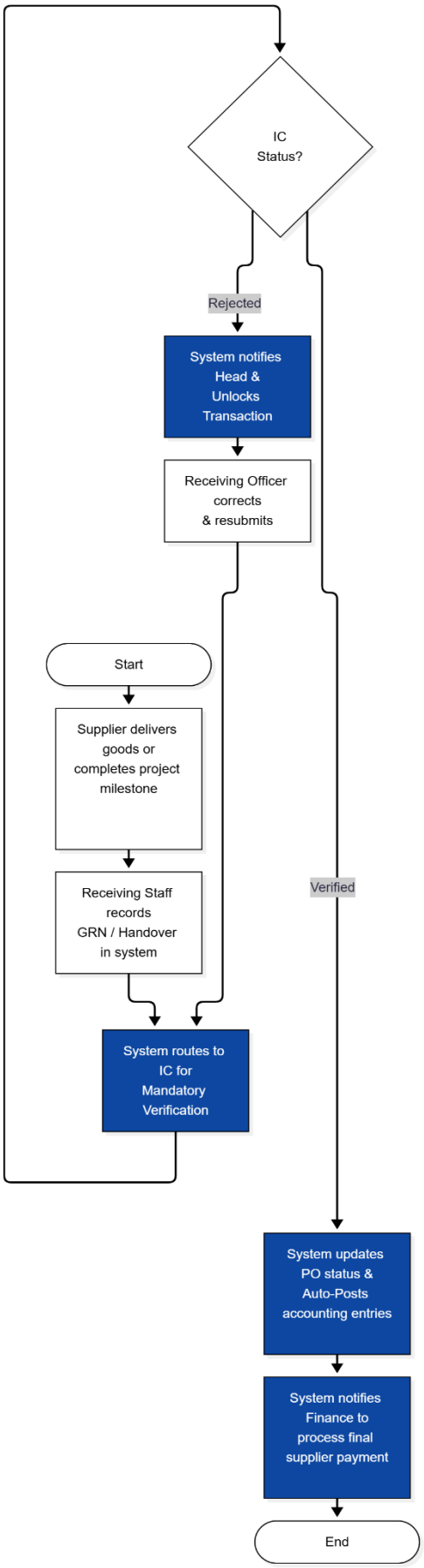
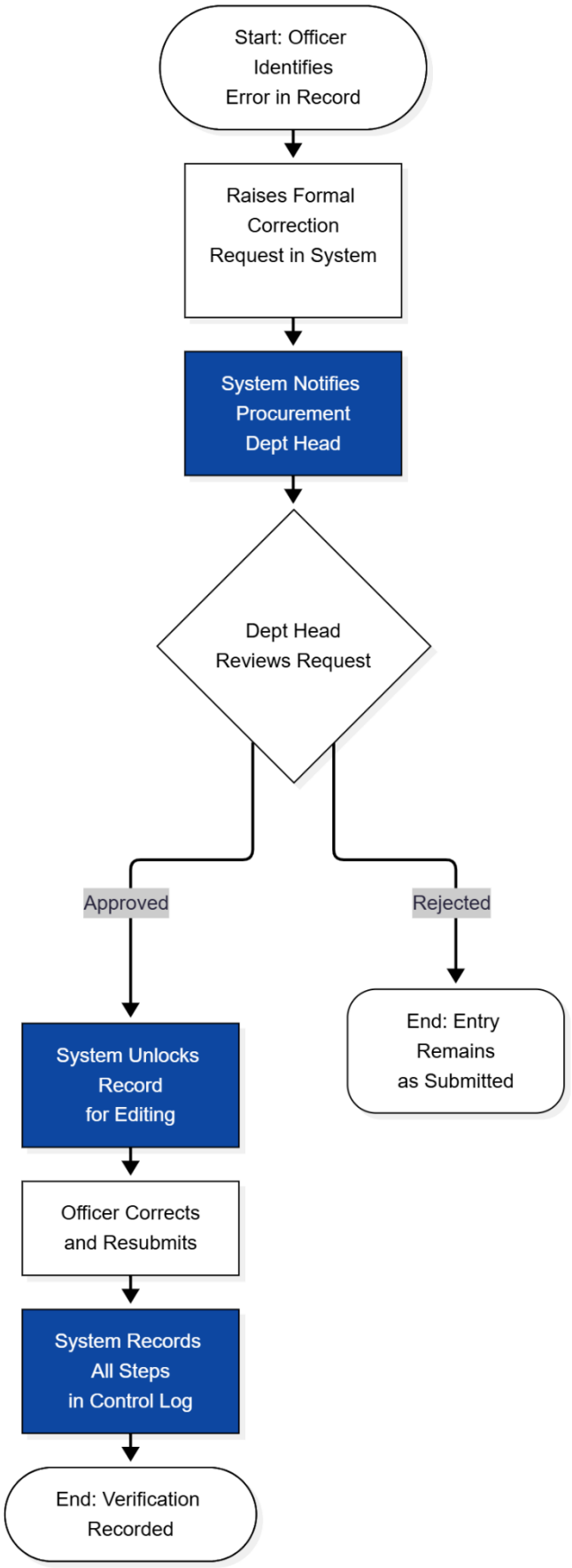


Figure A.6.5: Officer Correction Request Workflow



A.7. Planning and Policy Workflows

Figure A.7.1: Dynamic Plan Item Setup and Annual Plan Entry

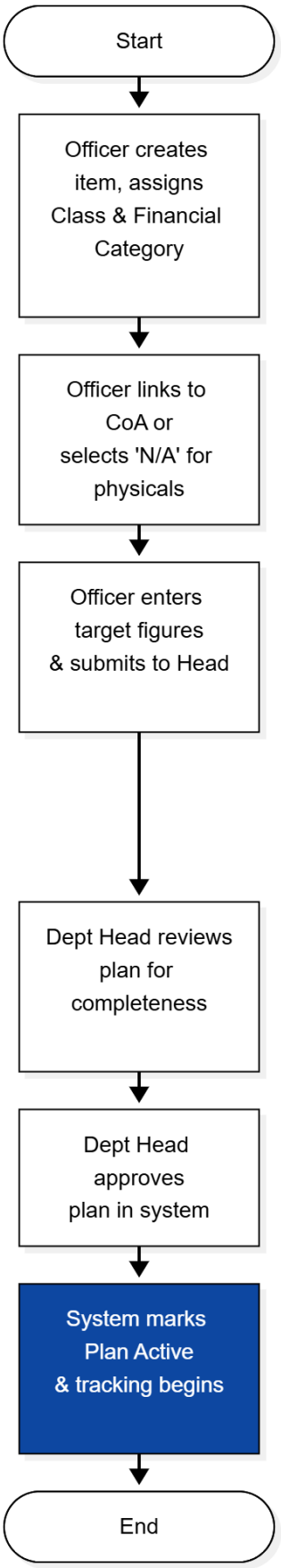


Figure A.7.2: Actuals Tracking and Real-Time Variance Monitoring

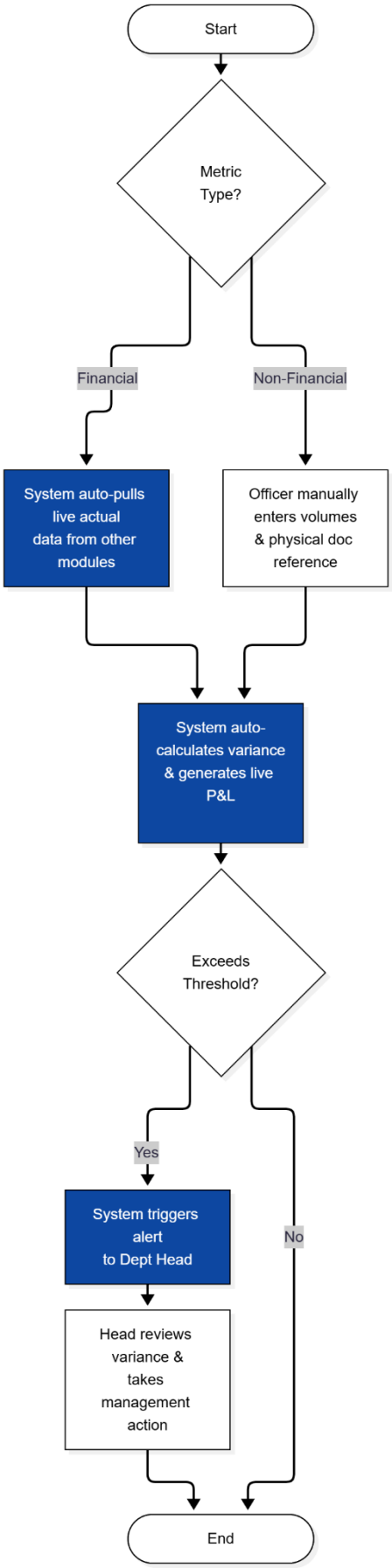


Figure A.7.3: Plan Amendment Protocol

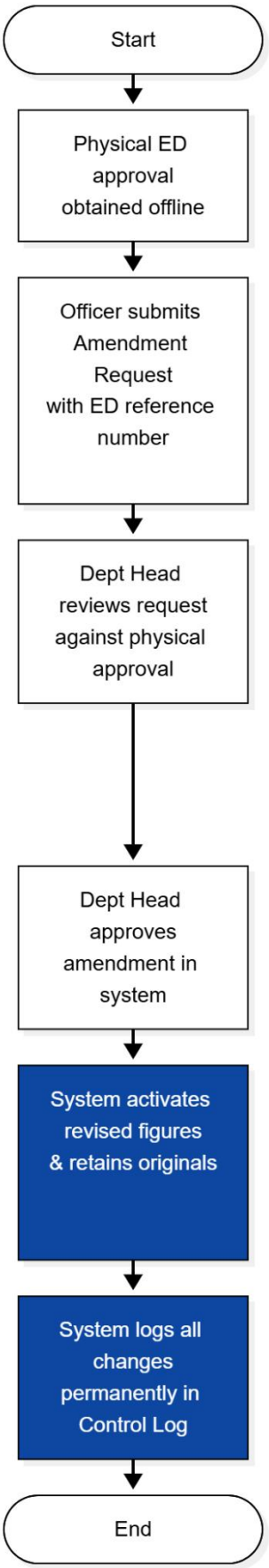
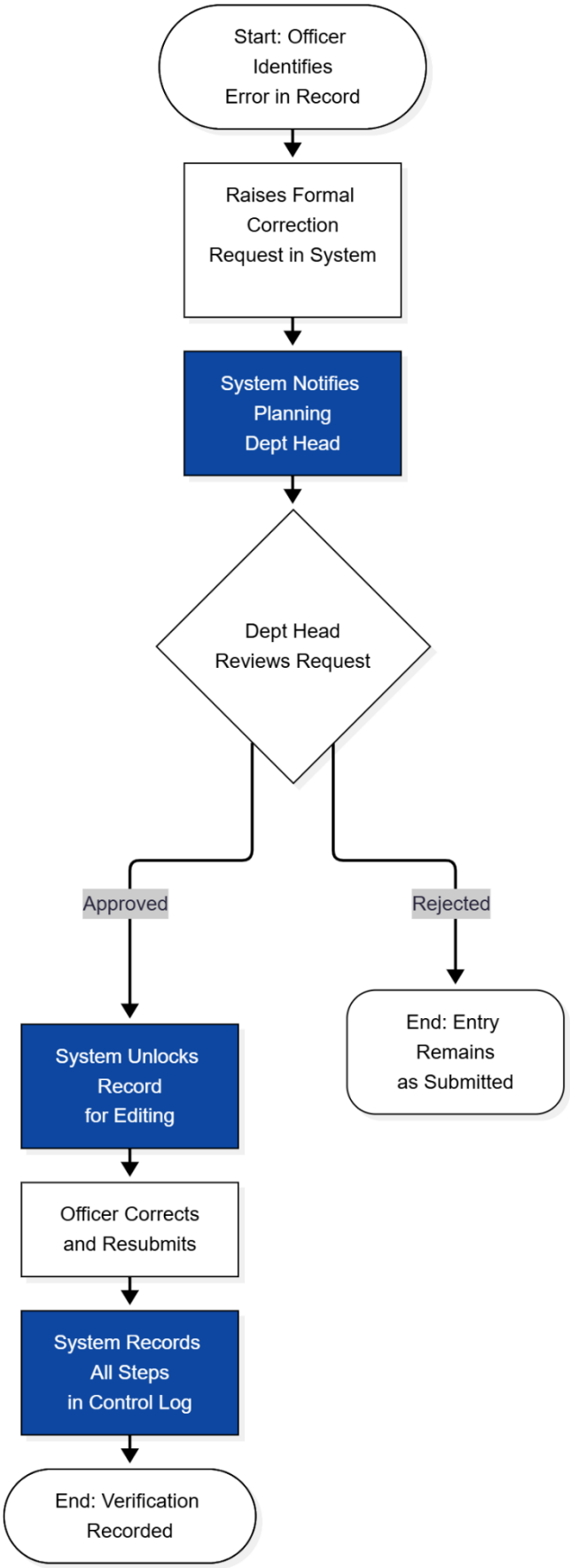


Figure A.7.4: Officer Correction Request





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